LARGE ERDŐS–STRAUS WITNESSES II: THE MULTIPLIER MODULUS IS INFINITELY OFTEN $\exp\left((\log p)^{1/4-o(1)}\right)$

ANONYMOUS

Date: Draft v6, October 2026.

Abstract. For a prime p let $W(p)$ be the least modulus $M \equiv 3 \pmod 4$ of a "multiplier" congruence that yields a solution of $4/p = 1/x+1/y+1/z$. We prove $W(p) \geq \exp\left(c(\log p)^{1/4}(\log\log p)^{-1/4}\right)$ for infinitely many primes p in Mordell's hard classes, assuming Gallagher's prime number theorem for the family of all characters and the Nair–Tenenbaum bound for short sums of multiplicative functions of polynomials; more precisely, the least hard prime with $W(p) > T$ is at most $\exp(O((\log T)^4 \log\log T))$. The proof quarantines n to a *random square class* modulo a growing modulus (the residues of the multiplier congruences are Jacobi non-residues, so no congruence is ever forced), controls the cost by a β-weighted local lemma and a supermartingale bookkeeping, and transfers the resulting Haar density to primes through the one-sided ℓ^2 sandwich of Bazzi and Razborov, an energy bound for avoidance indicators of cylinder events, and a linear transfer theorem based on Gallagher's theorem. We then describe the surrounding picture. The profinite density $\delta^*(T)$ of the integers with $W(n) > T$ satisfies $(\log T)^3 \ll \log(1/\delta^*(T)) \ll (\log T)^3(\log\log T)^5$ (upper bound modulo Nair–Tenenbaum; lower bound via a Janson-type inequality for one-hot product spaces with a $1/\log\log T$ loss, and without loss using the atoms of a companion note on the exceptional set). The same exponent governs the primes: $c(\log T)^3 \leq \log(\pi(x)/\#\{p \leq x : W(p) > T\}) \leq C(\log T)^3(\log\log T)^3$ for $\log T \leq c'(\log x/\log\log x)^{1/4}$; the lower bound for the count is proved by summing the transfer over all outcomes of the random quarantine. The exponent $1/4$ is the limit of the minorant architecture: on every fibre $n \equiv r \pmod Q$ with $\log Q \leq T^{0.05}$, no minorant of $\mathbf{1}[W(n) > T]$ of level $\exp(c(\log T)^4/\log\log T)$ (or $\exp(c(\log T)^4)$, using the companion note) has positive Haar mean. This follows from a planting argument, the dual of a lower-bound sieve of dimension $\asymp (\log T)^3$, and the planted law is exponentially pseudorandom against classes and characters of every modulus. Hence linear certificates with full-orbit uniform accounting cannot pass $1/4$ whatever the prime input (Bombieri–Vinogradov, Elliott–Halberstam, GRH below x, Siegel-model main terms, each through its consequences for prime counts at the scale x). Conversely, a Linnik-type hypothesis for sifted sets gives $W(p) \geq \exp((\log p)^{1/3-o(1)})$ infinitely often, the exponent suggested by the Haar density, while a Hardy–Littlewood-type hypothesis for product sets certifies only $W(p) \geq (\log p)^{2+o(1)}$. For the analogous witness W_m of m/n the event classes are Jacobi non-residues for every event exactly when $m \equiv 0 \pmod 4$; then the exponent $1/4$ and the Haar exponent 3 persist, while for every m, including Sierpiński's $5/n$, we get the exponent $1/5$, and $1/4$ under a bounded-drift hypothesis (these m/n results are proved in internally reviewed working notes and summarised here). All results are proved modulo the cited theorems listed in the text; the companion note is internally proved but not externally refereed.

1. Introduction

The Erdős–Straus conjecture asserts that $4/n = 1/x+1/y+1/z$ is solvable in positive integers for every $n \geq 2$. It suffices to treat primes. It has been verified for $n \leq 10^{18}$ ([32]; [42] for 10^{17}), and Mordell's identities settle every

prime outside the six classes $p \equiv 1^2, 11^2, 13^2, 17^2, 19^2, 23^2 \pmod{840}$ [31] (cf. [37]). We call the primes in these classes *hard*. For the exceptional set see Vaughan [50] and [38]; for the average number of solutions see [11].

For $M \equiv 3 \pmod 4$ put $A_M = (M+1)/4$. If $uvw = A_M$ and $n \equiv -uv^{-1} \pmod M$, then $s = (nv+u)/M$ is a positive integer and

$$\frac{4}{n} = \frac{1}{suw} + \frac{1}{nsvw} + \frac{1}{nuvw} \tag{1}$$

(direct verification; this is the classical multiplier family, cf. [50, 11]). Define

$$W(n) = \min\{M \equiv 3\ (4) :\ n \equiv -uv^{-1} \pmod M \text{ for some } uvw = A_M\}, \tag{2}$$

with $W(n) = +\infty$ if no such M exists. Thus $W(p) \le T$ provides an explicit representation with modulus at most T. Every prime $p \equiv 1 \pmod{\operatorname{lcm}\{M \le T\}}$ has $W(p) > T$ (Lemma 2.3), so Linnik's theorem gives $W(p) \gg \log p$ infinitely often (the classical template of Ω-results obtained by fixing a residue class by the Chinese remainder theorem and then applying Linnik's theorem, as for least quadratic non-residues [15]). In [2] we proved that for every fixed k, $W(p) \ge (\log p)^k e^{-C_k \log\log p/\log\log\log p}$ for infinitely many hard primes. Here we prove a sub-exponential rate.

Theorem 1.1 (Proved modulo Theorems 5.2 and 3.1)**.** *There is an absolute constant $c > 0$ such that for infinitely many hard primes p*

$$W(p) \ \ge\ \exp\Bigl(c\,\frac{(\log p)^{1/4}}{(\log\log p)^{1/4}}\Bigr).$$

More precisely, for every sufficiently large T there is a hard prime p with $p > T$, $W(p) > T$ and $\log p \le C(\log T)^4 \log\log T$.

Theorem 5.2 is Gallagher's prime number theorem for the family of all primitive characters of conductor at most Q_G [14], quoted from [27, Thm. 28.19] together with the standard facts on exceptional zeros (Theorem 5.1). No hypothesis on Siegel zeros is needed. Theorem 3.1 is the Nair–Tenenbaum bound [33] for sums of multiplicative functions of several polynomials, quoted in the form of Henriot [17]; it is used once, to bound Haar-weighted divisor sums over the moduli $M = 4n-1$ (Lemma 3.2). The Elsholtz–Tao estimates used in the third version are not needed. The constants in Theorem 1.1 are effective, except that we have not checked whether the implied constants in [33] are; the results proved modulo [46] below have ineffective constants. We make no numerical claim.

Typical size. Theorem 1.1 is about the extreme primes. For the typical prime, $W(p)$ is small: large values have a cubic-exponential tail.

Theorem 1.2 (Proved modulo [46] and Page's theorem)**.** *There are absolute $c, c_1 > 0$ such that, uniformly for $x \ge 3$ and $\log T \le c_1(\log x)^{1/4}$,*

$$\#\{p \le x :\ W(p) > T\} \ \ll\ \pi(x)\, e^{-c(\log T)^3}.$$

This is Theorem 11.2. It uses the atom family of the companion note [46] (internally proved, not externally refereed; inputs Bombieri–Vinogradov, Brun–Titchmarsh and Shiu) and, for small T, the prime number theorem in progressions with a Landau–Page term. At $\log T \asymp (\log x)^{1/4}$ it is the prime case of the exceptional-set bound $E(N) \ll N e^{-c(\log N)^{3/4}}$ of [46] (the same exponent was obtained earlier by Kandukoori, Ramesh and Kumar [45]). Together with Theorem 1.1 it describes both ends of the range: for $\log T \le c_1(\log x)^{1/4}$ the primes $p \le x$ with $W(p) > T$ have relative frequency at most $e^{-c(\log T)^3}$, and once $\log x \ge C(\log T)^4 \log\log T$ such primes exist. In fact, at the same threshold, they have relative frequency at least $e^{-C(\log T)^3(\log\log T)^3}$:

Theorem 1.3 (Tail exponent 3 over primes; proved modulo Theorems 5.2, 3.1, [46] and Page's theorem)**.** *There are absolute $c, C, c' > 0$ such that for large T and $\log T \le c'(\log x/\log\log x)^{1/4}$,*

$$\begin{aligned}\pi(x)\, e^{-C(\log T)^3(\log\log T)^3} \;&\le\; \#\{p \le x \text{ hard}:\ W(p) > T\}\\ &\le\; \#\{p \le x:\ W(p) > T\} \;\le\; \pi(x)\, e^{-c(\log T)^3}.\end{aligned}$$

The lower bound is proved modulo Theorems 5.2 and 3.1 alone, for all x with $\log x \ge C(\log T)^4 \log\log T$.

This is Theorem 12.1 and Corollary 12.2. The lower bound comes from the main term of the transfer theorem on each fibre of the random quarantine, summed over all outcomes of the quarantine process (§12); the losses are the Haar price of the square-class restriction, the residual mass, and a possible exceptional zero whose conductor divides the quarantine modulus.

The Haar exponent. Let $\delta^*(T)$ be the Haar measure of $\{n \in \hat{\mathbb{Z}}^\times :\ W(n) > T\}$ relative to the class $n \equiv 1 \pmod{24}$ (§3).

Theorem 1.4 (Haar exponent 3; proved modulo Theorem 3.1 and [46], see below)**.** *For large T,*

$$c\,(\log T)^3 \;\le\; \log\frac{1}{\delta^*(T)} \;\le\; C(\log T)^3(\log\log T)^5.$$

The upper bound is proved modulo Theorem 3.1 (Theorem 3.3). The lower bound is proved modulo [46] (Proposition 11.1); with $(\log T)^3/\log\log T$ in place of $(\log T)^3$ it is proved modulo the fundamental lemma of sieve theory alone (Theorem 4.4).

The ceiling. The exponent of $\log p$ in Theorem 1.1 is governed not by the Haar density but by the level of the minorant used to transfer it to primes, and this level cannot be lowered.

Theorem 1.5 (Proved modulo Theorems 5.1, 5.2 and the fundamental lemma; sharpened modulo [46])**.** *There are absolute $c, T_0 > 0$ such that for $T \ge T_0$, every modulus Q with $\log Q \le T^{0.05}$ and every unit r modulo Q: if B is a linear combination of functions $n \mapsto \phi(n \bmod qQ)$ with $\log q \le$*

$c(\log T)^4/\log\log T$, and $B(n) \le \mathbf{1}[W(n) > T]$ for all units $n \equiv r \pmod Q$, then the Haar mean of B on this class is ≤ 0. Modulo [46]*, the same holds with $\log q \le c(\log T)^4$.*

This is Theorem 10.6 and Proposition 11.3. Consequently (Corollary 10.7) every argument that certifies a hard prime with $W(p) > T$ through a minorant of positive Haar mean and a transfer theorem costing $\log x \gg \log(\text{level})$, as ours does, has $\log x \gg (\log T)^4/\log\log T$ (and $\gg (\log T)^4$ modulo [46]): such arguments cannot give $W(p) \ge \exp((\log p)^{1/4+\delta})$, and Theorem 1.1 is optimal among them up to a factor $(\log\log p)^{1/2}$ in $\log W(p)$, or $(\log\log p)^{1/4}$ modulo [46]. It also shows that the energy bound used for the sandwich cannot be sharpened for this purpose (Corollary 10.8). The level $(\log T)^4$ is the product of the log-size $\log T$ of the moduli and the dimension $(\log T)^3$ of the sieve formed by the events with one large prime factor. The same product, read at the budget $\log N$ of an integer sieve instead of the budget $\log x$ of a prime transfer, gives the exceptional-set exponent $3/4$ of [46] (Remark 11.4).

Beyond the ceiling. The planted law behind Theorem 1.5 is exponentially pseudorandom against residue classes, Dirichlet characters and additive characters of *every* modulus (Lemma 13.1). So every minorant B of any level has $\mathbb{E}B \le e^{-c(\log T)^4/\log\log T}\|B\|_\times$, where $\|B\|_\times$ is the ℓ^1 norm of its coefficients in classes and characters (Theorem 13.2). As a consequence, linear certificates with full-orbit uniform error accounting cannot beat the exponent $1/4$, whatever true prime input they use, including Bombieri–Vinogradov, Elliott–Halberstam, GRH restricted to moduli $\le x$, and Siegel-model main terms, each used through its consequences for the unweighted prime counts at the scale x (Corollary 13.3, Proposition 13.4). If primes behaved like a set of Haar density $\delta^*(T)$, Theorem 1.4 would suggest $\log W(p) \approx (\log p)^{1/3}$ for the extremal primes. We isolate a hypothesis that gives this.

Theorem 1.6 (Conditional on Hypothesis LS; proved modulo Theorem 3.1)**.** *Assume that every system of unit-class congruences $n \not\equiv a_E \pmod{m_E}$ with $m_E \le T$ whose solution set has Haar density $\delta > 0$ contains a prime $p > T$ with $\log p \le C(\log T + \log(1/\delta))$ (Hypothesis LS, "Linnik's theorem for sifted sets"). Then $W(p) \ge \exp\left(c_C(\log p)^{1/3}(\log\log p)^{-5/3}\right)$ for infinitely many hard primes p.*

This is Theorem 14.1. For a single residue class LS is Linnik's theorem. It is not reachable by the linear certificates above (§13), and the analogous hypothesis restricted to CRT product sets (a Hardy–Littlewood-type singular series) gives $W(p) \ge (\log p)^{2-o(1)}$ infinitely often and, applied to product subsets of the avoiding set, cannot certify more than $W(p) \ge (\log p)^{2+o(1)}$ (modulo the Barban–Davenport–Halberstam theorem) (Proposition 14.2). Altogether, with $\mathcal{L} = \log T$: the extremal primes have $\mathcal{L} \ge (\log p)^{1/4-o(1)}$ (proved), the minorant method stops at $(\log p)^{1/4+o(1)}$ (proved, within its scope), the Haar heuristic and LS give $(\log p)^{1/3-o(1)}$, and the primes with $W(p) > T$ have relative frequency $e^{-\mathcal{L}^{3+o(1)}}$ for $\mathcal{L} \le (\log p)^{1/4-o(1)}$.

Other numerators. The family (1) has an analogue (13) for every m/n, with witness W_m (§15). The only arithmetic input of the square-class quarantine that depends on m is the Jacobi symbol of the event classes, and it is -1 for every event exactly when $m \equiv 0 \pmod 4$ (Lemma 15.1, Proposition 15.2). Accordingly, for $m \equiv 0 \pmod 4$ the exponent $1/4$ and the Haar exponent 3 carry over (Theorem 15.3). For every m, including Sierpiński's $5/n$, the class-of-one architecture of the third version gives the exponent $1/5$ (Theorem 15.5), and the exponent $1/4$ follows from a bounded-drift hypothesis ADM_m for an admissible random process (Theorem 15.7), which we reduce to a hypothesis on the initial segment of the process but do not prove. These statements concern the family (13) only.

Versions. The first version of this note had exponent $1/14$; it used the Thorner–Zaman theorem [47] through the transfer theorem of [2]. The second had exponent $1/7$: it replaced Thorner–Zaman by the linear transfer theorem (Theorem 6.1 here), and bounded the approximation error by Håstad's switching lemma. The third had exponent $1/5$: it replaced the switching lemma by the energy bound (§7) and used a graded class-of-one quarantine, whose cost was bounded with four results of Elsholtz and Tao. The fourth version reached $1/4$ by a new quarantine: a random square class, a β-weighted local lemma and a supermartingale bookkeeping (§2) with Haar masses bounded by the Nair–Tenenbaum theorem (§3); it added the Haar lower bound (§4) and the ceiling (§10). The fifth version left the proof of Theorem 1.1 unchanged and added: the typical size, the Haar lower bound without logarithmic loss and the ceiling at level $(\log T)^4$, all via the atoms of [46] (§11); the Wiener-norm barrier and its consequences for linear certificates (§13); and the conditional exponent $1/3$ (§14). The present sixth version leaves all of this unchanged and adds the lower bound for the tail of W over primes, hence the two-sided tail exponent 3 (§12), and the section on other numerators (§15); the discussion of the literature was revised after a further novelty audit.

The method. Fix T. We restrict n to a class $n \equiv r \pmod Q$, with $Q = 8\prod_\ell \ell^{a_\ell}$ and r a quadratic residue modulo every prime of Q. Then $W(n) > T$ whenever n avoids a finite family of congruence *events* $n \equiv -4D \pmod m$, one for each surviving pair (M, D) with $M \le T$, $D \mid A_M^2$, where m is the part of M not absorbed by Q. The key observation (Lemma 2.2) is that the Jacobi symbol $(\frac{-4D}{M})$ is always -1; so if every prime of M divides Q, the event is excluded, and no event is ever forced, whatever square class r we choose. For $r = 1$ this is the class-of-one quarantine of [2]; Mordell's hard classes are the case $Q = 840$.

We build (Q, r) by a random process (Lemma 2.9). While the weighted mass $\tilde w_\ell = \sum_{E \ni \ell} \beta^{|\operatorname{supp} E|}\mathbb{P}(E)$ at a prime $\ell \le Y = \mathcal{L}^{O(1)}$ exceeds $\eta = \frac34 \log\beta$, we reveal one more digit of n at ℓ, uniformly among the squares. The β-weighted local lemma (Lemma 2.8) needs exactly $\tilde w_\ell \le \eta$ at every coordinate. With $\beta = 1 + 1/\log\mathcal{L}$ the threshold is $\asymp 1/\log\mathcal{L}$ at every prime and

every level, instead of $\asymp \log \ell/\mathcal{L}$ in the previous version. The masses form supermartingales (a step at a new prime at most doubles the mean of an event's probability, and this is paid by a factor $2^{\omega_Y(M)}$), so by optional stopping the expected cost is a Haar-weighted divisor sum, $\ll \mathcal{L}^3(\log \mathcal{L})^5$ by the Nair–Tenenbaum theorem. Primes $\ell > Y$ are never quarantined; a second-moment potential for pairs of events shows that their masses are small with high probability. The class of one, used before, inflates the masses by $\gcd(M, 4D+1)$, which costs a factor $\mathcal{L}$ and requires the Elsholtz–Tao estimates.

To find a *prime* in the avoiding set we need a function $B = \sum_i c_i \mathbf{1}[n \equiv b_i \ (d_i)]$ with $B \le \mathbf{1}[\text{avoid}]$ pointwise, mean $\gg \delta$ (the Haar density of the avoiding set on the fibre), Haar ℓ^1 norm $\mathbb{E}|B|$ comparable to its mean, and small moduli d_i. The linear transfer theorem (Theorem 6.1) then gives a prime $p \equiv r \pmod Q$ with $\log p \ll \log Z$, where Z is Q times the largest d_i. It is Linnik's theorem with weights: the function $B(n)\mathbf{1}[n \equiv r \ (Q)]$ is expanded in characters, every coefficient is bounded by the Haar ℓ^1 norm of B, whatever the number of cells, and Gallagher's theorem bounds the contribution of all characters of conductor at most a power of Z at once. Since r is a square, every real character that factors through $(\mathbb{Z}/Q)^\times$ is 1 at r, so the exceptional-zero analysis is the same as for the class of one.

As in the previous versions, B is the one-sided ℓ^2 sandwich of Bazzi [3], in the form of Razborov [40] (Lemma 8.2). For events $E_1, \dots, E_m$ with indicators A_i and arbitrary functions u_j,

$$B = 1 - \sum_i A_i \Big(1 - \sum_{j<i} A_j u_j\Big)^2 \ \le \ \mathbf{1}[\text{no } E_i \text{ occurs}],$$

and, if each u_j depends only on the coordinates not fixed by E_j, $\mathbb{E}[F - B] \le m^2 \sum_j \mathbb{P}(E_j)\, \mathbb{E}(F^{(j)} - u_j)^2$, where $F^{(j)}$ is the indicator that no earlier event occurs, conditioned on E_j. The u_j are truncations of the Efron–Stein decomposition of $F^{(j)}$, and the error is an Efron–Stein tail, which we bound by the energy bound of the previous version (Theorem 7.5): if F is the avoidance indicator of any family of cylinder events on a finite product probability space, and $\lambda_v \ge 1$ are weights with $\prod_{v \in \operatorname{supp} E} \lambda_v \le 2$ for every event E, then

$$\sum_U \Big(\prod_{v \in U} \lambda_v \Big) \|F^{=U}\|_2^2 \ \le \ 1.$$

Applied to the base-ℓ digits of the residues, with weights adapted to the moduli (Theorem 7.9), it shows that truncating $F^{(j)}$ to moduli at most e^τ costs $2^{-\tau/(2\log T)}$. So moduli of size $\log d_i \ll \mathcal{L} \cdot (S + \mathcal{L})$ suffice, where $S \ll \mathcal{L}^3 \log \mathcal{L}$ is the residual mass. Altogether $\log Z \ll \mathcal{L}^4 \log \mathcal{L}$. The ceiling (Theorem 10.6) shows that the level $\mathcal{L}S$ is forced up to a factor $(\log \mathcal{L})^2$: conditioned on the small primes, the events with one large prime factor form a sieve of dimension $\asymp \mathcal{L}^3/\log \mathcal{L}$ on the primes above $T^{0.6}$, and a planting

lemma (Lemma 10.1), the dual of a lower-bound sieve, shows that no function of lower level can be a minorant with positive mean.

Relation to the literature. *The energy bound.* The nearest prior art known to us is Lecomte and Tan [24], who bound the Fourier coefficients of a DNF on $\{\pm1\}^n$ by the probability that a set S is covered by the terms satisfied at a random point; this is the device of our Lemma 7.2. Their cover counts are unsigned, and their degree concentration still comes from Håstad's switching lemma, with an unspecified constant. In Lemma 7.2 the cover counts are signed, so that they vanish when a satisfied event misses the set; the polarization identity (Lemma 7.3) and the deletion–contraction bound (Lemma 7.4) then give the constant 1, on any finite product probability space. In Boolean language (Remark 7.8): every width-k DNF $g : \{\pm1\}^n \to \{\pm1\}$ has $\sum_{|S|>t} \hat{g}(S)^2 \le 4 \cdot 2^{-(t+1)/k}$, under any product measure, while the switching-lemma route gives $2 \cdot 2^{-t/(20k)}$ [34, §4.4]. The base $2^{-1/k}$ is sharp over general product spaces (Remark 7.7). Theorem 7.5 and its corollaries are new to us, but our literature search was partial (we checked [24], [34] and some lecture notes; we did not search the literature on switching-lemma variants, Fourier growth or hypercontractivity with $\rho > 1$ systematically), and we make no priority claim.

The local lemmas. The β-weighted local lemma (Lemma 2.8) is a standard per-coordinate form of the asymmetric local lemma of [1, Lemma 5.1.1] with the choice $x_E = \beta^{|\operatorname{supp} E|}\mathbb{P}(E)$ (size-exponential weights are the classical device for non-uniform hypergraph colouring); the conditional form for events outside the family is standard (cf. [18]). We claim no novelty for it; the point specific to this note is the choice $\beta = 1 + 1/\log \mathcal{L}$, which makes the threshold independent of the prime and of the level. Theorem 4.3 combines the lopsided local lemma of Erdős and Spencer [10] with the proof scheme of Janson's inequality [23], which needs increasing events and Harris' inequality; for atomic events on one-hot coordinates we use instead the correlation inequality of Lemma 4.1. This is a routine combination of the Boppana–Spencer proof of Janson's inequality with the inflation bound of [18]; we found no statement in this form, but did not search systematically (the work of Lu and Székely on the local lemma for random injections is the closest suspected prior art, unchecked). The planting lemma (Lemma 10.1) is the linear-programming dual of lower-bound sieves; constructions of k-wise independent laws with extremal all-zero or all-one probabilities of the same kind appear in [4, 39]. In the identical-marginal case it gives, after flipping the bits, $n_c(k,p) \le (k+1)p/(1-p) + 2k + 2$ for the threshold $n_c(k,p)$ of [4, Thm. 27] (the least n such that some k-wise independent law on n bits of mean p gives probability 0 to the all-ones vector). This removes the factor $\log(1/(1-p))$ and the restriction to $1/(1-p)$ a prime power from their upper bound, and is within a factor 3 of their lower bound for $p \ge 1/2$ (a factor $2 + O(1/k)$ as $p \to 1$); we have not checked the later literature. Its use to

bound the level of minorants of a concrete arithmetic indicator, on every fibre of modulus $\log Q \le T^{0.05}$, is new to us.

The exponent 3. Elsholtz and Tao [11, Remark 1.2] observe that their count of solutions suggests, heuristically, that a prime p has a representation with probability $1 - O(\exp(-c \log^3 p))$, and that Vaughan's large-sieve bound [50] is a rigorous analogue of this reasoning. Theorem 1.4 is a rigorous profinite version of this exponent 3 for the witness modulus; what is new there is the matching lower bound and the identification of the exponent up to logarithms. Theorem 11.2 is Vaughan's sieve method applied to W (new as a statement about W, not in method), and Corollary 12.2 shows that the exponent 3 is also the tail exponent over primes, in the range $\log T \le (\log x)^{1/4-o(1)}$ (far from the scale $\log T \asymp \log p$ of the heuristic). The classical obstruction that polynomial identities cannot cover quadratic residue classes ([31]; Schinzel and Yamamoto [49], cited via [11, Prop. 1.6]: there are no Type I or Type II solutions for odd perfect squares) is qualitative. Lemma 2.2 and its m/n version Lemma 15.1 are the form of it that we use; for the family (13) it holds for every event exactly when $m \equiv 0 \pmod 4$ (with Proposition 15.2).

The rest. The sandwich (Lemma 8.2) is Razborov's identity. To our knowledge tools from the theory of bounded independence ("polylogarithmic independence fools DNF and AC^0" [3, 40, 7]) have not been used before to construct sieve minorants for primes. Gallagher's theorem is classically the engine of Linnik's theorem on the least prime in a single progression. Here it serves as a transfer device for an arbitrary signed combination of progressions; the cost is governed by the Haar ℓ^1 norm of the minorant and its largest modulus, not by the number of cells. Theorem 6.1 is a convenient packaging of Gallagher's proof of Linnik's theorem, not a new transfer principle; inserting sieve weights into Linnik-range arguments is standard. The nearest classical analogues are bounds for the least prime in a union of residue classes, or in a Chebotarev class of an abelian extension (Lagarias–Montgomery–Odlyzko [28]; Thorner–Zaman [48]); these concern nonnegative indicators. We have not searched systematically for an earlier use of Gallagher's theorem in this role. Earlier uses of the switching lemma in number theory known to us run in the opposite direction: they show that the Möbius function is uncorrelated with bounded-depth circuits [16, 6]. The Haar side, a local-lemma lower bound for the density of the uncovered set of a congruence system, has relatives in the work on covering systems: Filaseta, Ford, Konyagin, Pomerance and Yu [12], who to our knowledge already use the local lemma for this purpose, Hough [20] and Balister, Bollobás, Morris, Sahasrabudhe and Tiba [5]. We do not use those results. The Jacobi-symbol property of the multiplier classes (Lemma 2.2) is implicit in the classical fact that the identities cover all non-residue classes; its use to quarantine to random square classes is ours. The divisor-sum bounds use Nair–Tenenbaum [33, 17] in a standard way. Hypothesis LS (§14) is a uniform version of Linnik's theorem for sifted sets; we did not search the literature for it systematically and claim no priority for the formulation.

Status conventions. Every theorem header carries a label. *Proved* means proved in full here. *Proved modulo X* means proved here assuming the cited statement X, whose statement we checked in the source but whose proof we did not check. *Cited* marks external statements; *Classical* marks standard results for which we include a proof or a precise textbook reference. The cited inputs are Gallagher's theorem (Theorem 5.2, used together with the classical Landau–Page facts of Theorem 5.1), the Nair–Tenenbaum theorem (Theorem 3.1) and, for Theorems 4.4 and 10.6, the fundamental lemma of sieve theory. In §§11–14 three further inputs appear: the companion note [46], which is *internally proved but not externally refereed* ("proved modulo [46]" means proved here assuming its stated results, with their inputs Bombieri–Vinogradov, Brun–Titchmarsh and Shiu); the prime number theorem in progressions with a Landau–Page term ("Page's theorem", (12)); and the Barban–Davenport–Halberstam theorem. Unlike the other cited inputs, the exact form of these last two was recalled from the textbooks [8, Chs. 20, 29], not re-checked against the source (no copy archived); our use needs only their standard forms. In §15 the Elsholtz–Tao estimates [11] and Henriot's coefficient-uniform form of the Nair–Tenenbaum theorem [17, Thm. 5] enter, through the working notes [29, 30]. Results quoted from the working notes [35, 36, 29, 30] and from [43] (internally reviewed, not externally refereed) are labelled as proved *in* those notes; their proofs are sketched or referenced here, not given in full. *Conditional on LS* marks a statement that assumes the conjectural Hypothesis LS of §14. In the labels, "modulo Theorem 5.2" includes Theorem 5.1. Classical results whose proof we include, or which are standard textbook facts (the local lemma, the Efron–Stein decomposition, Mertens' theorem, the divisor bound, the Rosser–Schoenfeld bound for $\vartheta(x)$, mean values of non-negative multiplicative functions via Euler products), are not listed in the labels. *Proved implication* marks a statement about a class of arguments. *Assessment* marks heuristic comparisons, which are not claims. Nothing here bears on the truth of the Erdős–Straus conjecture: a prime with large $W(p)$ may well have other representations.

Notation. T is large and $\mathcal{L} = \log T$; $\log_2$ always denotes the binary logarithm (iterated logarithms are written out as $\log\log$). τ is the divisor function (and, in §§8–9, also a truncation parameter; no confusion arises). $\omega(m)$ and $\Omega(m)$ count prime factors without and with multiplicity; $\omega_Y(m)$ counts the prime factors $\le Y$, and m_Y is the Y-smooth part of m. A y-rough integer has all prime factors $> y$. For $\lambda = (\lambda_v)$ and a finite set U we write $\lambda^U = \prod_{v\in U}\lambda_v$.

2. Atoms, square classes and the local lemma

For $M \equiv 3 \pmod 4$ put $\mathcal{R}(M) = \{-4D \bmod M : \ D \mid A_M^2\}$. Note that $\gcd(A_M, M) = 1$ (as $4A_M = M+1$), so every element of $\mathcal{R}(M)$ is a unit class.

Lemma 2.1 (Proved; cf. [2])**.** $\{-uv^{-1} \bmod M : \ uvw = A_M\} = \mathcal{R}(M)$. *Consequently, for every integer n, $W(n) > T$ if and only if $n \bmod M \notin \mathcal{R}(M)$ for every $M \le T$ with $M \equiv 3 \pmod 4$.*

Proof. Since $4A_M \equiv 1 \pmod M$, from $uvw = A_M$ we get $v^{-1} \equiv 4uw$, hence $-uv^{-1} \equiv -4u^2w$ and $D = u^2w$ divides A_M^2. Conversely let $D \mid A_M^2$. For each prime q with $q^a \| A_M$ and $q^d \| D$ ($d \le 2a$) put $u_q = \max(0, d-a)$, $w_q = d - 2u_q$, $v_q = a - u_q - w_q$; these are non-negative, so $u = \prod q^{u_q}$, $v = \prod q^{v_q}$, $w = \prod q^{w_q}$ satisfy $uvw = A_M$ and $u^2w = D$. □

An *atom* is a pair (M, D) with $M \le T$, $M \equiv 3 \pmod 4$ and $D \mid A_M^2$. There are at most T^2 atoms, since $\tau(A_M^2) \le 2A_M \le T$.

Lemma 2.2 (Event classes are Jacobi non-residues; proved)**.** *Let (M, D) be an atom and let d be the squarefree part of D. Then*

(a) *every prime $\ell \mid M$ satisfies $\ell \nmid 2D$ and $\left(\frac{-4D}{\ell}\right) = \left(\frac{-d}{\ell}\right)$;*
(b) *the Jacobi symbol $\left(\frac{-4D}{M}\right)$ equals -1.*

Consequently, if r is an integer coprime to M that is a quadratic residue modulo every prime dividing M, then $r \not\equiv -4D \pmod M$.

Proof. (a) $\gcd(M, 4A_M) = 1$ and every prime of D divides A_M, so $\ell \nmid 2D$; and $-4D = -d \cdot (2t)^2$ for an integer t. (b) By (a), $\left(\frac{-4D}{M}\right) = \left(\frac{-d}{M}\right) = \left(\frac{-1}{M}\right)\left(\frac{d}{M}\right) = -\left(\frac{d}{M}\right)$, as $M \equiv 3 \pmod 4$. Every prime $p \mid d$ has $v_p(D)$ odd and $v_p(D) \le 2v_p(A_M)$, so $p \mid A_M$ and $M = 4A_M - 1 \equiv -1 \pmod p$. For odd p, quadratic reciprocity, with $(M-1)/2$ odd, gives $\left(\frac{p}{M}\right) = \left(\frac{M}{p}\right)(-1)^{(p-1)/2} = \left(\frac{-1}{p}\right)(-1)^{(p-1)/2} = 1$. If $p = 2$, then $2 \mid A_M$, so $M \equiv 7 \pmod 8$ and $\left(\frac{2}{M}\right) = 1$. Hence $\left(\frac{d}{M}\right) = 1$. For the consequence, $\left(\frac{r}{M}\right) = \prod_{\ell \mid M} \left(\frac{r}{\ell}\right)^{v_\ell(M)} = 1 \ne \left(\frac{-4D}{M}\right)$. □

We checked (a) and (b) by computer on all 2 401 032 atoms with $M \le 2 \cdot 10^5$ (no failures; this check is not used). The lemma is the mechanism behind two classical facts: the identities (1) cover the non-residue classes, and Mordell's hard classes modulo 840 are exactly the six square classes. For $r = 1$ it contains Lemma 2.3 below.

Lemma 2.3 (Class of one; proved)**.** $1 \notin \mathcal{R}(M)$ *for every* $M \equiv 3 \pmod 4$.

Proof. Lemma 2.2 with $r = 1$. □

Definition 2.4 (Square-class quarantine and residual system)**.** For an odd prime $\ell \le T$ let $f_\ell = \lfloor \mathcal{L} / \log \ell \rfloor \ge 1$ be the largest exponent with $\ell^{f_\ell} \le T$. A *square-class quarantine* is a pair (Q, r), where $Q = 8 \prod_\ell \ell^{a_\ell}$ with integers $0 \le a_\ell \le f_\ell$ for the odd primes $\ell \le T$, and r is a unit modulo Q with $r \equiv 1 \pmod 8$ that is a quadratic residue modulo every odd prime dividing Q. The *coordinate primes* are the odd $\ell \le T$ with $a_\ell < f_\ell$; write $\mathcal{P}$ for their set. For $\ell \in \mathcal{P}$ let

$$\Omega_\ell = \{x \in (\mathbb{Z}/\ell^{f_\ell})^\times : \ x \equiv r \pmod{\ell^{a_\ell}}\}$$

(all units if $a_\ell = 0$), with the uniform measure, and let $\mathbb{P}$, $\mathbb{E}$ be the product measure on $\Omega = \prod_{\ell\in\mathcal{P}} \Omega_\ell$, with coordinates X_ℓ. An atom *survives* (Q, r) if $-4D \equiv r \pmod{\gcd(M, Q)}$ and $M \nmid Q$. Its *support* is $\operatorname{supp}(M, D) = \{\ell : v_\ell(M) > a_\ell\}$, a nonempty subset of $\mathcal{P}$, with $s(M, D) = |\operatorname{supp}(M, D)| \le \omega(M)$; its *modulus* is $\prod_{\ell\in\operatorname{supp}} \ell^{v_\ell(M)}$, and its *event* is

$$E_{M,D} = \{X_\ell \equiv -4D \pmod{\ell^{v_\ell(M)}} \text{ for all } \ell \in \operatorname{supp}(M, D)\}.$$

For $\beta \ge 1$ the *weighted fibre mass* at $\ell \in \mathcal{P}$ and the weighted total mass are

$$\tilde{w}_\ell = \sum_{\substack{(M,D) \text{ surviving} \\ \ell\in\operatorname{supp}(M,D)}} \beta^{s(M,D)}\mathbb{P}(E_{M,D}), \qquad S_\beta = \sum_{\substack{(M,D) \\ \text{surviving}}} \beta^{s(M,D)}\mathbb{P}(E_{M,D}),$$

where surviving atoms are counted individually (two atoms may give the same event). For an integer n we put $X_\ell(n) = n \bmod \ell^{f_\ell}$, and $F(n) = 1$ if n lies in no event of a surviving atom, $F(n) = 0$ otherwise; on Ω, F is the indicator that no event $E_{M,D}$ occurs.

Every event is nonempty: $-4D$ is a unit modulo ℓ, and $-4D \equiv r \pmod{\ell^{a_\ell}}$ by survival. The modulus is at most $M \le T$. The class-of-one quarantines of [2] and of the previous versions are the case $r = 1$.

Lemma 2.5 (Reduction and weights; proved)**.** *Let (Q, r) be a square-class quarantine.*

(i) *If $n \equiv r \pmod{Q}$ and $F(n) = 1$, then $W(n) > T$.*

(ii) *For every surviving atom,*

$$\mathbb{P}(E_{M,D}) = \frac{\gcd(M, Q)}{M} \prod_{\ell\in\operatorname{supp}(M,D),\ a_\ell=0} \frac{\ell}{\ell - 1}.$$

(iii) *If $105 \mid Q$, then $r \bmod 840$ is one of Mordell's six hard classes, so every prime $p \equiv r \pmod{Q}$ is hard.*

Proof. (i) Let $M \le T$, $M \equiv 3 \pmod 4$, and suppose $n \equiv -4D \pmod M$ with $D \mid A_M^2$. Reducing modulo $\gcd(M, Q)$ gives $-4D \equiv r$. If $M \mid Q$, then $r \equiv -4D \pmod M$, and every prime of M divides Q, so r is a quadratic residue modulo it; this contradicts Lemma 2.2. Otherwise the atom survives, and n lies in its event, so $F(n) = 0$. Conclude with Lemma 2.1. (ii) Let $\ell \in \operatorname{supp}$ and $v = v_\ell(M)$. If $a_\ell = 0$, the class of $-4D$ modulo ℓ^v has probability $1/\varphi(\ell^v) = \ell^{-v} \cdot \ell/(\ell - 1)$ in Ω_ℓ. If $a_\ell \ge 1$, then $|\Omega_\ell| = \ell^{f_\ell - a_\ell}$, and the class (which lies in Ω_ℓ by survival) has $\ell^{f_\ell - v}$ elements, so probability $\ell^{a_\ell - v}$. As M is odd, $\gcd(M, Q) = \prod_{\ell\mid M} \ell^{\min(v_\ell(M), a_\ell)}$, and $\min(v_\ell(M), a_\ell) = v_\ell(M)$ for $\ell \notin \operatorname{supp}$. (iii) A unit modulo 840 that is $\equiv 1 \pmod 8$ and a square modulo 3, 5 and 7 is a square modulo 840, and there are $1 \cdot 1 \cdot 2 \cdot 3 = 6$ such classes; they are the six listed classes. □

Remark 2.6 (The graded class-of-one quarantine)**.** In the previous version $r = 1$, and the exponents a_ℓ were raised while the fibre mass exceeded $c(a_\ell + 1)\log \ell/\mathcal{L}$. With $r = 1$ a surviving atom has $\gcd(M, Q) \mid g =$

$\gcd(M, 4D+1)$, so by (ii) its weight is up to g/M instead of the Haar weight $1/\varphi(M)$. Summed over the atoms, the Haar weights give $O(\mathcal{L}^3 \log \mathcal{L})$ (Lemma 3.2), while for the inflated weights we could only prove $O(\mathcal{L}^4)$, and that needed the Elsholtz–Tao divisor-sum estimates [11]. The threshold $\propto \log \ell/\mathcal{L}$ cost a further factor $\mathcal{L}$ in the quarantine. A *random* square class removes the inflation in expectation (Lemma 2.9), Lemma 2.2 guarantees that no event is ever forced, and the β-weighted local lemma (Lemma 2.8) replaces the log-weighted threshold by a constant one. The graded class-of-one quarantine is therefore superseded, and we omit it.

The local lemma.

Lemma 2.7 (Local lemma with conditioning; classical)**.** *Let $\mathcal{A}$ be a finite family of events in a product space, each depending on a finite set of coordinates; write $A' \sim A$ if they share a coordinate. Suppose $x_A \in [0,1)$ satisfy $\mathbb{P}(A) \le x_A \prod_{A' \sim A,\, A' \neq A}(1 - x_{A'})$. Then $\mathbb{P}(\bigcap_{\mathcal{A}} \bar{A}) \ge \prod_{\mathcal{A}}(1 - x_A)$, and $\mathbb{P}(A \mid \bigcap_{A' \in \mathcal{S}} \bar{A}') \le x_A$ for every $\mathcal{S} \subseteq \mathcal{A}$ with $A \notin \mathcal{S}$. Moreover, for every event B depending on a set J of coordinates and every $\mathcal{S} \subseteq \mathcal{A}$, $\mathbb{P}(B \mid \bigcap_{\mathcal{S}} \bar{A}) \le \mathbb{P}(B) \prod_{A \in \Gamma(B)} (1 - x_A)^{-1}$, where $\Gamma(B)$ is the set of $A \in \mathcal{S}$ depending on some coordinate in J.*

Proof. The first two statements are [1, Lemma 5.1.1] and its proof. For the third (cf. [18, Thm. 2.1]), B is independent of $\bigcap_{A \in \mathcal{S} \setminus \Gamma(B)} \bar{A}$, so

$$\mathbb{P}\Big(B \;\Big|\; \bigcap_{\mathcal{S}} \bar{A}\Big) \le \frac{\mathbb{P}(B)\, \mathbb{P}(\bigcap_{\mathcal{S} \setminus \Gamma(B)} \bar{A})}{\mathbb{P}(\bigcap_{\mathcal{S}} \bar{A})} = \frac{\mathbb{P}(B)}{\mathbb{P}(\bigcap_{\Gamma(B)} \bar{A} \mid \bigcap_{\mathcal{S} \setminus \Gamma(B)} \bar{A})},$$

and the chain rule with the second statement bounds the denominator below by $\prod_{\Gamma(B)}(1 - x_A)$. □

Lemma 2.8 (β-weighted local lemma; proved)**.** *Let $\mathcal{A}$ be a finite family of events on a product probability space, each determined by the coordinates in a nonempty set $\operatorname{supp} A$, and put $s(A) = |\operatorname{supp} A|$. Let $1 < \beta \le e^{1/3}$, $\eta = \frac{3}{4} \log \beta$, and $\tilde{w}_v = \sum_{A \ni v} \beta^{s(A)} \mathbb{P}(A)$, the sum over the A with $v \in \operatorname{supp} A$. Suppose that $\tilde{w}_v \le \eta$ for every coordinate v. Then*

$$\mathbb{P}\Big(\bigcap_{\mathcal{A}} \bar{A}\Big) \;\ge\; \exp\Big(-\frac{4}{3} \sum_{A \in \mathcal{A}} \beta^{s(A)} \mathbb{P}(A)\Big),$$

and for every event B determined by a coordinate set J and every $\mathcal{S} \subseteq \mathcal{A}$, $\mathbb{P}(B \mid \bigcap_{\mathcal{S}} \bar{A}) \le \beta^{|J|} \mathbb{P}(B)$.

Proof. Put $x_A = \beta^{s(A)} \mathbb{P}(A)$. Every A has a coordinate $v \in \operatorname{supp} A$, so $x_A \le \tilde{w}_v \le \eta \le 1/4$. For $0 \le x \le 1/4$, $-\log(1-x) \le x/(1-x) \le \frac{4}{3} x$. For any coordinate set J, each A' sharing a coordinate with J is counted in some $\tilde{w}_v$, $v \in J$, so

$$\prod_{A' \text{ meets } J} (1 - x_{A'}) \;\ge\; \exp\Big(-\frac{4}{3} \sum_{v \in J} \tilde{w}_v\Big) \;\ge\; \exp\Big(-\frac{4}{3} |J| \eta\Big) = \beta^{-|J|}. \quad (3)$$

With $J = \operatorname{supp} A$ this gives $\mathbb{P}(A) = x_A \beta^{-s(A)} \le x_A \prod_{A' \sim A} (1 - x_{A'})$, the hypothesis of Lemma 2.7. Its conclusions give $\mathbb{P}(\bigcap \bar{A}) \ge \prod (1 - x_A) \ge \exp(-\frac{4}{3} \sum x_A)$ and, with (3), the conditional bound. □

The criterion is per coordinate, with a threshold that depends neither on the coordinate nor on the level of its quarantine; the price is the weight $\beta^{s(A)}$. We shall take $\beta = 1 + 1/\log \mathcal{L}$, so that $\beta^{\omega(M)}$ is $O(1)$ on average, and $\eta \asymp 1/\log \mathcal{L}$.

The square-class process. Fix a parameter $Y \ge 7$ and $\beta = 1 + 1/\log \mathcal{L}$, $\eta = \frac{3}{4} \log \beta$. The *square-class process* is the following random sequence of square-class quarantines (Q_i, r_i). Start from $Q = 8$, $r = 1$. Then make three *forced* steps at $\ell = 3, 5, 7$ with $a = 0$, and afterwards, while some odd prime $\ell \le Y$ with $\ell \in \mathcal{P}$ has $\tilde{w}_\ell > \eta$ (computed for the current quarantine), make a step at the least such ℓ. A *step* at (ℓ, a), where $a = a_\ell < f_\ell$ is the current exponent, replaces a_ℓ by $a+1$ and r by a class r' mod $Q\ell$ with $r' \equiv r \pmod{Q}$, chosen uniformly among the classes of the current fibre at ℓ that are squares modulo ℓ: if $a = 0$ this is uniform among the $(\ell - 1)/2$ nonzero squares modulo ℓ, and if $a \ge 1$ among the ℓ lifts of r mod ℓ^a to ℓ^{a+1}, all of which are squares. Each step raises some $a_\ell < f_\ell$ with $\ell \le Y$, so the process stops after at most $\sum_{\ell \le Y} f_\ell$ steps.

Every (Q_i, r_i) is a square-class quarantine. After the forced steps, $840 \mid Q$ and $r \equiv 1 \pmod{24}$, since 1 is the only nonzero square modulo 3. Primes $\ell > Y$ are never stepped. By Lemma 2.2, no atom ever has all its coordinates quarantined and matched: the final system contains no event of probability one.

Notation. At stage i let μ_i be the uniform probability on the units n with $n \equiv r_i \pmod{Q_i}$ (on $\hat{\mathbb{Z}}^\times$, or modulo any multiple of $\operatorname{lcm}(Q_i, M)$), and for an atom $E = (M, D)$ put $p_i(E) = \mu_i(n \equiv -4D \pmod{M})$. So $p_i(E) = \mathbb{P}(E_{M,D})$ if the atom survives (Q_i, r_i), and $p_i(E) = 0$ otherwise (if $M \mid Q_i$ the class is excluded by Lemma 2.2). Let $P_H(E) = 1/\varphi(M)$ be the Haar probability, $s_i(E)$ the support size at stage i, and $u_i(E) = \#\{\ell \le Y : \ \ell \mid M,\ a_\ell(i) = 0\}$. Write $\omega_Y(M)$ for the number of primes $\ell \le Y$ dividing M, and M_Y for the Y-smooth part of M.

Lemma 2.9 (Bookkeeping for the square-class process; proved)**.** *Consider the square-class process with parameter Y.*

(a) *For every fixed $\phi(E) \ge 0$, $G_i = \sum_E p_i(E) 2^{u_i(E)} \phi(E)$ is a supermartingale, and $G_0 = \sum_E P_H(E) 2^{\omega_Y(M)} \phi(E)$.*

(b) *With Q_{end} the final modulus,*

$$\mathbb{E}[\log Q_{\mathrm{end}}] \ \le \ \log 840 + \frac{1}{\eta} \sum_E P_H(E)\, 2^{\omega_Y(M)}\, \beta^{\omega(M)} \log M_Y.$$

(c) $\mathbb{E}\big[\sum_E \beta^{s_{\mathrm{end}}(E)} p_{\mathrm{end}}(E)\big] \le S_H^\beta := \sum_E P_H(E) 2^{\omega_Y(M)} \beta^{\omega(M)}$.

(d) *For every prime $\ell > Y$, $\mathbb{E}[\tilde{w}_\ell(\mathrm{end})^2] \le B_2(\ell)$, where*

$$B_2(\ell) = \sum_{\substack{E,E' \\ \ell\mid M,\ \ell\mid M'}} \beta^{\omega(M)+\omega(M')} 2^{\omega_Y(M)+\omega_Y(M')} P_H(E)P_H(E') \times \varphi\Big(\frac{\gcd(M,M')}{\ell^{\min(v_\ell(M),v_\ell(M'))}}\Big).$$

Hence $\mathbb{P}(\exists\, \ell > Y :\ \tilde{w}_\ell(\mathrm{end}) > \eta) \le \eta^{-2} \sum_{\ell>Y} B_2(\ell)$.
All sums are over atoms $E = (M, D)$, $E' = (M', D')$.

Proof. (a) Consider a step at (ℓ, a). Atoms with $\ell \nmid M$, or with $v_\ell(M) \le a$, are unaffected: their class modulo $\ell^{v_\ell(M)}$ is already determined by the fibre, and the step only refines it at higher levels. Let $v_\ell(M) > a$. If $a \ge 1$, the new class is uniform among the ℓ classes of the fibre at level $a+1$, so $\mathbb{E}[p_{i+1}(E) \mid \text{past}] = p_i(E)$, and u is unchanged. If $a = 0$, the conditional probability of the event becomes ℓ^{1-v} (with $v = v_\ell(M)$) if $r' \equiv -4D \pmod{\ell}$ and 0 otherwise; the class $r' \bmod \ell$ is uniform on the $(\ell-1)/2$ squares, so

$$\mathbb{E}[p_{i+1}(E) \mid \text{past}] = p_i(E)\,\varphi(\ell^v)\,\ell^{1-v}\,\frac{2}{\ell-1}\,\mathbf{1}\Big[\Big(\frac{-4D}{\ell}\Big) = 1\Big] = \Big(1 + \Big(\frac{-d}{\ell}\Big)\Big)p_i(E) \le 2p_i(E)$$

by Lemma 2.2(a), while u drops by one if $\ell \le Y$ (forced steps and all other steps have $\ell \le Y$). So each term $p_i(E)2^{u_i(E)}\phi(E)$ is a supermartingale. At the start $Q = 8$, $r = 1$ and M is odd, so $p_0(E) = P_H(E)$; and $u_0(E) = \omega_Y(M)$.

(b) For a pair (ℓ, a) with $\ell \le Y$ let $\tau_{\ell,a}$ be the stage at which the (unforced) step at (ℓ, a) is taken, ∞ if it is never taken. At that stage $\tilde{w}_\ell > \eta$, so

$$\log\ell \cdot \mathbf{1}[\tau_{\ell,a} < \infty] \le \frac{\log\ell}{\eta}\, G^{(\ell,a)}_{\tau_{\ell,a}} \mathbf{1}[\tau_{\ell,a} < \infty], \qquad G^{(\ell,a)}_i = \sum_{E:\ v_\ell(M)\ge a+1} p_i(E) 2^{u_i(E)} \beta^{\omega(M)},$$

since at stage $\tau_{\ell,a}$ the atoms with $\ell \in \operatorname{supp} E$ are those with $v_\ell(M) \ge a+1$, and $\beta^s p \le \beta^\omega 2^u p$. The stopping times are bounded, and optional stopping for the nonnegative supermartingale $G^{(\ell,a)}$ (part (a) with $\phi = \beta^{\omega(M)}\mathbf{1}[v_\ell(M) \ge a+1]$) gives $\mathbb{E}[G^{(\ell,a)}_\tau \mathbf{1}[\tau < \infty]] \le G^{(\ell,a)}_0$. Summing over $\ell \le Y$ and $a < v_\ell(M)$, and using $\sum_{\ell\le Y} v_\ell(M)\log\ell = \log M_Y$, gives the claim; the forced steps cost $\log 105$, and $\log Q \le \log 8 + \log 105 + \sum_{\text{steps}} \log\ell$.

(c) Part (a) with $\phi = \beta^{\omega(M)}$, and $s_{\text{end}} \le \omega$.

(d) The prime ℓ is never stepped, so for $\ell \mid M$, $p_i(E) = p_i(E^{(\ell)})/\varphi(\ell^{v_\ell(M)})$, where $E^{(\ell)}$ is the event with its condition at ℓ removed, and

$$\tilde{w}_\ell^2 \le \sum_{E,E'} \beta^{\omega(M)+\omega(M')} p(E)p(E'),$$

the sum over atoms with $\ell \mid M$, $\ell \mid M'$. For two events F, F' (conditions at sets of primes, with given residues) put $\Pi_i = p_i(F)p_i(F')R_i$, where $R_i = \prod_{\ell'} \rho_i(\ell')$ over the primes ℓ' constrained by both, and $\rho_i(\ell')$ is the number of classes modulo ℓ'^j in the current fibre, $j \le \min(v_{\ell'}(F), v_{\ell'}(F'))$ being the depth to which the two prescribed residues agree ($\rho_i(\ell') = 1$ once $a_{\ell'} \ge j$). Then $p_i(F)p_i(F') \le \Pi_i$, and $\Pi_i 2^{u_i(F)+u_i(F')}$ is a supermartingale. Indeed, consider a step revealing level $a+1$ at ℓ'. If neither event constrains ℓ' at this level, nothing changes. If only one does, its factor behaves as in (a), and its ρ is already 1. If both do and they agree at this level, both probabilities are multiplied by the same factor $N \cdot \mathbf{1}[\text{match}]$, where N is the number of classes of the current fibre at this level ($N = \ell' - 1$ if $a = 0$, $N = \ell'$ if $a \ge 1$), and ρ is divided by N; so $\Pi \mapsto N\mathbf{1}[\text{match}]\Pi$, whose mean is Π if $a \ge 1$ and $(1 + (\frac{-d}{\ell'}))\Pi \le 2\Pi$ if $a = 0$, in which case both u's drop. If both constrain ℓ' and disagree at this level, they cannot both match, so $\Pi \mapsto 0$. At the start $R_0 \le \prod \varphi(\ell'^j) \le \varphi(\gcd(M_F, M_{F'}))$, so $\mathbb{E}[\Pi_{\text{end}}] \le 2^{\omega_Y(M_F)+\omega_Y(M_{F'})} P_H(F)P_H(F')\varphi(\gcd(M_F, M_{F'}))$. Apply this to $F = E^{(\ell)}$, $F' = E'^{(\ell)}$, multiply by $\varphi(\ell^{v_\ell(M)})^{-1}\varphi(\ell^{v_\ell(M')})^{-1}$ and sum. The last claim is Chebyshev's inequality and a union bound. □

Part (d) is the reason for the random class: the second moment of the mass at a prime that is never quarantined is controlled by a Haar pair-correlation sum, a deterministic divisor sum in which $\gcd(M, 4D+1)$ does not occur.

3. Haar masses and the upper bound on the Haar side

The only analytic input of this section is the following theorem. For fixed $k \ge 1$, $A, B \ge 1$ and $\varepsilon > 0$, let $\mathcal{M}_k(A, B, \varepsilon)$ be the class of non-negative functions F of k variables with

$$F(a_1 b_1, \dots, a_k b_k) \le \min\left(A^{\Omega(a_1 \cdots a_k)},\, B(a_1 \cdots a_k)^{\varepsilon}\right) F(b_1, \dots, b_k)$$

whenever $\gcd(a_1 \cdots a_k, b_1 \cdots b_k) = 1$.

Theorem 3.1 (Nair–Tenenbaum [33, Thm. 1]; cited in the form of [17, (1.1)])**.** *Let $Q_1, \dots, Q_k \in \mathbb{Z}[X]$ be pairwise coprime irreducible polynomials, $Q = Q_1 \cdots Q_k$ of degree g, without fixed prime divisor. Let $\rho_{Q_j}(n)$ and $\rho(n)$ be the numbers of zeros of Q_j and of Q modulo n. Let $0 < \alpha, \delta < 1$, $A, B \ge 1$, $0 < \varepsilon \le \alpha\delta/(12g^2)$ and $F \in \mathcal{M}_k(A, B, \varepsilon)$. Then, uniformly in $x \ge c_0 \|Q\|^{\delta}$ and $x^{\alpha} < y \le x$,*

$$\sum_{x < n \le x+y} F(|Q_1(n)|, \dots, |Q_k(n)|) \;\ll\; y \prod_{p \le x} \Big(1 - \frac{\rho(p)}{p}\Big) \times \sum_{n_1 \cdots n_k \le x} F(n_1, \dots, n_k) \frac{\rho_{Q_1}(n_1) \cdots \rho_{Q_k}(n_k)}{n_1 \cdots n_k},$$

where c_0 and the implied constant depend only on $k, g, A, B, \varepsilon, \alpha, \delta$ and the discriminant of Q.

We use it with $k = 2$, $Q_1(n) = n$, $Q_2(n) = 4n - 1$: these are coprime and irreducible, $Q = 4n^2 - n$ has discriminant 1 and no fixed prime divisor ($Q(1) = 3$, $Q(2) = 14$), $\rho(2) = 1$, $\rho(p) = 2$ for odd p, $\rho_{Q_1} \equiv 1$ and $\rho_{Q_2}(m) = \mathbf{1}[m \text{ odd}]$. We take $y = x$, $\alpha = \delta = 1/2$ and $\varepsilon = 1/200$. (Henriot notes that NT's own class $\mathcal{M}_k$ requires the inequality whenever $\gcd(a_i, b_i) = 1$ for each i, so the class above is slightly larger and Theorem 3.1 is Henriot's reading of NT's proof. Our functions F, products of multiplicative functions of separate variables, lie in NT's original class, and NT's own range ($\varepsilon < 1/(8g^2)$, $x^{4g^2\varepsilon} \le y \le x$) covers $\varepsilon = 1/200$, $g = 2$, $y = x$; so for our use the citation rests on [33, Thm. 1] alone.)

Throughout, $\beta = 1 + 1/\log \mathcal{L}$, $2 \le Y \le T$, $H(M) = \beta^{\omega(M)} 2^{\omega_Y(M)}$, and

$$w(M) = \tau(A_M^2)\, H(M)\, \frac{M}{\varphi(M)} \quad (M \equiv 3 \ (4)), \qquad w(M) = 0 \text{ otherwise.}$$

Summing over $D \mid A_M^2$ gives $\sum_D H(M) P_H(E_{M,D}) = w(M)/M$. Sums over M run over $M \le T$.

Lemma 3.2 (Haar masses; proved, (A) modulo Theorem 3.1)**.** *Let $2 \le Y \le T$ and $\beta = 1 + 1/\log \mathcal{L}$.*

(A) *We have*

$$S_H^\beta = \sum_M \frac{w(M)}{M} \ll \mathcal{L}^3 \log Y,$$
$$\sum_E P_H(E) 2^{\omega_Y(M)} \beta^{\omega(M)} \log M_Y \ll \mathcal{L}^3 (\log Y)^4.$$

(B) $\sum_{\ell > Y \text{ prime}} B_2(\ell) \le \frac{\mathcal{L}+1}{Y}\, \Xi$, *where* $\Xi = \sum_M w(M)^2 \tau(M) \omega(M)/M \ll \mathcal{L}^{C_0}$ *for an absolute constant C_0, uniformly in $2 \le Y \le T$.*

Proof. (A) Write $n = A_M$, so $M = 4n - 1$, and $f_2(m) = H(m)m/\varphi(m)$, a multiplicative function with $f_2(p^k) = \beta\, 2^{[p \le Y]} p/(p-1) \le 6$ for odd p (as $\beta \le 2$). The function $F(n_1, n_2) = \tau(n_1^2) f_2(n_2)$ satisfies $F(a_1 b_1, a_2 b_2) = \tau(a_1^2) f_2(a_2) F(b_1, b_2)$ when $\gcd(a_1 a_2, b_1 b_2) = 1$, and $\tau(a_1^2) f_2(a_2)$ is at most $8^{\Omega(a_1 a_2)}$ and at most $B(a_1 a_2)^{1/200}$ for an absolute B (divisor bound). So $F \in \mathcal{M}_2(8, B, 1/200)$, uniformly in T and Y. Theorem 3.1 on $x < n \le 2x$ gives

$$\sum_{x < n \le 2x} \tau(n^2) f_2(4n-1) \ll \frac{x}{(\log x)^2} \sum_{n_1 \le x} \frac{\tau(n_1^2)}{n_1} \sum_{n_2 \le x} \frac{f_2(n_2)}{n_2}$$
$$\ll \frac{x}{(\log x)^2} (\log x)^3 (\log x)^\beta (\log Y)^\beta,$$

by the Euler products $\prod_{p \le x}(1 + 3/p + \dots)$ and

$$\prod_{p \le Y} \Big(1 + \frac{2\beta}{p-1} + O(p^{-2})\Big) \prod_{Y < p \le x} \Big(1 + \frac{\beta}{p-1} + O(p^{-2})\Big).$$

Since $(\log x)^{\beta-1}, (\log Y)^{\beta-1} \le \mathcal{L}^{1/\log\mathcal{L}} = e$, this is $\ll x(\log x)^2 \log Y$. As $\tau(A_M^2)H(M)/\varphi(M) = \tau(n^2)f_2(M)/M$ with $M \asymp x$, the block contributes $\ll \mathcal{L}^2 \log Y$; the $O(\mathcal{L})$ dyadic blocks (and the bounded range $x < c_0$) give the first bound. For the second, use $\log M_Y \le \log Y \cdot \Omega(M_Y) \le \log Y \cdot \tau(M_Y)$ (as $\sum k_i \le \prod(k_i+1) - 1$), and repeat the argument with $f_2(m)\tau(m_Y)$ in place of f_2. Its values at p^k are at most $6(k+1) \le 12^k$ and $\ll_\varepsilon p^{k\varepsilon}$, so the class condition holds uniformly; its Euler factor at $p \le Y$ is $1+4\beta/(p-1)+O(p^{-2})$, so $\sum_{n_2\le x} f_2(n_2)\tau((n_2)_Y)/n_2 \ll (\log x)(\log Y)^3$, and the bound $\mathcal{L}^3(\log Y)^4$ follows.

(B) For an atom pair with $\ell \mid M$, $\ell \mid M'$ write $M = \ell m$, $M' = \ell m'$. Then $\gcd(M, M')/\ell^{\min(v_\ell(M), v_\ell(M'))}$ divides $\gcd(m, m')$, so its φ is at most $\gcd(m, m') = \sum_{e\mid(m,m')} \varphi(e)$. Summing over D, D',

$$B_2(\ell) \le \sum_e \varphi(e) V(e\ell)^2, \qquad V(q) = \sum_{q\mid M} \frac{w(M)}{M} = \sum_{k\le T/q} \frac{w(qk)}{qk}.$$

By Cauchy–Schwarz in k, $V(q)^2 \le (\mathcal{L}+1)\sum_k w(qk)^2/(q^2k)$. Put $N = e\ell k$: the term $\varphi(e)w(N)^2/(e^2\ell^2 k)$ equals $(w(N)^2/N)\varphi(e)/(e\ell) \le (w(N)^2/N)/\ell$, and for fixed ℓ there are at most $\tau(N)$ factorisations $N = e\ell k$. So

$$\sum_{\ell>Y} B_2(\ell) \le (\mathcal{L}+1)\sum_N \frac{w(N)^2\tau(N)}{N} \sum_{\ell\mid N,\ \ell>Y} \frac{1}{\ell} \le \frac{\mathcal{L}+1}{Y}\,\Xi.$$

For Ξ, use $\omega \le \tau$, $2^{\omega_Y} \le 2^\omega$ and Cauchy–Schwarz with $M = 4n-1 \ge 3n$:

$$\Xi \le \Big(\sum_{n\le T} \frac{\tau(n^2)^4}{n}\Big)^{1/2} \Big(\sum_{m\le 4T} \frac{g(m)^2}{m}\Big)^{1/2},$$

$$g(m) = \beta^{2\omega(m)} 4^{\omega(m)} \Big(\frac{m}{\varphi(m)}\Big)^2 \tau(m)^2.$$

Both are mean values of non-negative multiplicative functions f with $f(p^k) \le C(k+1)^C$ for an absolute C; bounding them by $\prod_{p\le 4T}(1+f(p)/p+f(p^2)/p^2+\dots)$ gives $\ll \mathcal{L}^{C_0}$, with C_0 absolute (since $\beta \le 2$) and independent of Y. □

The proof of (A) is the only place where Theorem 3.1 is used. All masses here are Haar masses, so the Elsholtz–Tao estimates are not needed.

Let $\delta^*(T)$ be the Haar measure of $\{n \in \hat{\mathbb{Z}}^\times : W(n) > T\}$ relative to the class $n \equiv 1 \pmod{24}$, that is, the proportion, among the units modulo $L = \mathrm{lcm}(24, M \le T)$ that are $\equiv 1 \pmod{24}$, of those with $W(n) > T$ (the condition $W(n) > T$ depends only on n mod L). On all of $\hat{\mathbb{Z}}^\times$ the proportion is exactly half of this, because the atom $(3, 1)$ excludes $n \equiv 2 \pmod 3$.

Theorem 3.3 (Haar side, upper bound; proved modulo Theorem 3.1)**.** *For large T there is a square-class quarantine (Q, r) with $840 \mid Q$, $r \equiv 1 \pmod{24}$, all prime factors of Q at most $Y := \mathcal{L}^{C_0+4}$ (with C_0 from Lemma 3.2), and, with $\beta = 1 + 1/\log\mathcal{L}$ and $\eta = \frac{3}{4}\log\beta$,*

$$\tilde{w}_\ell \le \eta \quad (\ell \in \mathcal{P}), \qquad S_\beta \le C\mathcal{L}^3 \log\mathcal{L}, \qquad \log Q \le C\mathcal{L}^3(\log\mathcal{L})^5, \tag{4}$$

where C is absolute. Consequently $\delta = \mathbb{E}F \geq \exp(-\frac{4}{3}S_\beta)$ and

$$\log \frac{1}{\delta^*(T)} \;\leq\; \log \varphi(Q) + \frac{4}{3}S_\beta \;\ll\; \mathcal{L}^3(\log \mathcal{L})^5.$$

Proof. Run the square-class process with this Y. As $1/\eta \asymp \log \mathcal{L}$ and $\log Y \asymp \log \mathcal{L}$, Lemma 2.9(b,c) and Lemma 3.2(A) give $\mathbb{E}[\log Q_{\mathrm{end}}] \ll \mathcal{L}^3(\log \mathcal{L})^5$ and $\mathbb{E}[S_\beta(\mathrm{end})] \ll \mathcal{L}^3 \log \mathcal{L}$. By Lemma 2.9(d) and Lemma 3.2(B), $\mathbb{P}(\exists \ell > Y : \tilde{w}_\ell > \eta) \leq \eta^{-2}(\mathcal{L}+1)\mathcal{L}^{C_0}/Y \leq 1/4$ for large T. By Markov's inequality each of the events $\log Q_{\mathrm{end}} > 4\mathbb{E}[\log Q_{\mathrm{end}}]$ and $S_\beta > 4\mathbb{E}[S_\beta]$ has probability at most $1/4$. So some realisation (Q, r) avoids all three. In it every coordinate $\ell \leq Y$ has $\tilde{w}_\ell \leq \eta$ because the process stopped, and every $\ell > Y$ by the choice of realisation; this is (4). Lemma 2.8 gives $\delta \geq \exp(-\frac{4}{3}S_\beta)$. By Lemma 2.5(i), the units $n \equiv r \pmod{Q}$ with $F(n) = 1$ have $W(n) > T$. Among the units $\equiv 1 \pmod{24}$ the class $r \bmod Q$ has relative Haar measure $\varphi(24)/\varphi(Q)$, as $r \equiv 1 \pmod{24}$, and within it the residues $n \bmod \ell^{f_\ell}$, $\ell \in \mathcal{P}$, are independent and uniform on Ω_ℓ. So $\delta^*(T) \geq (\varphi(24)/\varphi(Q))\,\delta$. □

Together with Theorem 4.4 below, this determines the Haar exponent, $\log\log(1/\delta^*(T)) = (3+o(1))\log \mathcal{L}$. The quarantine costs $\mathcal{L}^3(\log \mathcal{L})^5$; the class-of-one route of the previous version gave $\log(1/\delta^*(T)) \ll \mathcal{L}^5 \log \mathcal{L}$ modulo the Elsholtz–Tao estimates.

4. A Janson-type inequality and the lower bound on the Haar side

In this section (X_v) are independent random variables with finite or countable ranges, and an *atomic event* is an event $E = \{X_v = a_{E,v} \text{ for } v \in S_E\}$ with S_E finite and single values $a_{E,v}$. For a family $\mathcal{F}$ put $\mathrm{Av}(\mathcal{F}) = \bigcap_{E\in\mathcal{F}} \bar{E}$. Two atomic events *conflict* if they prescribe different values on some common coordinate (then $E \cap E' = \emptyset$), and *share a bit* if $S_E \cap S_{E'} \neq \emptyset$ and they agree on all common coordinates (then $\mathbb{P}(E\cap E') > 0$). Harris' inequality is not available here: for one-hot variables $\mathbb{P}(X = a \mid X \neq a') > \mathbb{P}(X = a)$. The following lemmas replace it.

Lemma 4.1 (Compatible events; proved)**.** *Let A be atomic and let $\mathcal{C}$ be a family of atomic events none of which conflicts with A. Then $\mathbb{P}(A\cap \mathrm{Av}(\mathcal{C})) \leq \mathbb{P}(A)\mathbb{P}(\mathrm{Av}(\mathcal{C}))$.*

Proof. Condition on A. For $C \in \mathcal{C}$, $C \cap A = A \cap C'$, where C' is C with the coordinates in S_A deleted; C' is independent of $(X_v)_{v\in S_A}$ and $C' \supseteq C$. So $\mathbb{P}(\mathrm{Av}(\mathcal{C}) \mid A) = \mathbb{P}(\bigcap \bar{C}') \leq \mathbb{P}(\bigcap \bar{C})$. □

Lemma 4.2 (Lopsided local lemma; proved, standard)**.** *Let $\mathcal{F}$ be a family of atomic events, $\Gamma(E)$ the set of $E' \in \mathcal{F}$ conflicting with E, and suppose $x_E \in (0,1)$ satisfy $\mathbb{P}(E) \leq x_E \prod_{E'\in\Gamma(E)}(1 - x_{E'})$. Then for every atomic event A and every $\mathcal{S} \subseteq \mathcal{F}$,*

$$\mathbb{P}(A \mid \mathrm{Av}(\mathcal{S})) \leq \mathbb{P}(A) \prod_{E'\in\mathcal{S} \text{ conflicting with } A} (1 - x_{E'})^{-1}.$$

Proof. By Lemma 4.1, $\mathbb{P}(E \mid \mathrm{Av}(\mathcal{S}')) \le \mathbb{P}(E)$ whenever no event of $\mathcal{S}'$ conflicts with E, so the lopsided local lemma of Erdős and Spencer [10] applies with the graph Γ; its proof gives $\mathbb{P}(E \mid \mathrm{Av}(\mathcal{S})) \le x_E$ for $E \notin \mathcal{S}$. For A, let $\mathcal{S}_1$ be the events of $\mathcal{S}$ conflicting with A and $\mathcal{S}_2 = \mathcal{S} \setminus \mathcal{S}_1$. Then $\mathbb{P}(A \mid \mathrm{Av}(\mathcal{S})) \le \mathbb{P}(A \mid \mathrm{Av}(\mathcal{S}_2))/\mathbb{P}(\mathrm{Av}(\mathcal{S}_1) \mid \mathrm{Av}(\mathcal{S}_2))$; the numerator is at most $\mathbb{P}(A)$ by Lemma 4.1, and the chain rule bounds the denominator below by $\prod_{\mathcal{S}_1}(1 - x_E)$. □

Theorem 4.3 (Janson-type inequality for atomic events; proved)**.** *Under the hypothesis of Lemma 4.2 put* $K = \max_{E \in \mathcal{F}} \prod_{E' \in \Gamma(E)} (1 - x_{E'})^{-1}$, $\mu = \sum_{E \in \mathcal{F}} \mathbb{P}(E)$, *and* $\Delta = \sum \mathbb{P}(E \cap E')$ *over the unordered pairs of distinct events of* $\mathcal{F}$ *that share a bit. Then*

$$-\log \mathbb{P}(\mathrm{Av}(\mathcal{F})) \;\ge\; \mu - K\Delta, \qquad -\log \mathbb{P}(\mathrm{Av}(\mathcal{F})) \;\ge\; \min\Big(\frac{\mu}{2}, \frac{\mu^2}{4K\Delta}\Big).$$

Proof. Order $\mathcal{F} = \{E_1, \dots, E_N\}$ and put $B_i = \mathrm{Av}(\{E_j : j < i\})$, so $-\log \mathbb{P}(\mathrm{Av}(\mathcal{F})) = -\sum_i \log(1 - \mathbb{P}(E_i \mid B_i)) \ge \sum_i \mathbb{P}(E_i \mid B_i)$. Let $B_i^d = \mathrm{Av}(\{E_j : j < i,\ S_j \cap S_i = \emptyset\}) \supseteq B_i$. Then $\mathbb{P}(E_i \mid B_i) \ge \mathbb{P}(E_i)\mathbb{P}(B_i \mid E_i)/\mathbb{P}(B_i^d)$. Given E_i, the events E_j $(j < i)$ conflicting with E_i are impossible, those sharing a bit become E_j' (E_j with the coordinates of S_i deleted, so $\mathbb{P}(E_i)\mathbb{P}(E_j') = \mathbb{P}(E_i \cap E_j)$), and B_i^d is independent of E_i. Hence

$$\mathbb{P}(B_i \mid E_i) \ge \mathbb{P}(B_i^d) - \sum_{j<i,\ j \text{ shares a bit with } i} \mathbb{P}(E_j' \cap B_i^d).$$

B_i^d is the avoidance event of a subfamily of $\mathcal{F}$, and every event conflicting with E_j' conflicts with E_j, so Lemma 4.2 gives $\mathbb{P}(E_j' \mid B_i^d) \le K\mathbb{P}(E_j')$. Therefore $\mathbb{P}(E_i \mid B_i) \ge \mathbb{P}(E_i) - K\sum_{j<i,\, j\sim i} \mathbb{P}(E_i \cap E_j)$, and summing over i gives the first bound. For the second, a subfamily $\mathcal{F}'$ satisfies the hypothesis with the same x_E and the same or smaller K, and $\mathbb{P}(\mathrm{Av}(\mathcal{F})) \le \mathbb{P}(\mathrm{Av}(\mathcal{F}'))$. For a random $\mathcal{F}'$ with inclusion probability p, $\mathbb{E}\mu' = p\mu$ and $\mathbb{E}\Delta' = p^2\Delta$, so some $\mathcal{F}'$ has $\mu' - K\Delta' \ge p\mu - p^2K\Delta$; take $p = \min(1, \mu/(2K\Delta))$. □

Δ counts only compatible pairs; conflicting pairs, which carry the structure "one coordinate, many residues", enter only through K. If $x_E = 2\mathbb{P}(E)$, $\mathbb{P}(E) \le 1/8$ and $\sum_{E' \in \Gamma(E)} \mathbb{P}(E') \le 1/8$ for all E, the hypothesis holds and $K \le e^{1/3}$. Theorem 4.3 was checked by exact enumeration on small random instances (not used). We did not find it in the literature in this form; the Janson inequality itself [23] needs positively correlated (increasing) events, and the lopsided local lemma alone gives upper bounds on $\mathbb{P}(E \mid \cdot)$, not lower bounds. We make no priority claim.

The lower bound. For the ES events the coordinates are $X_\ell = n \bmod \ell^f$ (f large) for odd primes ℓ, independent and uniform on units (on $X_3 \equiv 1 \pmod 3$ in the normalisation of $\delta^*(T)$), and $E_{M,D} = \bigcap_{\ell \mid M} \{X_\ell \equiv -4D \ (\ell^{v_\ell(M)})\}$. For squarefree M with $3 \nmid M$ these are atomic events in the

variables $X_\ell \bmod \ell$, and $\mathbb{P}(E_{M,D}) = 1/\varphi(M)$. Distinct pairs $(M, -4D \bmod M)$ are distinct events.

Theorem 4.4 (Haar side, lower bound; proved modulo the fundamental lemma)**.** *There are absolute constants $c > 0$ and T_0 such that for $T \geq T_0$*

$$\log \frac{1}{\delta^*(T)} \;\geq\; c\,\frac{\mathcal{L}^3}{\log \mathcal{L}}.$$

The only external input is the fundamental lemma of sieve theory [13, Lemma 6.3, Cor. 6.10] (cited), used for a lower bound for rough numbers in progressions to small moduli. T_0 is astronomically large.

The family. Put $y = \mathcal{L}^5$, $N_0 = \mathcal{L}^2$, and for an integer D write $D^* = \prod_p p^{\lceil v_p(D)/2 \rceil}$. Then $D \mid A^2 \iff D^* \mid A$, and each $n \geq 1$ equals D^* for exactly $2^{\omega(n)}$ integers D (write $D = \kappa t^2$ with κ squarefree; then $D^* = \kappa t$). Let $\mathcal{F}$ be the set of events $E_{M,D}$ with

$$\sqrt{T} \leq M \leq T, \quad M \text{ squarefree with all prime factors} > y, \quad M \equiv 3 \ (4),$$
$$n := D^* \in [N_0, T^{1/10}], \quad n \mid A_M.$$

We use: (F1) $D \leq n^2 \leq T^{1/5} < M/4$, so different D give different residues $-4D \bmod M$, and $\mathbb{P}(E_{M,D}) = 1/\varphi(M)$; (F2) for y-rough squarefree $m \leq T$, $\omega(m) \leq \mathcal{L}/\log y$ and $\varphi(m) \geq m/2$; (F3) $\sum_{n \leq x} 2^{\omega(n)}/n \leq (1 + \log x)^2$ and $\sum_{n \leq x} 2^{\omega(n)} \leq x(1 + \log x)$; (F4) for a progression $v \equiv c \pmod d$ in $[v_0, V]$ with least element $v_1 \geq v_0$, $\sum 1/v \leq 1/v_1 + \log(V/v_0)/d$; (F5) if $\omega(m) y^{-1} \leq 1$, then $\sum_{g \mid m,\ g>1,\ g \text{ squarefree and } y\text{-rough}} g^{-1} \leq (1 + 1/y)^{\omega(m)} - 1 \leq 2\omega(m)/y$.

Lemma 4.5 (Mass; proved modulo the fundamental lemma)**.** *We have $\mu = \sum_{E \in \mathcal{F}} \mathbb{P}(E) \geq c_1 \mathcal{L}^3/\log y$.*

Proof. By (F1), $\mu \geq \sum_{N_0 \leq n \leq T^{1/10}} 2^{\omega(n)} \sum_M 1/M$, over y-rough squarefree $M \in [\sqrt{T}, T]$ with $M \equiv -1 \pmod{4n}$; this congruence implies $M \equiv 3 \pmod 4$ and $n \mid A_M$. For a dyadic block $(X, 2X] \subseteq [\sqrt{T}, T]$, sift the progression (of length $X/(4n) \geq X^{4/5}/4$) by the primes $p < y$, $p \nmid 2n$, at level $X^{1/2}$; as $s = \log X/(2 \log y) \to \infty$, the fundamental lemma gives at least $c_2 X/(n \log y)$ y-rough elements. Those divisible by p^2 with $p \geq y$ number at most $X/(4n(y-1)) + \pi(\sqrt{2X})$, which is negligible as $n \leq X^{1/5}$. So each of the $\gg \mathcal{L}$ blocks contributes $\geq c_2/(4n \log y)$, and $\sum_{N_0 \leq n \leq T^{1/10}} 2^{\omega(n)}/n \gg \mathcal{L}^2$. □

Lemma 4.6 (Local-lemma hypothesis; proved)**.** *For $T \geq T_0$, every $E \in \mathcal{F}$ has $\mathbb{P}(E) \leq 1/8$ and $\sum_{E' \in \Gamma(E)} \mathbb{P}(E') \leq 1/8$. Hence Theorem 4.3 applies with $x_E = 2\mathbb{P}(E)$ and $K \leq e^{1/3}$.*

Proof. $\mathbb{P}(E) \leq 2/\sqrt{T}$. Put $w_q = \sum_{E' \in \mathcal{F}:\, q \mid M'} \mathbb{P}(E')$. Events conflicting with E share a prime $q \mid M$, so $\sum_{\Gamma(E)} \mathbb{P} \leq (\mathcal{L}/\log y) \max_{q > y} w_q$. Writing $M' = qv$ with $qv \equiv -1 \pmod{4n'}$ and $qv \geq \sqrt{T}$, (F2) and (F4) give $w_q \leq (2/q) \sum_{D'} (\mathcal{L}/(4n') + \min(1, q/\sqrt{T}))$. By (F3) the first part is at most

$\mathcal{L}^3/(2q)$, and the second at most $4\mathcal{L}T^{-2/5}$. So $\sum_{\Gamma(E)}\mathbb{P} \le (\mathcal{L}/\log y)(\mathcal{L}^3/(2y)+4\mathcal{L}T^{-2/5}) = O(1/(\mathcal{L}\log\mathcal{L}))$. □

Lemma 4.7 (Pair sum; proved)**.** $\Delta = \sum_{\textit{bit-sharing pairs}}\mathbb{P}(E\cap E') \le C\mathcal{L}^2$.

Proof. Let $E = E_{M,D}$, $E' = E_{M',D'}$ be distinct and share a bit; put $g = \gcd(M,M') > 1$, $M = gv$, $M' = gv'$. Then g is y-rough, so $g > y$, and compatibility means $g \mid D - D'$. Also $M \ne M'$: otherwise $M \mid D - D'$ and $E = E'$ by (F1). By (F2), $\mathbb{P}(E\cap E') = 1/(\varphi(g)\varphi(v)\varphi(v')) \le 8/(gvv')$. For fixed g and D the admissible v satisfy $gv \equiv -1 \pmod{4n}$ and $gv \ge \sqrt{T}$, so by (F4)

$$\sigma(g,n) := \sum_{v}\frac{1}{v} \le \frac{\mathcal{L}}{4n} + \min\Big(1, \frac{g}{\sqrt{T}}\Big). \tag{5}$$

(a) $D = D'$. Then $v \ne v'$ and $v \equiv v' \equiv -g^{-1} \pmod{4n}$. For fixed (v,n), g runs over a progression modulo $4n$ with $g > y$, so $\sum_g 1/g \le 1/y + \mathcal{L}/(4n)$; and $\sum_{v<v',\ v'\equiv v\ (4n)} 1/(vv') \le (1+\mathcal{L})^2/(4n)$. So $\Delta_a \le 8\sum_{N_0\le n\le T^{1/10}} 2^{\omega(n)}(1/y + \mathcal{L}/(4n))(1+\mathcal{L})^2/(4n) \ll \mathcal{L}^4/y + \mathcal{L}^3\log N_0/N_0 \ll \mathcal{L}\log\mathcal{L}$.

(b) $D \ne D'$. Now $g \mid m = |D - D'|$ with $0 < m < T^{1/5}$, so $g < T^{1/5}$ and (5) gives $\sigma(g,n) \le \mathcal{L}/(4n) + T^{-3/10}$. Summing over g with (F5),

$$\Delta_b \le \frac{2\mathcal{L}}{y}\Big[\frac{\mathcal{L}^2}{2}\Big(\sum_D\frac{1}{n}\Big)^2 + 4T^{-3/10}\mathcal{L}\,\#\mathcal{D}\sum_D\frac{1}{n} + 8T^{-3/5}(\#\mathcal{D})^2\Big],$$

where $\mathcal{D}$ is the set of D with $D^* \le T^{1/10}$, so $\sum_D 1/n \le (1+\mathcal{L})^2$ and $\#\mathcal{D} \le T^{1/10}(1+\mathcal{L})$ by (F3). So $\Delta_b \ll \mathcal{L}^7/y + T^{-1/10}\mathcal{L}^5 \ll \mathcal{L}^2$. □

Proof of Theorem 4.4. The events of $\mathcal{F}$ involve only primes $> y > 3$, so they are independent of n mod 24, and $\delta^*(T) \le \mathbb{P}(\mathrm{Av}(\mathcal{F}))$. By Theorem 4.3 and Lemmas 4.5–4.7, $\log(1/\delta^*(T)) \ge \mu - e^{1/3}\Delta \ge c_1\mathcal{L}^3/(5\log\mathcal{L}) - C\mathcal{L}^2$. □

Two restrictions make the pair sum small: $M \ge \sqrt{T} \ge (D^*)^5$ makes the first terms of the progressions in (5) harmless, and $D^* \ge \mathcal{L}^2$ removes the small-D hubs ($D = 1$ has residue -4 at every prime), whose same-D pairs would make $\Delta_a \asymp \mu$. The factor $\log\mathcal{L}$ is the price of y-roughness. A Monte Carlo estimate of $\delta^*(T)$ for $1023 \le T \le 32767$ fits $\log(1/\delta^*(T)) \approx 0.07\,\mathcal{L}^3/\log\mathcal{L}$ (evidence only, not used; at these T it cannot distinguish $\mathcal{L}^3/\log\mathcal{L}$ from, say, $\mathcal{L}^{2.5}$).

5. The analytic input: Gallagher's theorem

For a Dirichlet character χ put $\vartheta(x;\chi) = \sum_{p\le x}\chi(p)\log p$, and let $E_0(\chi) = 1$ if χ is the trivial character modulo 1 and $E_0(\chi) = 0$ otherwise. $\sum^*_{\chi \bmod q}$ is a sum over primitive characters.

Theorem 5.1 (Landau–Page; classical, [8, Ch. 14], [26, Cor. 11.10])**.** *There are absolute effective positive constants c_1, c_2 such that, for every $Y \ge 3$, $\prod_{q\le Y}\prod^*_{\chi \bmod q} L(s,\chi)$ has at most one zero in the region $\Re s > 1 - c_1/\log Y$, $|\Im s| \le Y$. If this zero β_1 exists, it is real and simple, and its character χ_1*

is quadratic. Moreover, every real zero β of $L(s,\chi)$, for χ a real primitive character of conductor $q \geq 3$, satisfies $1-\beta \geq c_2^{-1} q^{-1/2} (\log q)^{-2}$.

The first part is the classical zero-free region with the Landau–Page theorem. The height restriction matters: without it the statement would be a zero-free region of width $\asymp 1/\log Y$ uniformly in $|\Im s|$, which is not known. The "Exceptional Zero Statement" (28.61)–(28.62) of [27], which refers to [26, Cor. 11.10], omits the height restriction. The second part is the effective Page-type bound [8, Ch. 14], [27, (28.62)]. We use Theorem 5.1 only for *real* zeros: for the uniqueness of a real zero close to 1, for the quadraticity of its character, and for the lower bound for $1-\beta$.

Theorem 5.2 (Gallagher [14]; cited in the form of [27, Thm. 28.19])**.** *There are absolute constants $c \geq 1$ and $\kappa_0 \geq 3$ with the following property. Let $\kappa \geq \kappa_0$ be fixed and $1 < Q_G^{6c} \leq x$. Then*

$$\sum_{q \leq Q_G} \sideset{}{^*}\sum_{\chi \bmod q} |\vartheta(x;\chi) - E_0(\chi) x| \ \ll_\kappa \ x \exp\Big(-\frac{\log x}{\kappa \log Q_G}\Big) + \Big(\frac{\log x}{\log Q_G}\Big)^2 \frac{x}{Q_G},$$

unless $\prod_{q \leq Q_G} \prod^*_{\chi \bmod q} L(s,\chi)$ has a real zero β_1 with $1-\beta_1 < 1/(\kappa \log Q_G)$. In that case the term of the exceptional character χ_1 on the left is replaced by $|\vartheta(x;\chi_1) + x^{\beta_1}/\beta_1|$, and the right-hand side by

$$(1-\beta_1)(\log x)\Big(x \exp\Big(-\frac{\log x}{\log Q_G}\Big) + \frac{x \log x}{Q_G \log Q_G}\Big).$$

Source caveat. We quote [27], an unpublished draft (third volume of Montgomery–Vaughan), archived by us. We have not seen Gallagher's paper [14], so we cite it without a theorem number. The printed proof of [27, Thm. 28.19] has visible slips. Its case split is at $1-\beta_1 < 1/(3\kappa_0 \log Q_G)$, so it establishes the statement for $\kappa = 3\kappa_0$. The statement for every larger κ follows from that case. Let $\kappa \geq 3\kappa_0$, and suppose a real zero satisfies $1/(\kappa \log Q_G) \leq 1-\beta_1 < 1/(3\kappa_0 \log Q_G)$. Then $x^{\beta_1}/\beta_1 \leq 2x \exp(-\log x/(\kappa \log Q_G))$, and $(1-\beta_1)(\log x)\, x \exp(-\log x/\log Q_G) \ll x \exp(-\log x/(2\log Q_G))$. So the exceptional form of the bound implies the non-exceptional one with parameter κ. We therefore fix

$$\kappa := \max(3\kappa_0,\ 1/c_1).$$

An exceptional zero as in Theorem 5.2 is then a real zero in the region of Theorem 5.1 with $Y = Q_G$, so it is unique and χ_1 is real. Write C_G for the implied constant in Theorem 5.2 with this κ.

Gallagher's original statement may carry a range condition of the form $\exp(\sqrt{\log x}) \leq Q_G \leq x^b$; we have not checked this. Our use is robust to such a condition. In the proof of Theorem 6.1, $\log x/\log Q_G = \kappa L \geq 6c$, and $\log x \geq (\kappa L)^2$ by the hypothesis of that theorem, so $Q_G \geq \exp(\sqrt{\log x})$. In Lemma 10.4, $Q_G = 4T^\varepsilon$ and $x \leq T^{0.7}$, so again $\exp(\sqrt{\log x}) \leq Q_G \leq x$. The statement itself is the standard form of Gallagher's prime number theorem for the whole family of characters of conductor $\leq Q_G$, which underlies the usual proofs of Linnik's theorem.

6. A linear transfer theorem

Theorem 6.1 (Linear transfer; proved modulo Theorems 5.1 and 5.2)**.** *Let $T \ge 1$ and $Q \ge 2$ with $8 \mid Q$, and let r be a unit modulo Q with $r \equiv 1 \pmod 8$ that is a quadratic residue modulo every odd prime dividing Q. Let $B(n) = \sum_{i \in I} c_i \mathbf{1}[n \equiv b_i \ (d_i)]$ be a finite real combination of* consistent unit cells*: $\gcd(b_i, d_i) = 1$ and $b_i \equiv r \pmod{\gcd(d_i, Q)}$. Assume that $B(n) \le \mathbf{1}[W(n) > T]$ for every $n \equiv r \pmod Q$ coprime to all d_i. Put $D = \operatorname{lcm}_i d_i$, $N_0 = \operatorname{lcm}(Q, D)$, $H = \{n \in (\mathbb{Z}/N_0)^\times : \ n \equiv r \pmod Q\}$, and let $\mathbb{E}_H$ be the mean over H. Put*

$$Z = Q \max_i d_i, \quad \mu = \mathbb{E}_H B, \quad A = \frac{\mathbb{E}_H |B|}{\mu},$$

and assume $\mu > 0$, $A \le Z^{1/4}$, and the twist condition*: for every real primitive character ψ of conductor $f > 1$ with $\gcd(f, Q) = 1$ and $f \mid d_i$ for some i,*

$$|\mathbb{E}_H[B\psi]| \le \mu/4.$$

There is an absolute effective constant C_2 such that, if

$$\log x \ \ge \ C_2(1 + \log A) \log Z,$$

then some prime $p \le x$ with $p \equiv r \pmod Q$ and $p \nmid D$ has $W(p) > T$.

In the second version $r = 1$ and the cells were coprime to Q; then $\mathbb{E}_H$ is the mean over the units modulo D, and $\mu = \sum_i c_i/\varphi(d_i)$. Partial quarantines produce cells whose moduli share primes with Q; consistency means that each cell meets the class r mod Q. The hypotheses on r are used only in (c) below: every real character that factors through $(\mathbb{Z}/Q)^\times$ takes the value 1 at r, exactly as at 1.

Neither the ℓ^1 mass $\sum_i |c_i|/\varphi(d_i)$ nor the number of cells enters; only the Haar quantity $A \ge 1$ does.

Proof. Parameters. Put $L = 2c + \log(400 C_G(A+1))$ and $Q_G = x^{1/(\kappa L)}$. Since $A \ge 1$ we have $L \le C_3(1 + \log A)$ for an absolute C_3. If C_2 is large enough, the hypothesis gives the following, using $A \le Z^{1/4}$ and $Z \ge 2$: $\log Q_G \ge (C_2/(\kappa C_3)) \log Z$, hence $Q_G \ge Z$ and $Q_G \ge 10^4 C_G(A+1)(\kappa L)^2$; $Q_G^{6c} \le x$ (as $\kappa L \ge 6c$); $\log x \ge 16$; $\log x \ge (\kappa L)^2$ (as $(1+\log A)^2 \le (1+\log A)(1+\frac{1}{4}\log Z)$); and $x \ge C_4 A Z^4$, with C_4 as large as needed below.

Character expansion. The function $f(n) = B(n)\mathbf{1}[n \equiv r \ (Q)]$ on units modulo N_0 satisfies $f(n) = \sum_{\chi \bmod N_0} c(\chi)\chi(n)$. The reduction map $(\mathbb{Z}/N_0)^\times \to (\mathbb{Z}/Q)^\times$ is onto, and H is the fibre over r, so $|H| = \varphi(N_0)/\varphi(Q)$ and

$$c(\chi) = \frac{1}{\varphi(N_0)} \sum_{n \in H} B(n)\bar\chi(n) = \frac{\mathbb{E}_H[B\bar\chi]}{\varphi(Q)}.$$

Consequently:

(a) $c(\chi) = 0$ unless the conductor of χ divides $\operatorname{lcm}(d_i, Q)$ for some i. Indeed, by consistency and the Chinese remainder theorem,

$\mathbf{1}[n \equiv b_i \ (d_i)]\mathbf{1}[n \equiv r \ (Q)]$ is the indicator of one unit class modulo $\operatorname{lcm}(d_i, Q)$, and its χ-coefficient vanishes unless χ is trivial on $\{n \equiv 1 \ (\operatorname{lcm}(d_i, Q))\}$. So $c(\chi) \neq 0$ implies that the conductor of χ is at most $Qd_i \le Z \le Q_G$.

(b) $|c(\chi)| \le \mathbb{E}_H|B|/\varphi(Q) = A\mu/\varphi(Q)$, and $c(\chi_0) = \mu/\varphi(Q)$ for the principal character χ_0.

(c) Let χ be real with $c(\chi) \neq 0$, and let ψ^* be the real primitive character inducing it, of conductor f. Write $f = f_1 f_2$, where f_1 is composed of primes dividing Q and $\gcd(f_2, Q) = 1$, and $\psi^* = \psi_1\psi_2$ accordingly. A real primitive conductor is $2^e f'$ with $e \in \{0, 2, 3\}$ and f' odd and squarefree; as $8 \mid Q$, $f_1 \mid Q$, and ψ_1 is a product of a character modulo 8 and Legendre symbols modulo odd primes of Q. On H we have $n \equiv r \pmod{Q}$, so $\psi_1(n) = \psi_1(r) = 1$, as $r \equiv 1 \pmod 8$ and r is a quadratic residue modulo every odd prime of Q. So $\chi = \psi_2$ on H. If $f_2 = 1$, then $\chi = 1$ on H and $c(\chi) = \mu/\varphi(Q)$. If $f_2 > 1$, then $c(\chi) = \mathbb{E}_H[B\psi_2]/\varphi(Q)$. A cell with $f_2 \nmid d_i$ has $\mathbb{E}_H[\mathbf{1}[n \equiv b_i \ (d_i)]\psi_2] = 0$: there is a prime $p \mid f_2$ with $p \nmid d_i Q$; on H the residue of n modulo $p^{v_p(N_0)}$ is uniform among units and independent of n modulo $N_0/p^{v_p(N_0)}$, and the p-component of ψ_2 is a nontrivial character modulo p. So $f_2 \mid d_i$ for some i, $\gcd(f_2, Q) = 1$, and the twist condition gives $|c(\chi)| \le \mu/(4\varphi(Q))$.

Let χ^* be the primitive character inducing χ. For a fixed modulus the map $\chi \mapsto \chi^*$ is injective, so by (a) at most $\sum_{r \le Z} \varphi(r) \le Z^2$ characters have $c(\chi) \neq 0$.

The prime sum. Let $S(x) = \sum_{p \le x,\ p \nmid N_0} f(p) \log p = \sum_\chi c(\chi)\vartheta(x; \chi)$. Since $|\vartheta(x;\chi) - \vartheta(x;\chi^*)| \le \log N_0 \le \log Q + \log D \le 2Z$ (as $D \mid \operatorname{lcm}(1, \dots, \max d_i)$, so $\log D \le \psi(\max d_i) < 1.04 \max d_i$ by Rosser–Schoenfeld [41]),

$$S(x) = \frac{\mu x}{\varphi(Q)} + \sum_{\chi:\, c(\chi) \neq 0} c(\chi)(\vartheta(x;\chi^*) - E_0(\chi^*)x) + R_1, \tag{6}$$

where $|R_1| \le Z^2 \cdot (A\mu/\varphi(Q)) \cdot 2Z = 2AZ^3\mu/\varphi(Q)$. Here $E_0(\chi^*) = 1$ only for $\chi = \chi_0$, and $|R_1| \le \mu x/(400\varphi(Q))$ as $x \ge 800AZ^3$.

Case 0: no exceptional zero in the sense of Theorem 5.2. The χ^* in (6) are distinct primitive characters of conductor $\le Q_G$. By (b) and Theorem 5.2, the middle sum is at most

$$\frac{A\mu}{\varphi(Q)} C_G x\Big(e^{-L} + \frac{(\kappa L)^2}{Q_G}\Big) \le \frac{\mu x}{\varphi(Q)}\Big(\frac{1}{400} + \frac{1}{10^4}\Big) \le \frac{\mu x}{200\,\varphi(Q)},$$

using $\log x/\log Q_G = \kappa L \ge L$. So $S(x) > 0$.

Exceptional case: a zero β_1 with character χ_1 and $1 - \beta_1 < 1/(\kappa \log Q_G)$. Put $u = (1 - \beta_1)\log x$; then $u < \log x/(\kappa \log Q_G) = L$, and $\beta_1 > 1/2$. The replaced right-hand side of Theorem 5.2 is $C_G u x(e^{-\kappa L} + \kappa L/Q_G)$. Since $u \le \min(u, 1)\, L$ (recall $L \ge 2$), its product with $A\mu/\varphi(Q)$ is at most

$$\frac{\mu x}{\varphi(Q)} \min(u, 1) \cdot C_G A L\Big(e^{-\kappa L} + \frac{\kappa L}{Q_G}\Big) \le \frac{\mu x}{200\,\varphi(Q)} \min(u, 1).$$

Here $C_G A L e^{-3L} \le C_G A e^{-L} \le 1/400$, and $C_G A \kappa L^2/Q_G \le 10^{-4}$. If no χ with $c(\chi) \neq 0$ has $\chi^* = \chi_1$, this bounds the middle sum of (6), and $S(x) > 0$ as in Case 0. Otherwise, by injectivity, exactly one such χ exists. It is real, because χ_1 is; hence $c(\chi)$ is real, as B and χ are. Let f_2 be as in (c). Its term is $c(\chi)(\vartheta(x;\chi_1) + x^{\beta_1}/\beta_1) - c(\chi)x^{\beta_1}/\beta_1$. The first part is covered by the bound just proved.

Case A: $f_2 = 1$. Then $c(\chi) = \mu/\varphi(Q)$, and the conductor q_1 of χ_1 divides Q, so $q_1 \le Z$. The main terms combine to $\lambda\mu x/\varphi(Q)$ with $\lambda = 1 - x^{\beta_1 - 1}/\beta_1$. Since $1/\beta_1 \le 1 + 2(1-\beta_1)$, $1 - e^{-u} \ge (1 - e^{-1})\min(u, 1)$ and $2(1-\beta_1)e^{-u} = 2ue^{-u}/\log x \le 2\min(u,1)/\log x$,

$$\lambda \;\ge\; 1 - e^{-u} - 2(1-\beta_1)e^{-u} \;\ge\; (0.63 - 2/16)\min(u,1) \;\ge\; \min(u,1)/2.$$

So the error from Theorem 5.2 is at most $\lambda\mu x/(100\varphi(Q))$. By Theorem 5.1, $u \ge 16(1-\beta_1) \ge 16/(c_2 Z^{1/2}(\log Z)^2)$, so $1/\lambda \le 2 + c_2 Z^{1/2}(\log Z)^2/8 \le C_5 Z$. Hence $|R_1| \le 2AZ^3\mu/\varphi(Q) \le \lambda\mu x/(100\varphi(Q))$ once $x \ge 200 C_5 A Z^4$. Altogether $S(x) \ge (1 - 2/100)\lambda\mu x/\varphi(Q) > 0$.

Case B: $f_2 > 1$. By (c), $|c(\chi)x^{\beta_1}/\beta_1| \le (\mu/(4\varphi(Q))) \cdot 2x = \mu x/(2\varphi(Q))$, and $S(x) \ge (1 - 1/2 - 1/200 - 1/400)\mu x/\varphi(Q) > 0$.

In all cases $S(x) > 0$. So some prime $p \le x$ with $p \nmid N_0$ and $p \equiv r \pmod Q$ has $B(p) > 0$, and then $W(p) > T$. □

Remark 6.2 (Assessment)**.** This is Linnik's theorem with weights. Every character coefficient is bounded by $\mathbb{E}_H|B|/\varphi(Q)$, not by the ℓ^1 mass of a cell representation. Theorem 5.2 controls all characters of conductor $\le Q_G$ simultaneously, so the number of moduli d_i never multiplies the error. In [2] and in the first version of this note, each progression was instead treated separately with the Thorner–Zaman theorem [47]. That costs $\log x \gg K\max(\log Z, K)$ with $K = 1 + \log(\sum |c_i|/\varphi(d_i)\,/\,\mu)$, which here is of order $(\log T)^{14}$ instead of $(\log T)^7$ in the setting of the second version. No lower bound for $1 - \beta_1$ is needed in Case B. In Case A we use the effective bound of Theorem 5.1 only to absorb the term R_1.

7. An energy bound for avoidance indicators

This section is self-contained and does not use arithmetic.

Setting. Let $\Omega = \prod_{v \in V_0} \Omega_v$ be a finite product of finite sets with a product probability measure $\pi = \bigotimes_v \pi_v$; write $\mathbb{P}$, $\mathbb{E}$ and $\|\cdot\| = \|\cdot\|_{L^2(\pi)}$. A *cylinder event* is a set $E = \{X_v = \sigma_v \ (v \in \operatorname{supp} E)\}$ with $\operatorname{supp} E \subseteq V_0$ and $\sigma \in \prod_{\operatorname{supp} E} \Omega_v$. For a finite family $\mathcal{E}$ of cylinder events (repetitions allowed) let $F = F_{\mathcal{E}} = \prod_{E \in \mathcal{E}}(1 - \mathbf{1}_E)$ be its *avoidance indicator*.

Efron–Stein decomposition (classical; [9, 22], [34, §8.3]). For $\varphi : \Omega \to \mathbb{R}$ and $W \subseteq V_0$ let $\mathbb{E}[\varphi \mid X_W]$ be the conditional expectation given the coordinates in W, and

$$\varphi^{=U} = \sum_{W \subseteq U} (-1)^{|U \setminus W|}\, \mathbb{E}[\varphi \mid X_W].$$

Then $\varphi = \sum_U \varphi^{=U}$, the components are mutually orthogonal, $\varphi^{=U}$ depends only on X_U, and $\mathbb{E}_u \varphi^{=U} = 0$ for $u \in U$, where $\mathbb{E}_u$ averages over the coordinate u. Put $L_u = I - \mathbb{E}_u$ and $L_V = \prod_{u \in V} L_u$; then $L_V \varphi = \sum_{U \supseteq V} \varphi^{=U}$. For a down-closed family $\mathcal{D}$ of subsets of V_0, the function $\varphi^{\mathcal{D}} = \sum_{U \in \mathcal{D}} \varphi^{=U}$ is a linear combination of the $\mathbb{E}[\varphi \mid X_W]$, $W \in \mathcal{D}$ (by Möbius inversion), and

$$\mathbb{E}(\varphi - \varphi^{\mathcal{D}})^2 = \sum_{U \notin \mathcal{D}} \|\varphi^{=U}\|^2. \tag{7}$$

For $\mathcal{D} = \{U : |U| \le t\}$ this is the *energy above level* t, $\mathrm{En}(\varphi; t)$; it is the least $\mathbb{E}(\varphi - g)^2$ over sums g of functions of at most t coordinates.

Definition 7.1 (Energy generating function)**.** For weights $\lambda_v \ge 1$ put $\mu_v = \lambda_v - 1$ and

$$G_\varphi(\lambda) = \sum_{U \subseteq V_0} \lambda^U \, \|\varphi^{=U}\|^2 = \sum_{V \subseteq V_0} \mu^V \, \|L_V \varphi\|^2.$$

The second form follows from the identity $\lambda^U = \sum_{V \subseteq U} \mu^V$ and $\|L_V \varphi\|^2 = \sum_{U \supseteq V} \|\varphi^{=U}\|^2$. For every τ,

$$\sum_{U:\ \lambda^U > e^\tau} \|\varphi^{=U}\|^2 \ \le \ e^{-\tau} G_\varphi(\lambda). \tag{8}$$

For a single event E with $\pi_E = \mathbb{P}(E)$ one computes $G_{1-\mathbf{1}_E} = 1 - 2\pi_E + \pi_E \prod_{v \in \operatorname{supp} E}(\lambda_v - \mu_v \pi_v(\sigma_v))$, so for small $\pi_v(\sigma_v)$ the condition $\lambda^{\operatorname{supp} E} \le 2$ below is necessary for $G \le 1$.

Hypergraphs. A hypergraph $\mathcal{H}$ is a finite set of subsets (*edges*) of V_0. For $V \subseteq V_0$ put

$$N_{\mathcal{H}}(V) = \sum_{\mathcal{J} \subseteq \mathcal{H}:\ \bigcup_{J \in \mathcal{J}} (J \cap V) = V} (-1)^{|\mathcal{J}|}, \qquad Q_\mu(\mathcal{H}) = \sum_{V \subseteq V_0} \mu^V \, N_{\mathcal{H}}(V)^2.$$

$N_{\mathcal{H}}(V)$ depends only on the traces $J \cap V$, and it is unchanged if edges are repeated or replaced by edges with the same trace: a class of $m \ge 1$ edges with equal traces contributes $\sum_{j \ge 1} \binom{m}{j} (-1)^j = -1$, exactly like one edge. If some edge misses V, then $N_{\mathcal{H}}(V) = 0$ (toggle that edge).

Lemma 7.2 (Cover bound; proved)**.** *In the Setting, for every $V \subseteq V_0$,*

$$\|L_V F\|^2 \ \le \ \mathbb{E}_x\big[N_{\mathcal{H}(x)}(V)^2\big], \qquad \mathcal{H}(x) = \{\operatorname{supp} E : \ E \in \mathcal{E}, \ x \in E\}.$$

Consequently $G_F(\lambda) \le \mathbb{E}_x\big[Q_\mu(\mathcal{H}(x))\big]$ for all weights $\lambda_v \ge 1$.

Proof. Write $x = (\sigma, y)$ with $\sigma \in \Omega_V$, $y \in \Omega_{V_0 \setminus V}$. The operator L_V acts on the V-coordinates only, so $\|L_V F\|^2 = \mathbb{E}_y \|L_V g_y\|^2_{L^2(\Omega_V)}$, where $g_y = F(\cdot, y)$. Now g_y is the avoidance indicator on Ω_V of the restricted events: those $E \in \mathcal{E}$ whose conditions off V hold at y, restricted to $\operatorname{supp} E \cap V$. Expand $g_y = \sum_{\mathcal{J}} (-1)^{|\mathcal{J}|} \mathbf{1}_{\mathcal{J}}$ over subfamilies $\mathcal{J}$ of restricted events, where $\mathbf{1}_{\mathcal{J}}$ is the indicator of their intersection, a cylinder on Ω_V (or empty). L_V kills every function that does not depend on all coordinates of V, hence every cylinder

whose support is a proper subset of V. A cylinder with support V is a point indicator $\mathbf{1}_{\{\sigma\}}$. So $L_V g_y = L_V c_y$, where

$$c_y(\sigma) = \sum_{\mathcal{J}:\ \operatorname{supp}\mathbf{1}_{\mathcal{J}}=V,\ \mathbf{1}_{\mathcal{J}}=\mathbf{1}_{\{\sigma\}}} (-1)^{|\mathcal{J}|}.$$

As L_V is an orthogonal projection in $L^2(\pi_V)$, $\|L_V c_y\|^2 \le \mathbb{E}_\sigma c_y(\sigma)^2$. The subfamilies $\mathcal{J}$ counted in $c_y(\sigma)$ are exactly the subfamilies of events that hold at $x = (\sigma, y)$ whose traces on V cover V. Two events that hold at x and have the same support are the same cylinder, so by the remark above $c_y(\sigma) = N_{\mathcal{H}(x)}(V)$. Taking $\mathbb{E}_y$ gives the first claim. Multiply by μ^V and sum over V for the second. □

The suppression is built in: $N_{\mathcal{H}(x)}(V) = 0$ as soon as some event that holds at x misses V.

For a finite multiset $\mathcal{C}$ of subsets of V_0 (possibly empty ones) put

$$\Theta_\lambda(\mathcal{C}) = \sum_{\mathcal{J}\subseteq\mathcal{C}} (-1)^{|\mathcal{J}|}\lambda^{\bigcup\mathcal{J}} \qquad (\Theta_\lambda(\emptyset) = 1).$$

Lemma 7.3 (Polarization; proved)**.** *Let $\lambda_v \ge 1$, $\mu_v = \lambda_v - 1$, and let $P \subseteq V_0$ be random with $\mathbb{P}(v \in P) = \mu_v/\lambda_v$ independently. For every hypergraph $\mathcal{H}$,*

$$Q_\mu(\mathcal{H}) = \mathbb{E}_P[\Theta_\lambda(\mathcal{H}_P)^2], \qquad \mathcal{H}_P = \{J \in \mathcal{H} :\ J \cap P = \emptyset\}.$$

Proof. Let $\tau(R) = \mathbf{1}[R \cap J \ne \emptyset\ \forall J \in \mathcal{H}]$ for $R \subseteq V_0$, and $\hat\tau(V) = \sum_{R\subseteq V}(-1)^{|V\setminus R|}\tau(R)$. Expanding $\tau(R) = \sum_{\mathcal{J}\subseteq\mathcal{H}}(-1)^{|\mathcal{J}|}\mathbf{1}[R\cap\bigcup\mathcal{J} = \emptyset]$, and using that the Möbius transform of $R \mapsto \mathbf{1}[R \cap U = \emptyset]$ is $V \mapsto (-1)^{|V|}\mathbf{1}[V \subseteq U]$, we get $\hat\tau(V) = (-1)^{|V|}N_{\mathcal{H}}(V)$. Let

$$\mathcal{T}(p) = \sum_V \hat\tau(V)p^V = \sum_R \tau(R)\prod_{v\in R} p_v \prod_{v\notin R}(1-p_v)$$

be the multilinear extension of τ. If the p_v are independent with mean 0 and variance μ_v, distinct monomials are orthogonal, and $\mathbb{E}[\mathcal{T}(p)^2] = \sum_V \mu^V\hat\tau(V)^2 = Q_\mu(\mathcal{H})$. Take $p_v = 1$ with probability μ_v/λ_v and $p_v = -\mu_v$ otherwise (mean $\mu_v/\lambda_v - \mu_v/\lambda_v = 0$, variance $\mu_v/\lambda_v + \mu_v^2/\lambda_v = \mu_v$), and $P = \{v : p_v = 1\}$. Only the $R \supseteq P$ contribute; writing $R = P \cup B$ with $B \subseteq P^c$,

$$\mathcal{T}(p) = \sum_{B\subseteq P^c} (-\mu)^B\lambda^{P^c\setminus B}\,\tau(P\cup B).$$

Now $\tau(P\cup B) = \sum_{\mathcal{J}\subseteq\mathcal{H}_P}(-1)^{|\mathcal{J}|}\mathbf{1}[B\cap\bigcup\mathcal{J} = \emptyset]$, and for $U \subseteq P^c$,

$$\sum_{B\subseteq P^c\setminus U}(-\mu)^B\lambda^{P^c\setminus B} = \lambda^U\prod_{v\in P^c\setminus U}(\lambda_v - \mu_v) = \lambda^U.$$

Hence $\mathcal{T}(p) = \Theta_\lambda(\mathcal{H}_P)$. □

Lemma 7.4 (Matching bound; proved)**.** *If every $J \in \mathcal{C}$ has $w_J := \lambda^J \le 2$, then for every matching $\mathcal{M} \subseteq \mathcal{C}$ (pairwise disjoint members)*

$$|\Theta_\lambda(\mathcal{C})| \le \prod_{J\in\mathcal{M}} (w_J - 1); \qquad \text{in particular } |\Theta_\lambda(\mathcal{C})| \le 1.$$

Proof. For $v \in V_0$ let $\mathcal{C} - v$ be the members not containing v and $\mathcal{C}/v = \{J \setminus \{v\} : J \in \mathcal{C}\}$. Splitting $\mathcal{J}$ according to whether it contains a member through v,

$$\Theta(\mathcal{C}) = \Theta(\mathcal{C} - v) + \lambda_v[\Theta(\mathcal{C}/v) - \Theta(\mathcal{C} - v)] = \lambda_v\Theta(\mathcal{C}/v) - \mu_v\Theta(\mathcal{C} - v).$$

Both families again have all weights at most 2. We induct on $|\bigcup\mathcal{C}|$. If $\emptyset \in \mathcal{C}$, then $\Theta(\mathcal{C}) = 0$ (toggling the empty member pairs the terms), and the claim holds. If $\mathcal{C} = \emptyset$, then $\Theta = 1$ and $\mathcal{M} = \emptyset$. This covers the base case $\bigcup\mathcal{C} = \emptyset$. If $\mathcal{M} = \emptyset$ and $\mathcal{C} \ne \emptyset$, it suffices to prove the bound for $\mathcal{M} = \{J\}$ with any $J \in \mathcal{C}$, as $w_J - 1 \le 1$. So let $J_0 \in \mathcal{M}$ be nonempty, $v \in J_0$, and $\Pi' = \prod_{J\in\mathcal{M}\setminus\{J_0\}}(w_J - 1)$. In $\mathcal{C}/v$ the family $(\mathcal{M} \setminus \{J_0\}) \cup \{J_0 \setminus \{v\}\}$ is a matching, so $|\Theta(\mathcal{C}/v)| \le (w_{J_0}/\lambda_v - 1)\Pi'$. In $\mathcal{C} - v$ the family $\mathcal{M} \setminus \{J_0\}$ is a matching, so $|\Theta(\mathcal{C} - v)| \le \Pi'$. Therefore $|\Theta(\mathcal{C})| \le (w_{J_0} - \lambda_v)\Pi' + \mu_v\Pi' = (w_{J_0} - 1)\Pi'$. □

Theorem 7.5 (Energy bound; proved)**.** (a) *If every edge of a hypergraph $\mathcal{H}$ has $\lambda^J \le 2$, then $Q_\mu(\mathcal{H}) \le \prod_{J\in\mathcal{M}}(\lambda^J - 1)$ for every matching $\mathcal{M} \subseteq \mathcal{H}$; in particular $Q_\mu(\mathcal{H}) \le 1$.* (b) *In the Setting, if $\lambda_v \ge 1$ and $\lambda^{\operatorname{supp} E} \le 2$ for every $E \in \mathcal{E}$, then*

$$G_F(\lambda) = \sum_U \lambda^U \|F^{=U}\|^2 \;\le\; 1.$$

Proof. (a) By Lemma 7.3 and Lemma 7.4 applied to $\mathcal{H}_P$ with the matching $\mathcal{M} \cap \mathcal{H}_P$, $Q_\mu(\mathcal{H}) \le \mathbb{E}_P \prod_{J\in\mathcal{M}\cap\mathcal{H}_P}(\lambda^J - 1)^2$. The members of $\mathcal{M}$ are disjoint, so the events $\{J \cap P = \emptyset\}$ are independent, with probability $\prod_{v\in J}\lambda_v^{-1} = 1/w_J$. Hence $Q_\mu(\mathcal{H}) \le \prod_{J\in\mathcal{M}}[(1 - 1/w_J) + (w_J - 1)^2/w_J] = \prod_{J\in\mathcal{M}}(w_J - 1)$. (b) Lemma 7.2 and (a), since $\mathcal{H}(x)$ consists of supports of events. □

Corollary 7.6 (Energy tail; proved)**.** *In the Setting, if every event of $\mathcal{E}$ has $|\operatorname{supp} E| \le k$ ($k \ge 1$), then for $f \in \{F, 1 - F\}$ and every integer $t \ge 0$,*

$$\mathrm{En}(f; t) = \sum_{|U|>t} \|f^{=U}\|^2 \;\le\; 2^{-(t+1)/k}.$$

Proof. Theorem 7.5(b) with $\lambda_v = 2^{1/k}$ and (8); $(1 - F)^{=U} = -F^{=U}$ for $U \ne \emptyset$. □

Remark 7.7 (Sharpness; proved)**.** Over general product spaces the base $2^{-1/k}$ cannot be improved. Take m events on pairwise disjoint sets of k coordinates, each fixing at every coordinate of its support a value of probability p; put $\pi = p^k$. As the factors $1 - \mathbf{1}_{E_i}$ are independent, $F^{=U} = \prod_i(1 - \mathbf{1}_{E_i})^{=U\cap\operatorname{supp} E_i}$. For U the union of the supports of j of the events

this gives $\|F^{=U}\|^2 = [\pi(1-p)^k]^j(1-\pi)^{2(m-j)}$, and $|U| = jk$. Hence

$$\mathrm{En}(F; jk-1) \;\geq\; \binom{m}{j}[\pi(1-p)^k]^j(1-\pi)^{2(m-j)} \;\longrightarrow\; e^{-2s}\frac{s^j}{j!}$$

as $p \to 0$ with $m = \lfloor s/\pi \rfloor$. With $s = j/2$ and $j! \leq e\, j^{j+1/2}e^{-j}$, the limit is at least $2^{-j}/(e\sqrt{j}) = 2^{-(t+1)/k}/(e\sqrt{j})$ for $t = jk-1$. So no bound $C\rho^{-(t+1)/k}$ with $\rho > 2$ holds for all such systems (with alphabets of size about $1/p$), and at most a factor of order $\sqrt{t/k}$ could be gained. The condition $\lambda^{\operatorname{supp} E} \leq 2$ in Theorem 7.5 is necessary, by the single-event formula above.

Remark 7.8 (Boolean form and comparison; proved, except the cited constant)**.** A width-k DNF $g : \{\pm1\}^n \to \{\pm1\}$ is $1 - 2F$ for the family of its terms (a term with a literal set is a cylinder event). As $g^{=U} = -2F^{=U}$ for $U \neq \emptyset$, Corollary 7.6 gives, for every product measure on $\{\pm1\}^n$ (Efron–Stein weights; Fourier weights for the uniform measure),

$$\sum_{|S|>t} \hat{g}(S)^2 \;\leq\; 4 \cdot 2^{-(t+1)/k},$$

i.e. g is ε-concentrated up to degree $k\log_2(4/\varepsilon)$. The switching-lemma route of Linial, Mansour and Nisan gives $\sum_{|S|\geq t}\hat{g}(S)^2 \leq 2\cdot 2^{-t/(20k)}$ [34, §4.4], [25]. Also $\lambda^{|U|} \geq 1 + |U|\ln\lambda$ gives total influence $\sum_U |U|\,\|F^{=U}\|^2 \leq (k/\ln 2)\mathbb{P}(F = 0)$. The same bounds hold for q-ary DNFs with literals $x_v \in S_v$ (split each term into single-value cylinders on the same support; F and the supports do not change). We have not found Theorem 7.5 in the literature; see the introduction for what we checked.

Digit filtrations. For the application we need weights that grow along a filtration of each coordinate. Suppose that each coordinate X_ℓ ($\ell \in \mathcal{P}$, a finite index set) is a tuple of independent *digits* $(\delta_{\ell,i})_{i_0(\ell)\leq i<f_\ell}$, with arbitrary distributions, and that the coordinates are independent; the Efron–Stein decomposition $F = \sum_U F^{=U}$ below is taken over the digits, so U is a set of pairs (ℓ, i). For such U write $U_\ell = \{i : (\ell, i) \in U\}$. We say that an event E is a *digit-prefix event* if, for some integers $v_\ell(E) \geq 0$, it fixes exactly the digits $\delta_{\ell,i}$ with $i_0(\ell) \leq i < v_\ell(E)$, for each ℓ (to arbitrary values). Such an event is a cylinder event on the digits.

Theorem 7.9 (Filtration energy bound; proved)**.** *In this situation let $\lambda_\ell \geq 1$ ($\ell \in \mathcal{P}$), and suppose that every event of $\mathcal{E}$ is a digit-prefix event with $\prod_\ell \lambda_\ell^{2v_\ell(E)} \leq 2$. Then*

$$\sum_U \Big(\prod_{\ell:\, U_\ell\neq\emptyset} \lambda_\ell^{1+\max U_\ell}\Big)\|F^{=U}\|^2 \;\leq\; 1.$$

Proof. Write $\Lambda_{\ell,i} = \lambda_\ell^{i+1}$, $\Lambda_{\ell,i_0(\ell)-1} = 1$, and $\mu'_{\ell,j} = \Lambda_{\ell,j} - \Lambda_{\ell,j-1} \geq 0$ for $i_0(\ell) \leq j < f_\ell$. For such j let $P_{\ell,j} = I - \mathbb{E}_{B_{\ell,j}}$, where $\mathbb{E}_{B_{\ell,j}}$ averages over the *tail block* $B_{\ell,j} = (\delta_{\ell,i})_{i\geq j}$. Averaging over a digit in U kills $F^{=U}$, and $F^{=U}$ does not depend on digits outside U; hence $P_{\ell,j}F^{=U} = F^{=U}$ if $U_\ell \neq \emptyset$

and $\max U_\ell \geq j$, and $P_{\ell,j}F^{=U} = 0$ otherwise. Call a set V of pairs (ℓ, j) a *selection* if it contains at most one pair for each ℓ, and put $L'_V = \prod_{(\ell,j)\in V} P_{\ell,j}$. Then $\|L'_V F\|^2 = \sum_{U \succ V} \|F^{=U}\|^2$, where $U \succ V$ means $\max U_\ell \geq j$ for all $(\ell, j) \in V$. For fixed U, the sum of μ'^V over the selections $V \prec U$ factorises as $\prod_\ell (1 + \sum_{i_0(\ell)\leq j\leq \max U_\ell} \mu'_{\ell,j}) = \prod_{\ell:\, U_\ell\neq\emptyset} \Lambda_{\ell,\max U_\ell}$. So the left side of the claim equals $\sum_V \mu'^V \|L'_V F\|^2$, the sum over selections.

Fix a selection V and pass to the coarser product space whose coordinates are, for $(\ell, j) \in V$, the head $(\delta_{\ell,i})_{i<j}$ and the tail block $B_{\ell,j}$, and, for the other ℓ, the whole X_ℓ. There L'_V is the operator $L_{V'}$ of the Setting for the set V' of tail blocks. Each event E is a disjoint union of cylinder events on the coarse coordinates with the same coarse support (enumerate the unfixed digits of each coarse coordinate that E involves), and splitting every event in this way does not change F. Exactly one piece of E holds at x if E does. The trace of the coarse support of E on V' is the set of $B_{\ell,j}$ with $(\ell, j) \in V$ and $j < v_\ell(E)$. Identify $B_{\ell,j}$ with the pair (ℓ, j) and put $\hat{E} = \{(\ell, j) :\ i_0(\ell) \leq j < v_\ell(E)\}$. By Lemma 7.2 and the remark on traces,

$$\|L'_V F\|^2 \leq \mathbb{E}_x\big[N_{\hat{\mathcal{H}}(x)}(V)^2\big], \qquad \hat{\mathcal{H}}(x) = \{\hat{E} :\ E \in \mathcal{E},\ x \in E\}.$$

Summing over selections and adding the nonnegative terms of all other sets V of pairs, the left side of the claim is at most $\mathbb{E}_x[Q_{\mu'}(\hat{\mathcal{H}}(x))]$. By Theorem 7.5(a) it remains to show $\prod_{(\ell,j)\in\hat{E}}(1 + \mu'_{\ell,j}) \leq 2$ for every event. Fix ℓ and write $\lambda = \lambda_\ell$, $v = v_\ell(E)$, $i_0 = i_0(\ell)$, and assume $v > i_0$. Here $1 + \mu'_{\ell,i_0} = \lambda^{i_0+1}$ and $1 + \mu'_{\ell,j} = 1 + \lambda^j(\lambda - 1)$ for $j > i_0$. The hypothesis gives $\lambda^{2v} \leq 2$, so $\lambda^j \leq \lambda^v \leq \sqrt{2} < \lambda + 1$ for $j < v$, i.e. $1 + \lambda^j(\lambda - 1) \leq \lambda^2$. The factor of ℓ is therefore at most $\lambda^{i_0+1}\lambda^{2(v-1-i_0)} \leq \lambda^{2v}$, and the product over ℓ is at most $\prod_\ell \lambda_\ell^{2v_\ell(E)} \leq 2$. □

Corollary 7.10 (Modulus-weighted tail; proved)**.** *In Theorem 7.9 let the $\ell \in \mathcal{P}$ be primes and $\lambda_\ell = 2^{\log \ell/(2\mathcal{L})}$, and suppose $\prod_\ell \ell^{v_\ell(E)} \leq T$ for every event. For a set U of digits put $m_U = \prod_{\ell:\, U_\ell\neq\emptyset} \ell^{1+\max U_\ell}$. Then for every $\tau \geq 0$*

$$\sum_{U:\ m_U > e^\tau} \|F^{=U}\|^2 \ \leq\ 2^{-\tau/(2\mathcal{L})}.$$

Proof. $\prod_\ell \lambda_\ell^{2v_\ell(E)} = 2^{\log \prod_\ell \ell^{v_\ell(E)}/\mathcal{L}} \leq 2$, and the weight of U in Theorem 7.9 is $2^{\log m_U/(2\mathcal{L})}$. □

The cost of a junta is thus measured by the modulus m_U a function of the digits in U needs, not by the number of coordinates. For the Erdős–Straus events (digits in base ℓ) this removes both the binary width and the support size from the junta modulus.

8. The residual system and the sandwich minorant

Throughout this section T is large, $\beta = 1 + 1/\log \mathcal{L}$, $\eta = \frac{3}{4}\log\beta$, and (Q, r) is the square-class quarantine of Theorem 3.3; $E_1, \ldots, E_m$ ($m \leq T^2$) are the

events of the surviving atoms, in some order, $A_i = \mathbf{1}_{E_i}$, $F_{<i} = \prod_{j<i}(1-A_j)$, $F = F_{<m+1}$, $S = \sum_i \mathbb{P}(E_i) \le S_\beta$ and $\delta = \mathbb{E}F$. By (4) and Lemma 2.8, every subfamily of $\{E_i\}$ satisfies the hypothesis of Lemma 2.8, $\delta \ge e^{-4S_\beta/3}$, and for T large

$$\eta \le 0.19 \qquad \text{(as soon as } \mathcal{L} \ge 33\text{)}. \tag{9}$$

Digits. For $\ell \in \mathcal{P}$ write $x \in \Omega_\ell$ in base ℓ as $x = \sum_{i<f_\ell} \delta_{\ell,i}\ell^i$. The digits $\delta_{\ell,i}$ with $i < a_\ell$ are those of r, hence constant on Ω_ℓ. The others are independent and uniform: $\delta_{\ell,0}$ on $\{1,\dots,\ell-1\}$ if $a_\ell = 0$, and $\delta_{\ell,i}$ on $\{0,\dots,\ell-1\}$ for $\max(a_\ell,1) \le i < f_\ell$; indeed $x \mapsto (\delta_{\ell,i})_{i\ge a_\ell}$ is a bijection from Ω_ℓ onto the product of these ranges. So the free digits are those with $i \ge i_0(\ell) := a_\ell$, and $E_{M,D}$ is a digit-prefix event with $v_\ell(E_{M,D}) = v_\ell(M)$ for $\ell \in \operatorname{supp}(M,D)$ and $v_\ell(E_{M,D}) = 0$ otherwise. A function of the digits $\delta_{\ell,i}$ with $i \le i_\ell$ depends only on X_ℓ modulo ℓ^{1+i_ℓ}.

Cells. A *consistent unit cell* is a class $b \bmod d$ with d composed of primes of $\mathcal{P}$, $d \mid \prod_{\ell\in\mathcal{P}} \ell^{f_\ell}$, $\gcd(b,d) = 1$ and $b \equiv r \pmod{\gcd(d,Q)}$; on Ω it is the event $\{X_\ell \equiv b \pmod{\ell^{v_\ell(d)}})\ \forall \ell \mid d\}$, and on $\mathbb{Z}$ the class itself. The events $E_{M,D}$ are consistent unit cells (by survival). The intersection of two consistent unit cells is empty or a consistent unit cell whose modulus is the lcm of the two, both on Ω and on $\mathbb{Z}$. A function on Ω that depends only on $(X_\ell \bmod \ell^{b_\ell})_\ell$ is a linear combination of consistent unit cells of modulus $\prod_\ell \ell^{b_\ell}$ (the indicators of the points of its level sets).

Lemma 8.1 (Haar means; proved)**.** *Let $Q' = Q\ell'$ with ℓ' a prime not dividing $Q\prod_{\ell\in\mathcal{P}}\ell$, and r' the class modulo Q' with $r' \equiv r \pmod Q$ and $r' \equiv 1 \pmod{\ell'}$ (or $Q' = Q$, $r' = r$). Let φ be a combination of consistent unit cells with* lcm *of moduli D. With H as in Theorem 6.1 for Q', r' and D, $\mathbb{E}_H\varphi = \mathbb{E}\varphi$. The same holds for $\varphi\psi$, where ψ is a character modulo a squarefree $f \mid D$ coprime to Q, viewed on Ω through $X_\ell \bmod \ell$ for $\ell \mid f$.*

Proof. Let $N_0 = \operatorname{lcm}(Q', D)$. By the Chinese remainder theorem the uniform measure on H is the product of the uniform measures on $H_p = \{x \in (\mathbb{Z}/p^{v_p(N_0)})^\times : x \equiv r' \ (p^{v_p(Q')})\}$. For $\ell \in \mathcal{P}$ with $\ell \mid D$, $v_\ell(Q') = a_\ell$ and $v_\ell(N_0) \le f_\ell$, and reduction maps the uniform measure on Ω_ℓ to the uniform measure on H_ℓ (all fibres have the same size). The other primes of N_0 do not divide D, and φ, ψ do not depend on them. □

For $j \le m$ let $F^{(j)}$ be the function of the digits not fixed by E_j obtained from $F_{<j}$ by giving the digits fixed by E_j their values on E_j. Then $A_jF_{<j} = A_jF^{(j)}$. At ℓ, the digits not fixed by E_j are those with $i \ge i_0^{(j)}(\ell) := \max(a_\ell, v_\ell(E_j))$, and $F^{(j)}$ is the avoidance indicator of the restrictions of the events E_i $(i<j)$ compatible with E_j. Each restriction fixes the digits $i_0^{(j)}(\ell) \le i < v_\ell(E_i)$, so it is a digit-prefix event for the filtration starting at $i_0^{(j)}$, with the same $v_\ell(E_i)$ and $\prod_\ell \ell^{v_\ell(E_i)} \le M_i \le T$. (A restriction may fix nothing; then $F^{(j)} \equiv 0$.)

Lemma 8.2 (Bazzi–Razborov sandwich [3, 40]; proved)**.** *Let $u_1, \dots, u_m$ be arbitrary real functions, $v_i = \sum_{j<i} A_j u_j$, and*

$$B = 1 - \sum_{i \le m} A_i (1 - v_i)^2.$$

Then $B \le F$ pointwise, and with $e_j = F_{<j} - u_j$,

$$F - B = \sum_i A_i \Big(\sum_{j<i} A_j e_j \Big)^2, \qquad \mathbb{E}[F - B] \le m^2 \sum_{j \le m} \mathbb{E}[A_j e_j^2].$$

If u_j is a function of the digits not fixed by E_j, then $\mathbb{E}[A_j e_j^2] = \mathbb{P}(E_j)\, \mathbb{E}(F^{(j)} - u_j)^2$.

Proof. The identity $1 - F = \sum_i A_i F_{<i}$ holds pointwise: the sum counts the first occurring event. Put $f_{<i} = 1 - F_{<i} = \sum_{j<i} A_j F_{<j}$. If $F_{<i} = 1$, then $A_j = 0$ for all $j < i$, so $v_i = 0$ and $A_i(1 - v_i)^2 = A_i = A_i F_{<i}$. If $F_{<i} = 0$, then $f_{<i} = 1$, and $A_i(1 - v_i)^2 - A_i F_{<i} = A_i(f_{<i} - v_i)^2$. In both cases $A_i(1 - v_i)^2 - A_i F_{<i} = A_i(f_{<i} - v_i)^2$, and summing over i,

$$F - B = \sum_i A_i \big[(1 - v_i)^2 - F_{<i} \big] = \sum_i A_i (f_{<i} - v_i)^2 \ \ge 0,$$

with $f_{<i} - v_i = \sum_{j<i} A_j (F_{<j} - u_j)$. By Cauchy–Schwarz, $(\sum_{j<i} A_j e_j)^2 \le m \sum_{j<i} A_j e_j^2$. Drop $A_i \le 1$ and sum over i. For the last claim, $A_j e_j^2 = A_j (F^{(j)} - u_j)^2$, and $F^{(j)} - u_j$ is independent of A_j. □

This is Razborov's simplification [40] of Bazzi's argument [3]. The choice of u_j as an ℓ^2-approximation of the "conditional survival function" $F^{(j)}$ is the one used there. We include the short proof for completeness. The identity and the inequality $B \le F$ use only that each A_i takes values in $\{0, 1\}$, so they also hold for the functions on $\mathbb{Z}$ obtained from cell expansions.

Choice of u_j. Fix $\tau \ge 0$. For $j \le m$ let $\mathcal{D}_j$ be the family of sets U of digits not fixed by E_j with $m_U \le e^\tau$, where $m_U = \prod_{\ell : U_\ell \ne \emptyset} \ell^{1 + \max U_\ell}$; it is down-closed. Put

$$u_j = (F^{(j)})^{\mathcal{D}_j} = \sum_{U \in \mathcal{D}_j} (F^{(j)})^{=U}, \tag{10}$$

the Efron–Stein decomposition being taken over the digits not fixed by E_j.

Lemma 8.3 (Approximation error; proved)**.** *With u_j as in (10), $\mathbb{E}[F - B] \le m^2 S\, 2^{-\tau/(2\mathcal{L})}$.*

Proof. By (7) and Corollary 7.10, applied to $F^{(j)}$ with the filtration starting at $i_0^{(j)}$ (the digits are indexed by their absolute position, so m_U is the modulus needed), $\mathbb{E}(F^{(j)} - u_j)^2 \le 2^{-\tau/(2\mathcal{L})}$. Conclude with Lemma 8.2. □

Lemma 8.4 (Cell structure of B; proved)**.** *With u_j as in (10), B is a finite real combination of consistent unit cells of modulus at most $T^3 e^{2\tau}$, composed of primes of $\mathcal{P}$. The identity $B = 1 - \sum_i A_i(1 - v_i)^2$ also holds for the expanded cell combination evaluated at any integer n.*

Proof. By (7), u_j is a combination of the functions $\mathbb{E}[F^{(j)} \mid \delta_W]$, $W \in \mathcal{D}_j$. Such a function depends only on $X_\ell \bmod \ell^{1+\max W_\ell}$ for the ℓ with $W_\ell \neq \emptyset$, so it is a combination of consistent unit cells of modulus $m_W \le e^\tau$. Expanding,

$$B = 1 - \sum_i A_i + 2\sum_i \sum_{j<i} A_i A_j u_j - \sum_i \sum_{j,j'<i} A_i A_j A_{j'} u_j u_{j'}.$$

Each term is a product of at most three events (moduli $\le T$) and at most two u's, so every cell is an intersection of consistent unit cells of total modulus at most $T^3 e^{2\tau}$. The last claim holds because the intersection identities hold on $\mathbb{Z}$. □

Lemma 8.5 (ℓ^1-tightness; proved)**.** *If $B \le F$ pointwise, $F \ge 0$ and $\mathbb{E}[F - B] \le \eta\,\mathbb{E}B$, then $\mathbb{E}|B| \le (1 + 2\eta)\mathbb{E}B$.*

Proof. $B^- := \max(-B, 0) \le F - B$ since $F \ge 0$, and $\mathbb{E}|B| = \mathbb{E}B + 2\mathbb{E}B^-$. □

Lemma 8.6 (Twist; proved)**.** *Assume $\mathbb{E}[F - B] \le \delta/100$, and write B as in Lemma 8.4, with $\mu = \mathbb{E}B$. Let ψ be a real primitive character of conductor $f > 1$ with $\gcd(f, Q) = 1$ and $f \mid d_i$ for some cell modulus d_i. Then $\mu \ge 0.99\,\delta$ and $|\mathbb{E}[B\psi]| \le \mu/4$, where ψ is viewed on Ω through $X_\ell \bmod \ell$, $\ell \mid f$.*

Proof. $\mu = \mathbb{E}F - \mathbb{E}[F - B] \ge 0.99\,\delta$. As $2 \nmid f$, f is squarefree. Its prime factors divide d_i, so they lie in $\mathcal{P}$, and $a_\ell = 0$ for them as $\gcd(f, Q) = 1$. So $\psi = \prod_{\ell \mid f} \psi_\ell(X_\ell \bmod \ell)$ with nontrivial characters ψ_ℓ of the uniformly distributed digits $\delta_{\ell,0}$. Now

$$|\mathbb{E}[B\psi]| \le \mathbb{E}|B - F| + |\mathbb{E}[F\psi]| = \mathbb{E}[F - B] + |\mathbb{E}[F\psi]|.$$

Fix $\ell_0 \mid f$ and let F' be the avoidance indicator of the events E_i with $\ell_0 \notin \operatorname{supp} E_i$. Given the other coordinates $X_{-\ell_0}$, we have $F = F' \cdot \mathbf{1}[X_{\ell_0} \notin \mathrm{Forb}]$, where Forb is the union of the classes $X_{\ell_0} \equiv -4D_i \pmod{\ell_0^{v_{\ell_0}(M_i)}}$ over the events $E_i \ni \ell_0$ whose other conditions hold. Write $E_i \setminus \ell_0$ for the event given by those other conditions, and π_i for the Ω_{ℓ_0}-probability of the class at ℓ_0, so that $\mathbb{P}(E_i) = \mathbb{P}(E_i \setminus \ell_0)\pi_i$. Since $\mathbb{E}_{X_{\ell_0}} \psi_{\ell_0} = 0$,

$$\begin{aligned} |\mathbb{E}[F\psi]| &= \Big|\mathbb{E}\Big[F' \textstyle\prod_{\ell \neq \ell_0} \psi_\ell \; \mathbb{E}_{X_{\ell_0}}\big[\psi_{\ell_0}\, \mathbf{1}[X_{\ell_0} \in \mathrm{Forb}]\big]\Big]\Big| \\ &\le \sum_{i:\, \ell_0 \in \operatorname{supp} E_i} \mathbb{P}(F' = 1,\; E_i \setminus \ell_0)\, \pi_i. \end{aligned}$$

The event $E_i \setminus \ell_0$ is determined by the $s(E_i) - 1$ coordinates of $\operatorname{supp} E_i \setminus \{\ell_0\}$, so Lemma 2.8, applied to the family defining F', gives $\mathbb{P}(E_i \setminus \ell_0 \mid F' = 1) \le \beta^{s(E_i)-1}\mathbb{P}(E_i \setminus \ell_0)$. Hence

$$|\mathbb{E}[F\psi]| \le \beta^{-1} \sum_{i:\, \ell_0 \in \operatorname{supp} E_i} \beta^{s(E_i)} \mathbb{P}(E_i)\, \mathbb{E}F' = \beta^{-1} \tilde{w}_{\ell_0} \mathbb{E}F' \le \eta\, \mathbb{E}F'.$$

The same bound gives $\mathbb{E}F' - \mathbb{E}F = \mathbb{E}[F'\mathbf{1}[X_{\ell_0} \in \mathrm{Forb}]] \le \eta\,\mathbb{E}F'$, so $\mathbb{E}F' \le \delta/(1 - \eta)$. By (9), $|\mathbb{E}[B\psi]| \le (0.01 + \eta/(1 - \eta))\delta \le 0.245\,\delta \le 0.245\,\mu/0.99 < \mu/4$. □

9. Proof of the main theorems

Theorem 9.1 (Assembly; proved modulo Theorems 5.2 and 3.1)**.** *Let T be large, let (Q, r) be the square-class quarantine of Theorem 3.3, put $S_1 = C\mathcal{L}^3 \log \mathcal{L}$ (the bound for S_β in (4)) and*

$$\tau = 2\mathcal{L}\lceil \log_2 (100\, T^4 (S_1 + 1) e^{3S_1}) \rceil.$$

Then there is a hard prime p with $p > T$, $W(p) > T$ and $\log p \le C_6 \log Z$, where C_6 is absolute and

$$\log Z \le \log Q + 4\tau + 6\mathcal{L} + 1.$$

Proof. Take the residual system of (Q, r); $m \le T^2$. If $m = 0$, then $F \equiv 1$; we put $B = 1$ (one cell of modulus 1), so that $\mu = A = 1$ and the twist condition is vacuous, and go directly to the transfer step below. Otherwise define B by Lemma 8.2 with u_j as in (10). By Lemma 8.3, the choice of τ, $S \le S_\beta \le S_1$ and $\delta \ge e^{-4S_\beta/3} \ge e^{-3S_1}$,

$$\mathbb{E}[F - B] \le m^2 S\, 2^{-\tau/(2\mathcal{L})} \le \frac{S\, e^{-3S_1}}{100(S_1 + 1)} \le \frac{e^{-3S_1}}{100} \le \frac{\delta}{100}.$$

Lemma 8.6 gives $\mu = \mathbb{E}B \ge 0.99\,\delta > 0$, and, as $\mathbb{E}[F - B] \le \mu/99$, Lemma 8.5 gives $\mathbb{E}|B| \le 1.03\,\mu$. By Lemma 8.4, B is a combination of consistent unit cells of moduli $d_i \le T^3 e^{2\tau}$.

Pointwise inequality on integers. For every integer n the values $A_i(n) \in \{0, 1\}$ satisfy the hypotheses of Lemma 8.2, and the cell expansion of B is valid on $\mathbb{Z}$ (Lemma 8.4). So $B(n) \le F(n)$, and by Lemma 2.5(i), $B(n) \le \mathbf{1}[W(n) > T]$ for every $n \equiv r \pmod{Q}$.

Transfer. Let $R = \max(T, \max_i d_i)$, choose a prime $\ell_{\mathrm{aux}} \in (R, 2R]$, put $Q' = Q\ell_{\mathrm{aux}}$, and let r' be the class modulo Q' with $r' \equiv r \pmod{Q}$ and $r' \equiv 1 \pmod{\ell_{\mathrm{aux}}}$. Then $8 \mid Q'$, $r' \equiv 1 \pmod 8$, and r' is a quadratic residue modulo every odd prime of Q'. As $\ell_{\mathrm{aux}} > \max d_i$ is prime, $\gcd(d_i, Q') = \gcd(d_i, Q)$, so the cells are consistent for (Q', r'), and no cell or event involves ℓ_{aux}. The inequality $B \le \mathbf{1}[W > T]$ holds on $n \equiv r' \pmod{Q'}$. The functions B and $|B|$ depend only on $X \bmod D$, so both are combinations of consistent unit cells (point cells modulo D), and by Lemma 8.1, $\mathbb{E}_H B = \mu$ and $\mathbb{E}_H|B|/\mu \le 1.03 \le Z^{1/4}$. A real primitive character of conductor $f > 1$ coprime to Q' is coprime to Q, so Lemmas 8.6 and 8.1 give the twist condition. Here $\log Z \le \log Q + \log(2R) + \log \max d_i \le \log Q + \log 2 + 2(3\mathcal{L} + 2\tau)$. Theorem 6.1 with $\log x = C_2(1 + \log 1.03) \log Z$ gives a prime $p \equiv r' \pmod{Q'}$ with $W(p) > T$ and $\log p \le C_6 \log Z$. Since $\ell_{\mathrm{aux}} \mid p - 1$, $p > \ell_{\mathrm{aux}} > T$; and $p \equiv r \pmod{Q}$ with $840 \mid Q$, so p is hard by Lemma 2.5(iii). □

The prime ℓ_{aux} is needed for infinitely many *distinct* primes: $W(p) > T$ alone does not force $p > T$.

Proof of Theorem 1.1. By (4), $\tau \ll \mathcal{L}(\mathcal{L} + S_1) \ll \mathcal{L}^4 \log \mathcal{L}$ and $\log Q \ll \mathcal{L}^3 (\log \mathcal{L})^5$. So $\log Z \ll \mathcal{L}^4 \log \mathcal{L}$, and Theorem 9.1 gives a hard prime $p > T$ with $W(p) > T$ and $\log p \le C\mathcal{L}^4 \log \mathcal{L}$. As $\mathcal{L} < \log p$, this gives

$\mathcal{L}^4 \geq \log p/(C \log\log p)$, i.e. $W(p) > T \geq \exp\left(c(\log p)^{1/4}(\log\log p)^{-1/4}\right)$. Applying this to a sequence $T_1 < T_2 < \dots$ with T_{j+1} larger than the prime found for T_j gives infinitely many distinct primes. □

Remark 9.2 (Where the exponent comes from; Assessment)**.** The certified bound is $\log p \ll \log Z \ll \log Q + \tau$. The quarantine term is $\log Q \ll \mathcal{L}^3(\log \mathcal{L})^5$, of the order of the Haar density (Theorems 3.3 and 4.4). The bottleneck is the junta term $\tau \asymp \mathcal{L}(S_1 + \mathcal{L}) \asymp \mathcal{L}^4 \log \mathcal{L}$: the cells of the sandwich have modulus about $e^{\mathcal{L}S}$. In the previous version both terms were of order $\mathcal{L}^5$ (up to $\log \mathcal{L}$). Section 10 shows that the junta term cannot be reduced within the present architecture.

10. The ceiling of the minorant architecture

The proof of Theorem 1.1 passes through a minorant $B \leq F$ of positive Haar mean whose cells have modulus $e^{O(\mathcal{L}S)}$. Here we show that no minorant of smaller level exists, dense or sparse, on any fibre whose modulus satisfies $\log Q \leq T^{0.05}$ (for larger Q the transfer itself is too expensive, Corollary 10.7).

Setting. Fix a modulus Q and a unit r modulo Q, and consider the *fibre* $\{n \in \hat{\mathbb{Z}}^\times : \ n \equiv r \ (Q)\}$ with its Haar probability. Let $F(n) = \mathbf{1}[W(n) > T]$. A function on the fibre has *level* at most D if it is a linear combination of functions $n \mapsto \phi(n \bmod qQ)$ with $q \leq D$. Every cell of modulus at most D and every character of conductor at most D (restricted to the fibre) has level at most D; the minorant of Theorem 9.1 has level at most $\max_i d_i \leq T^3 e^{2\tau}$. All functions below depend on n modulo a fixed finite modulus, so all probability spaces are finite.

Lemma 10.1 (Planting; proved)**.** *Let $b_1, \dots, b_N$ be independent bits with $\mathbb{P}(b_i = 1) = p_i \in [0, 1)$, and law ν_0. Put $r_i = p_i/(1 - p_i)$, $R = \sum_i r_i$ and $r^* = \max_i r_i$. Let $k \geq 0$ with*

$$R \ \geq \ (k + 1) + (2k + 1)r^*. \tag{11}$$

Then there is a probability law ν on $\{0, 1\}^N$ with the same marginals as ν_0 on every set of at most k coordinates and $\nu(b = 0) = 0$.

Proof. Write 1_y for the configuration with ones exactly on $y \subseteq [N]$, and $P_0 = \nu_0(1_\emptyset) = \prod(1-p_i)$, so $\nu_0(1_y) = P_0 \prod_{i \in y} r_i$. Let e_a denote the elementary symmetric functions. If $e_{k+1}(r) = 0$, at most k of the r_i are positive and $R \leq k r^*$, contradicting (11). For $|J| = k + 1$ put $w_J = \prod_{i \in J} r_i / e_{k+1}(r)$, so $\sum_J w_J = 1$, and

$$\nu = \nu_0 + P_0 \sum_{|J| = k+1} w_J \sigma_J, \qquad \sigma_J = \sum_{y \subseteq J} (-1)^{|y|+1} \delta_{1_y}.$$

Marginals. For $|K| \leq k$ and $z \subseteq K$, the σ_J-mass of $\{x : \ x_K = 1_z\}$ is $[z \subseteq J] \sum_{y' \subseteq J \setminus K} (-1)^{|z|+|y'|+1} = 0$, as $J \setminus K \neq \emptyset$. In particular $(K = \emptyset)$ ν has total mass 1. *Zero.* $\nu(1_\emptyset) = P_0 - P_0 \sum_J w_J = 0$. *Positivity.* Only $y \subseteq J$ with $|y| = j \geq 2$ even receives negative charge, of total

$P_0 \sum_{J \supseteq y} w_J = P_0 \prod_{i \in y} r_i \, e_{k+1-j}(s)/e_{k+1}(r)$, where $s = (r_i)_{i \notin y}$. So it suffices that $e_{k+1-j}(s) \le e_{k+1}(r)$. Now $e_{k+1}(r) \ge e_{k+1}(s)$, and $e_a(s) \ge e_{a-1}(s)$ for $1 \le a \le k+1$: each a-set arises a times as $I \cup \{i\}$ with $|I| = a-1$, $i \notin I$, so $a\, e_a(s) \ge e_{a-1}(s)(\sum s - (a-1)r^*)$, and $\sum s - (a-1)r^* \ge R - (j+a-1)r^* \ge R - (2k+1)r^* \ge k+1 \ge a$. Chain $a = k+2-j, \dots, k+1$. □

This is the dual (linear programming) side of lower-bound sieves: minorants of degree k of $\mathbf{1}[b=0]$ with positive mean exist only if no such ν exists (the duality itself is standard, cf. [44, Thm. 5]). It is close in spirit to work on k-wise independent laws with extremal probabilities [4, 39]. The construction is elementary; in the identical-marginal case it improves the upper bound of [4, Thm. 27] (see the discussion of the literature in the introduction). In the Poisson-like model with R of order S, Bonferroni inequalities of order $\asymp S$ do give positive minorants, so the order $k \asymp R$ is right.

Lemma 10.2 (Level barrier; proved)**.** *Let independent coordinates be split into small ones X_s and big ones $(X_b)_{b \in \mathfrak{B}}$. Let $\mathcal{F}$ be a family of events, each of the form $E = \{X_s \in \Sigma_E\} \cap \{X_{b(E)} \in \Gamma_E\}$ with exactly one big coordinate $b(E)$, and let F be the indicator that no event of a family $\mathcal{E} \supseteq \mathcal{F}$ occurs. For a value x_s put $\Omega_b(x_s) = \bigcup_{E:\, b(E)=b,\, x_s \in \Sigma_E} \Gamma_E$, $p_b(x_s) = \mathbb{P}(X_b \in \Omega_b(x_s))$, $r_b = p_b/(1-p_b)$ and $R(x_s) = \sum_b r_b(x_s)$. Suppose $p_b(x_s) \le p^* < 1$ for all b, x_s, put $r^* = p^*/(1-p^*)$, and suppose $R(x_s) \ge (k+1) + (2k+1)r^*$ for* every *x_s. Let B be a linear combination of functions each depending on X_s and on at most k big coordinates. If $B \le F$ pointwise, then $\mathbb{E}B \le 0$.*

Proof. Define a law ν: draw x_s from its true law; draw bits $(b_b)_b$ from the law of Lemma 10.1 for the independent bits $\mathbf{1}[X_b \in \Omega_b(x_s)]$ (bits with $p_b = 0$ have $w_J = 0$ and are never set); given the bits, draw each X_b independently from its true law conditioned on $X_b \in \Omega_b(x_s)$ (bit 1) or $X_b \notin \Omega_b(x_s)$ (bit 0). For a function ϕ of x_s and of the big coordinates in a set K with $|K| \le k$: given x_s, the bits on K are independent with the true marginals, so $(X_b)_{b \in K}$ has its true product law, and $\mathbb{E}_\nu \phi = \mathbb{E}\phi$. Hence $\mathbb{E}B = \mathbb{E}_\nu B \le \mathbb{E}_\nu F$. Under ν some bit is 1, so some event of $\mathcal{F}$ occurs and $F = 0$. □

The ES instance. Fix a small absolute $\varepsilon \in (0, 1/20]$ (chosen in Lemma 10.4), and put $X = T^\varepsilon$, $V = X^{1/3}$, $L = \log X = \varepsilon \mathcal{L}$ and $y = \mathcal{L}^6$. With D^* as in §4, let $\mathcal{F}^*$ be the set of events $E_{v\ell,D}$ with

$$n_D := D^* \in [V, X], \quad v \le V \text{ squarefree and } y\text{-rough},$$
$$\ell \in (T^{0.6}, T^{0.7}] \text{ prime}, \quad v\ell \equiv -1 \pmod{4n_D}.$$

The congruence gives $v\ell \equiv 3 \pmod 4$ and $n_D \mid A_{v\ell}$, i.e. $D \mid A_{v\ell}^2$; and $D \le n_D^2 \le T^{2\varepsilon}$. On a fibre $n \equiv r \pmod Q$ the *big* coordinates are the $X_\ell = n \bmod \ell$ for primes $\ell > T^{0.6}$ with $\ell \nmid Q$. Everything else is small: the residues of n modulo the primes $\le T^{0.6}$ and modulo 2, the coordinates fixed by the fibre (constants), and, for big ℓ, the higher ℓ-adic digits of n beyond $n \bmod \ell$. On the units these higher digits are uniform and independent of

X_ℓ, so all coordinates are independent, as Lemma 10.2 requires. A modulus $q \le D$ has at most $\log D/(0.6\mathcal{L})$ *distinct* prime factors $> T^{0.6}$, and a function of n mod qQ depends on X_ℓ only for these ℓ (whatever their exponent in q). So a function of level at most D is a combination of functions of the small coordinates and of at most $k = \lfloor \log D/(0.6\mathcal{L}) \rfloor$ big ones.

Lemma 10.3 (Unique small part; proved)**.** *For each pair (ℓ, D) there is at most one v with $E_{v\ell,D} \in \mathcal{F}^*$. Every event of $\mathcal{F}^*$ has exactly one big coordinate (if $\ell \nmid Q$), $p_\ell(x) \le p^* := T^{-1/2}$ for all ℓ and x, and, for every small configuration x,*

$$\sum_{\ell > T^{0.6}} p_\ell(x) = \sum_{v} \sum_{\substack{D:\ n_D \in [V,X] \\ 4D \equiv -x \ (v)}} c(v, n_D), \qquad c(v,n) = \sum_{\substack{\ell \in (T^{0.6}, T^{0.7}] \\ \ell \equiv -v^{-1} \ (4n)}} \frac{1}{\ell - 1},$$

with v over the y-rough squarefree $v \le V$, $c(v,n) = 0$ if $\gcd(v, 2n) > 1$, and $p_\ell(x)$ computed with X_ℓ uniform on the units modulo ℓ.

Proof. $v \equiv -\ell^{-1} \pmod{4n_D}$ and $1 \le v \le V \le n_D < 4n_D$, so v is determined by (ℓ, n_D). As $v < T^{0.6} < \ell$, ℓ is the only prime factor of $v\ell$ above $T^{0.6}$. Given x, the events at ℓ whose small part holds are the $E_{v\ell,D}$ with $x \equiv -4D \pmod v$, and distinct D give distinct classes $-4D$ mod ℓ, as $0 < |D - D'| < T^{2\varepsilon} < \ell$. So $p_\ell(x) = (\ell-1)^{-1} \#\{D :\ v(\ell, D) \text{ exists},\ x \equiv -4D\ (v)\}$; this is at most $X(1+L)/(\ell-1) \le T^{-1/2}$ by (F3). Sum over ℓ and group by v. □

Lemma 10.4 (Class-uniform primes; proved modulo Theorems 5.1 and 5.2)**.** *There is an absolute $\varepsilon_0 > 0$ such that for $\varepsilon \le \varepsilon_0$ and T large the following holds. Apply Theorem 5.2 with $Q_G = 4X$, let q_1 be the conductor of the exceptional character if there is one, and put $\mathcal{N}_{\mathrm{exc}} = \{n :\ q_1 \mid 4n\}$ if q_1 exists and $q_1 > \mathcal{L}^{1.9}$, and $\mathcal{N}_{\mathrm{exc}} = \emptyset$ otherwise. Then $c(v,n) \ge 0.13/\varphi(4n)$ for every $n \in [V, X] \setminus \mathcal{N}_{\mathrm{exc}}$ and every $v \le V$ with $\gcd(v, 2n) = 1$.*

Proof. Put $q = 4n \le Q_G$ and $b \equiv -v^{-1} \pmod q$. Each χ mod q is induced by a primitive χ^*, distinct χ give distinct χ^*, and $|\vartheta(x;\chi) - \vartheta(x;\chi^*)| \le \log q$. By orthogonality and Theorem 5.2, for $x \in [T^{0.6}, T^{0.7}]$,

$$\Big|\vartheta(x;q,b) - \frac{x}{\varphi(q)} + [q_1 \mid q]\,\chi_1(b)\frac{x^{\beta_1}}{\beta_1 \varphi(q)}\Big| \le \frac{\mathrm{Err}_G(x)}{\varphi(q)} + \log q,$$

where $\mathrm{Err}_G(x)$ is the right-hand side of Theorem 5.2 (times C_G), and the bracket is present only in the exceptional case. The range condition $Q_G^{6c} \le x$ holds for $\varepsilon \le 0.05/c$, and $\log x/\log Q_G \ge 0.6/(\varepsilon + o(1))$. Without an exceptional zero, $\mathrm{Err}_G(x) \le C_G x(e^{-0.6/(\kappa(\varepsilon+o(1)))} + o(1))$. With one, $(1-\beta_1)\log x \le \log x/(\kappa \log Q_G) \le 0.7/(\kappa(\varepsilon + o(1)))$, so $\mathrm{Err}_G(x) \le C_G x(0.7/(\kappa\varepsilon))(e^{-0.6/(\varepsilon+o(1))} + o(1))$. Both are at most $x/16$ for ε small, and $\log q = o(x/\varphi(q))$. If $q_1 \mid q$ and $q_1 \le \mathcal{L}^{1.9}$, Theorem 5.1 gives $1 - \beta_1 \gg \mathcal{L}^{-0.95}(\log \mathcal{L})^{-2}$, so $x^{\beta_1 - 1} = o(1)$; if $q_1 \mid q$ and $q_1 > \mathcal{L}^{1.9}$, then $n \in \mathcal{N}_{\mathrm{exc}}$. So $\vartheta(t;q,b) \in [\frac{7}{8} - o(1), \frac{9}{8} + o(1)]\, t/\varphi(q)$ on $[T^{0.6}, T^{0.7}]$, and partial summation

against $1/(t\log t)$ gives $c(v,n)\ge\sum 1/\ell\ge\frac{7}{8}\log\frac{7}{6}/\varphi(q)-O(1/(\mathcal{L}\varphi(q)))\ge 0.13/\varphi(q)$. □

If $q_1>\mathcal{L}^{1.9}$, put $q''=q_1/\gcd(q_1,4)\ge q_1/4$; then $n\in\mathcal{N}_{\mathrm{exc}}$ if and only if $q''\mid n$, and by the divisor bound $\tau(q'')/q''\le q''^{-0.95}\le 4\mathcal{L}^{-1.8}$ for T large.

Lemma 10.5 (Class-uniform D; proved)**.** *For T large, every odd squarefree y-rough $v\le V$ and every unit a modulo v,*

$$W(v,a):=\sum_{\substack{D:\ n_D\in[V,X]\setminus\mathcal{N}_{\mathrm{exc}}\\ D\equiv a\ (v)}}\frac{1}{\varphi(4n_D)}\ \ge\ \frac{L^2}{200\,v}.$$

Proof. Write $D=\kappa t^2$ with κ squarefree, so $n_D=\kappa t$. Keep only $t\le X^{1/6}$ with $(t,v)=1$ and $\kappa\in(K_1,K_2]$, $K_1=X^{2/3}$, $K_2=X/t$; then $n_D\in[V,X]$, and $D\equiv a\pmod v$ means $\kappa\equiv b:=at^{-2}\pmod v$. As $\varphi(4n)\le 2n$, $W(v,a)\ge H_{\mathrm{all}}-H_{\mathrm{exc}}$, where H_{all} and H_{exc} are the sums of $1/(2\kappa t)$ over the kept pairs, with all n and with $n\in\mathcal{N}_{\mathrm{exc}}$ respectively. For a unit b modulo v, the number of squarefree $\kappa\le K$ with $\kappa\equiv b\pmod v$ is $\sigma_v K/v+O(\sqrt{K})$ with $\sigma_v=\sum_{(d,v)=1}\mu(d)/d^2\ge 6/\pi^2$ (write $\mu^2(\kappa)=\sum_{d^2\mid\kappa}\mu(d)$). Partial summation gives $\sum_{\kappa\in(K_1,K_2],\,\kappa\equiv b}1/\kappa\ge(\sigma_v/v)\log(X^{1/3}/t)-O(K_1^{-1/2})$. For y-rough v, $\sum_{t\le X^{1/6},(t,v)=1}t^{-1}\log(X^{1/3}/t)\ge L^2/24-O(\omega(v)L^2/y)-O(L)\ge L^2/48$. So $H_{\mathrm{all}}\ge\frac12\cdot\frac{6}{\pi^2}\cdot\frac{L^2}{48v}-O(LX^{-1/3})$. If $\mathcal{N}_{\mathrm{exc}}\ne\emptyset$: $q''\mid\kappa t$ forces $q''/(q'',t)\mid\kappa$, which with $\kappa\equiv b\pmod v$ is one class modulo $vq''/(q'',t)$ or empty (empty if a prime of v divides $q''/(q'',t)$, since b is a unit modulo v; otherwise the Chinese remainder theorem gives one class), so by (F4) the $1/\kappa$-sum is at most $1/K_1+(q'',t)L/(vq'')$. Since $\sum_{t\le X}(q'',t)/t\le\tau(q'')(1+L)$, we get $H_{\mathrm{exc}}\le L^2\tau(q'')/(vq'')+O(LX^{-2/3})\le 4L^2\mathcal{L}^{-1.8}/v$. As $X^{-1/3}\le 1/v$, $W(v,a)\ge(L^2/v)(6/(96\pi^2)-o(1))\ge L^2/(200v)$. □

Theorem 10.6 (No positive low-level minorant; proved modulo Theorems 5.1, 5.2 and the fundamental lemma)**.** *There are absolute constants $c,\varepsilon>0$ such that for T large, every modulus Q with $\log Q\le T^{0.05}$ and every unit r modulo Q the following holds. If B is a function on the fibre $n\equiv r\pmod Q$ of level at most D, with $B\le F$ pointwise and $\log D\le c\,\mathcal{L}^4/\log\mathcal{L}$, then $\mathbb{E}B\le 0$.*

Proof. Use $\mathcal{F}^*$ without the events whose ℓ divides Q; each of its events has exactly one big coordinate, and $p^*\le T^{-1/2}$ (Lemma 10.3). Fix a small configuration x; its entries are units. By Lemma 10.3, $R(x)\ge\sum_\ell p_\ell(x)\ge\sum_v\sum_{D\equiv -x/4\ (v),\ n_D\notin\mathcal{N}_{\mathrm{exc}}}c(v,n_D)-\omega_{\mathrm{big}}(Q)p^*$, where $\omega_{\mathrm{big}}(Q)\le\log Q/(0.6\mathcal{L})\le T^{0.05}$ counts the prime factors of Q above $T^{0.6}$. For $D\equiv -x/4\pmod v$, D is a unit modulo v, so $\gcd(v,2n_D)=1$, and Lemmas 10.4 and 10.5 give

$$R(x)\ \ge\ 0.13\cdot\frac{L^2}{200}\sum_{\substack{v\le V\\ y\text{-rough, squarefree}}}\frac{1}{v}-T^{0.05}T^{-1/2}\ \ge\ \mu^*:=c_9\,\varepsilon^3\frac{\mathcal{L}^3}{\log\mathcal{L}},$$

where the v-sum is $\geq c_3 \log V/\log y$ by the fundamental lemma (as in Lemma 4.5; squarefreeness costs $O(\mathcal{L}/y)$). With $k = \lfloor \log D/(0.6\mathcal{L})\rfloor$ we have $(k+1)+(2k+1)r^* \leq (k+1)(1+4p^*) \leq \mu^*$ if $c \leq 0.3c_9\varepsilon^3$. Lemma 10.2, applied with $\mathcal{E}$ the set of all events with $M \leq T$, gives $\mathbb{E}B \leq 0$. □

The mechanism: conditioned on the coordinates below $T^{0.6}$, what remains is a sieve on the big primes with removed classes of total density $\gg \mathcal{L}^3/\log\mathcal{L}$ (the bulk of the ES mass comes from moduli with one large prime factor), so a sieve of dimension $\kappa \asymp \mathcal{L}^3/\log\mathcal{L}$ with all moduli above $T^{0.6}$. Lower-bound sieves of dimension κ need level about z^κ, here $e^{\asymp \mathcal{L}^4/\log\mathcal{L}}$. Lemma 10.1 is the elementary form of this, and the uniqueness family makes $R(x)$ large for *every* small configuration, not only on average.

Corollary 10.7 (The 1/4 ceiling; proved implication)**.** *Consider any argument that produces a hard prime p with $W(p) > T$ from a function $B \leq F$ on a fibre $n \equiv r \pmod Q$, of level at most Z and with $\mathbb{E}B > 0$, through a transfer theorem whose hypothesis requires $\log x \geq C \log Z$ for the search bound x (with $Z \geq Q$). Every such argument has $\log x \gg \mathcal{L}^4/\log\mathcal{L}$. Theorem 6.1, and hence the proof of Theorem 1.1, is of this type, whatever the minorant (sandwich, Selberg, Bonferroni, dense or sparse, any value of A). So such arguments cannot give $W(p) \geq \exp((\log p)^{1/4+\delta})$ for any $\delta > 0$, and Theorem 1.1 is optimal among them up to a factor $(\log\log p)^{1/2}$ in $\log W(p)$. Modulo* [46]*, Proposition 11.3 improves $\mathcal{L}^4/\log\mathcal{L}$ to $\mathcal{L}^4$ and the factor to $(\log\log p)^{1/4}$.*

Proof. If $\log Q > T^{0.05}$, then $\log x \geq C \log Z \geq C \log Q > e^{0.05\mathcal{L}}$. Otherwise Theorem 10.6 forces $\log Z > c\mathcal{L}^4/\log\mathcal{L}$. In both cases $\log x \gg \mathcal{L}^4/\log\mathcal{L}$, so the certified statement "some hard $p \leq x$ has $W(p) > T$" holds at best with $\mathcal{L} \ll (\log x \log\log x)^{1/4}$. □

Scope. Corollary 10.7 says nothing about the true size of the least such prime; nothing about prime input other than low-conductor minorants (bilinear or Type II information, parity-sensitive input, or mixtures of majorants and minorants); nothing about transfers whose positivity comes from an exceptional-character term rather than from $\mathbb{E}B > 0$ (such transfers can only work at Siegel scales); and nothing about functions that are not of bounded level in the above sense. Section 13 extends the obstruction to minorants of unbounded level, to Siegel-model main terms and to equidistribution hypotheses, for linear certificates with full-orbit uniform accounting; Section 14 gives a hypothesis outside that class which reaches 1/3.

Corollary 10.8 (The energy tail cannot be improved; proved modulo Theorems 5.1, 5.2, 3.1 and the fundamental lemma)**.** *On the fibre of Theorem 3.3, there is no bound $\mathbb{E}(F^{(j)} - u_j)^2 \leq 2^{-\tau/\rho}$ for all j and all τ, with u_j as in (10) and $\rho \leq c'\mathcal{L}/(\log\mathcal{L})^2$ (c' small absolute). So the weights of Theorem 7.9 are optimal for this purpose up to a factor $(\log\mathcal{L})^2$.*

Proof. Such a bound gives $\mathbb{E}[F - B] \le m^2 S 2^{-\tau/\rho} \le \delta/100$ for some $\tau \ll \rho(S_1 + \mathcal{L}) \ll \rho\mathcal{L}^3 \log \mathcal{L}$, so $\mathbb{E}B \ge 0.99\delta > 0$ (Lemma 8.6), and B has level at most $T^3 e^{2\tau}$ (Lemma 8.4). For c' small this contradicts Theorem 10.6. □

Remark 10.9 (The heuristic exponent 1/3; Assessment)**.** By Theorems 3.3 and 4.4, $\log(1/\delta^*(T)) = \mathcal{L}^{3+o(1)}$. If the least prime in the avoiding set behaved like the least prime in a set of Haar density δ^* (as for a single progression, by Linnik's theorem), one would expect a hard prime with $W(p) > T$ and $\log p \ll \mathcal{L}^{3+o(1)}$, i.e. $\log W(p) \approx (\log p)^{1/3}$ infinitely often. This is a heuristic, not a claim. By Corollary 10.7 it cannot be reached by a level-Z minorant with a transfer costing $\log x \gg \log Z$: one would need prime-detecting input for a sifted set of sieve dimension $\asymp \mathcal{L}^3$ at level well below its sieving limit, that is, information beyond minorants of bounded level, such as bilinear (Type II) estimates or other parity-sensitive input adapted to the ES system. We do not know how to provide it. Section 13 makes this precise for linear certificates, and Section 14 shows that a Linnik-type hypothesis for sifted sets (Theorem 14.1) gives the exponent 1/3 up to logarithms.

11. One-big-prime atoms: the Haar exponent without loss, typical size, and the ceiling at level $\mathcal{L}^4$

This section imports one input from the companion note [46], which proves the exceptional-set bound $E(N) \ll N \exp(-c(\log N)^{3/4})$, obtained earlier, by a related argument, in [45]. That note is *internally proved, not externally refereed*; results below that use it are labelled "modulo [46]". Its analytic inputs are the Bombieri–Vinogradov theorem, the Brun–Titchmarsh inequality and Shiu's theorem, and its constants are not asserted to be effective.

The atoms of [46]. Fix $\varkappa = 1/480$ (any fixed $\varkappa \in (0, 1/240)$ would do). For X large put $t = \log X$, $K_X = \lfloor X^{\varkappa} \rfloor$, $\mathcal{K} = \{k \le K_X : k \equiv 1 \ (4)\}$ and $L_K = \operatorname{lcm} \mathcal{K}$. The family $\mathcal{A}_X$ of [46, (4)] consists of quadruples (k, ℓ, u, v) with $k \in \mathcal{K}$, ℓ a prime in a dyadic block $I_j \subseteq (X^{1/2}, X]$, $H < u, v \le z_j = x_j^{1/6}$ (with $H = K_X^{10}$ and x_j the left end of I_j), $(u, v) = (uv, k) = 1$, a bound on $\omega(uv)$, and $4uv \mid k\ell + 1$; the atom is the class $E_A = \{n \equiv -uv^{-1} \pmod{k\ell}\}$, and $H_X(n) = \sum_A \mathbf{1}_{E_A}(n)$. With $M = k\ell \equiv 3 \pmod 4$ and $w = (M+1)/(4uv)$ we have $uvw = A_M$, so E_A is one of the multiplier congruences in (2): if $n \in E_A$ then $W(n) \le k\ell \le K_X X$. We use three facts from [46]:

(A1) [46, Lemma 2.2]: at a fixed ℓ the atoms have pairwise distinct residues modulo ℓ, and there are at most $z_j^2 \le \ell^{1/3}$ of them; the residue $-uv^{-1} \bmod \ell$ is a unit, since $uv < \ell$. An atom is *active* on the class $c = n \bmod L_K$ iff $k \mid u + cv$.

(A2) [46, Cor. 4.3 and Lemma 3.2]: with $f_c(\ell)$ the number of atoms at ℓ active on c and $\mu_c = \sum_\ell f_c(\ell)/\ell$, there are constants $a, C_u > 0$ with $a\,t^3 \le \mu_c \le C_u t^3$ for every unit c modulo L_K and all large X.

(A3) [46, Lemma 8.1 and Thm. 8.2]: with $y = Bt^3$, r the least even integer $\geq D_B t^3$ (B, D_B as there; D_B may be enlarged), $S_y(n) = \mathbf{1}[(n, P_y) = 1]$ and $Q_r(h) = \sum_{j \leq r} (-1)^j \binom{h}{j}$, the function $\nu_X = S_y\, Q_r(H_X)$ is ≥ 0 on $\mathbb{Z}$, $Q_r(0) = 1$, $Q_r(h) \leq \mathbf{1}[h = 0] + \binom{h}{r+1}$, and ν_X is a combination $\sum_i c_i \mathbf{1}[n \equiv a_i \ (q_i)]$ with $\log \max q_i \leq C' t^4$ and $\sum_i |c_i| \leq e^{C' t^4}$; moreover $\sum_{n \leq N} \nu_X(n) \ll N e^{-c_a t^3}$ whenever $\log N \geq C_0 t^4$.

Throughout, $X = T^{1/(1+\varkappa)}$, so that $K_X X \leq T$ and $t \asymp \mathcal{L}$; and $\mathbb{P}_*$, $\mathbb{E}_*$ denote Haar measure on $\hat{\mathbb{Z}}^\times$. Under $\mathbb{P}_*$ the coordinate $c = n \bmod L_K$ is a unit, and the residues $n \bmod \ell$, $\ell \in (X^{1/2}, X]$ (all coprime to L_K), are independent of c and of each other, each uniform on $(\mathbb{Z}/\ell)^\times$. By (A1), given c, $H_X = \sum_\ell I_\ell$ with independent Bernoulli variables I_ℓ of mean $f_c(\ell)/(\ell - 1)$.

Proposition 11.1 (Haar lower bound without loss; proved modulo [46])**.** *For large T, $\mathbb{P}_*(H_X = 0) \leq e^{-at^3}$ and $\delta^*(T) \leq e^{-at^3}$. Hence $\log(1/\delta^*(T)) \gg \mathcal{L}^3$, and with Theorem 3.3, $\mathcal{L}^3 \ll \log(1/\delta^*(T)) \ll \mathcal{L}^3 (\log \mathcal{L})^5$.*

Proof. Given c, $\mathbb{P}_*(H_X = 0 \mid c) = \prod_\ell (1 - f_c(\ell)/(\ell - 1)) \leq e^{-\mu_c} \leq e^{-at^3}$ by (A2). This holds for every unit c, hence also after conditioning on $n \equiv 1 \pmod{24}$, which restricts only $c \bmod 3$ (note $9 \in \mathcal{K}$) and the independent coordinate $n \bmod 8$. If $W(n) > T$ then n lies in no atom, since atoms are multiplier congruences of modulus $\leq K_X X \leq T$; so $\delta^*(T) \leq \mathbb{P}_*(H_X = 0 \mid n \equiv 1 \ (24)) \leq e^{-at^3}$. □

This removes the factor $1/\log \mathcal{L}$ of Theorem 4.4, at the price of the inputs of [46] and ineffective constants; Theorem 4.4 uses only the fundamental lemma. The reason is structural: each atom of [46] has one large prime and a small coordinate c that is revealed, so the events are conditionally independent and an exact product replaces the Janson-type inequality.

Theorem 11.2 (Typical size; proved modulo [46] and Page's theorem)**.** *There are absolute $c, c_1 > 0$ such that, uniformly for $x \geq 3$ and $3 \leq T \leq \exp(c_1 (\log x)^{1/4})$,*

$$\#\{p \leq x \text{ prime} : \ W(p) > T\} \ \ll \ \pi(x)\, e^{-c(\log T)^3}.$$

Here "Page's theorem" is the classical prime number theorem in progressions, uniform for $q \leq \exp(c_2 \sqrt{\log x})$, with the possible exceptional term [8, Ch. 20]: for $(a, q) = 1$,

$$\psi(x; q, a) = \frac{x}{\varphi(q)} - \mathbf{1}[q_1 \mid q]\, \frac{\chi_1(a) x^{\beta_1}}{\varphi(q) \beta_1} + O\big(x e^{-c_3 \sqrt{\log x}}\big), \tag{12}$$

where χ_1 (primitive modulo q_1) is the possible exceptional real character for this range. We recalled this form without re-checking it against the source; any version with this range and error suffices. The Siegel–Walfisz theorem would not: the moduli below have size $\exp(C(\log \log x)^{4/3})$.

Proof. We may assume $T \geq T_0$. For a prime $p > y$ with $W(p) > T$ we have $S_y(p) = 1$ and $H_X(p) = 0$, so $\nu_X(p) = 1$; hence the count is at most $y + \sum_{p \leq x} \nu_X(p)$, and $y = Bt^3$ is negligible. Choose c_1 so that $\log x \geq C_0 t^4$.

Case A: $c_a t^3 \geq 2 \log\log x$. As $\nu_X \geq 0$, (A3) gives $\sum_{p \leq x} \nu_X(p) \leq \sum_{n \leq x} \nu_X(n) \ll x e^{-c_a t^3} \leq x e^{-c_a t^3/2}/\log x$.

Case B: $c_a t^3 < 2 \log\log x$. Then $t^4 \ll (\log\log x)^{4/3}$, so every modulus q_i and the sum $\sum_i |c_i|$ are at most $\exp(O((\log\log x)^{4/3}))$, inside the range of (12) for large x. Since $0 \leq \nu_X \leq \sum |c_i|$,

$$\sum_{p \leq x} \nu_X(p) \leq \sqrt{x} \sum_i |c_i| + \frac{2}{\log x} \sum_{n \leq x} \nu_X(n) \Lambda(n),$$

$$\sum_{n \leq x} \nu_X(n) \Lambda(n) = \sum_i c_i \psi(x; q_i, a_i).$$

Terms with $(a_i, q_i) > 1$ count only powers of the primes dividing (a_i, q_i) and contribute $O(\sum |c_i| \log q_i (\log x)^2)$. For the others insert (12). Put $\mathbb{E}_* \nu_X = \sum_{(a_i,q_i)=1} c_i/\varphi(q_i)$ and $\tilde{\chi}_1(n) = \chi_1(n \bmod q_1)$ on $\hat{\mathbb{Z}}^\times$. Since $\int_{n \equiv a \ (q)} \tilde{\chi}_1 \, d\mathbb{P}_*$ is $\chi_1(a)/\varphi(q)$ if $q_1 \mid q$ and 0 otherwise (χ_1 is primitive), the main terms add up to $x\, \mathbb{E}_*[\nu_X(1 - \epsilon \tilde{\chi}_1)]$ with $\epsilon = x^{\beta_1 - 1}/\beta_1 \in [0, 2]$. As $\nu_X \geq 0$ on $\hat{\mathbb{Z}}^\times$, this is at most $3x \mathbb{E}_* \nu_X$. So $\sum_{p \leq x} \nu_X(p) \leq (6x/\log x) \mathbb{E}_* \nu_X + O(x e^{-c_3 \sqrt{\log x}/2})$. On units $S_y = 1$. Given c, $\mathbb{E}_*[(H_X)_m \mid c] = m!\, e_m((f_c(\ell)/(\ell - 1))_\ell) \leq (2\mu_c)^m \leq (2C_u t^3)^m$ for the falling factorials, so by (A3) and Proposition 11.1,

$$\mathbb{E}_* \nu_X \leq \mathbb{P}_*(H_X = 0) + \frac{\mathbb{E}_*(H_X)_{r+1}}{(r+1)!} \leq e^{-at^3} + \Big(\frac{2eC_u t^3}{r+1}\Big)^{r+1} \leq 2e^{-at^3}$$

once $D_B \geq \max(2e^2 C_u, a)$. The error term is $\ll \pi(x) e^{-at^3}$ because $e^{at^3} \leq (\log x)^{2a/c_a}$. In both cases the count is $\ll \pi(x) e^{-\min(a, c_a/2) t^3}$, and $t \asymp \mathcal{L}$. □

The normalisation by $\pi(x)$ rather than x matters when $\mathcal{L}^3 \lesssim \log\log x$. At $T = \exp(c_1 (\log x)^{1/4})$ the theorem gives the prime case of the exceptional-set bound of [46]; it is a reproof, not an improvement. It is the opposite inequality to Theorem 1.1: an upper bound for the number of primes with $W(p) > T$ needs a *majorant*, and for majorants the exceptional character costs only the factor $1 + \epsilon \leq 3$; Theorem 1.1 needs a minorant. Heuristically (Assessment) the count is $\pi(x) e^{-\mathcal{L}^{3+o(1)}}$ up to $\mathcal{L} \approx (\log x)^{1/3}$. A matching lower bound $\pi(x) e^{-C\mathcal{L}^3 (\log \mathcal{L})^3}$ for $\mathcal{L} \leq c' (\log x/\log\log x)^{1/4}$ is proved in §12 (Theorem 12.1), from the main term of Theorem 6.1; an exceptional character whose conductor divides the quarantine modulus is paid for by a factor $e^{-O(\mathcal{L}^3 (\log \mathcal{L})^3)}$ there.

The same atoms sharpen the ceiling of Theorem 10.6.

Proposition 11.3 (Ceiling at level $c\mathcal{L}^4$; proved modulo [46])**.** *There are absolute $c, T_0 > 0$ such that for $T \geq T_0$, every modulus Q with $\log Q \leq T^{0.05}$ and every unit r modulo Q: if B is a function on the fibre $n \equiv r \pmod Q$ of level at most D, with $B \leq F$ pointwise and $\log D \leq c\mathcal{L}^4$, then $\mathbb{E}B \leq 0$.*

Proof. Apply Lemma 10.2 with big coordinates $X_\ell = n \bmod \ell$ for the primes $\ell \in (X^{1/2}, X]$ with $\ell \nmid Q$, and all other coordinates small (including $c = n \bmod L_K$, the residues at all other primes, and the higher ℓ-adic digits at big ℓ, as in §10). As there, a function of level $\le D$ reads at most $k = \lfloor 2\log D/t \rfloor$ big coordinates, since each big prime exceeds $X^{1/2}$. Let $\mathcal{F}$ be the set of atoms with $\ell \nmid Q$; each reads c (small, as $k \mid L_K$) and one big X_ℓ, and $\mathcal{F} \subseteq \mathcal{E}$, the set of all multiplier congruences with $M \le T$. Fix a small configuration; its c is a unit. By (A1), $p_\ell = f_c(\ell)/(\ell-1) \le \ell^{1/3}/(\ell-1) \le 2X^{-1/3} =: p^*$. At most $2\log Q/t \le T^{0.05}$ big primes divide Q, so by (A2)

$$R \;\ge\; \sum_{\ell \mid Q} p_\ell \;\ge\; \mu_c - T^{0.05}p^* \;\ge\; at^3/2$$

for $T \ge T_0$, for *every* small configuration. Since $r^* \le 4X^{-1/3}$, $(k+1)+(2k+1)r^* \le 2(k+1) \le at^3/2$ once $\log D \le c\mathcal{L}^4$ with c small. Lemma 10.2 gives $\mathbb{E}B \le 0$. □

Compared with Theorem 10.6, the factor $1/\log\mathcal{L}$ disappears, and the inputs (Gallagher's theorem, Landau–Page, the fundamental lemma) are replaced by those of [46] (with an ineffective T_0). In Corollary 10.7 one may therefore replace $\mathcal{L}^4/\log\mathcal{L}$ by $\mathcal{L}^4$, and Theorem 1.1 is optimal among the arguments described there up to a factor $(\log\log p)^{1/4}$ in $\log W(p)$.

Remark 11.4 (One sieve limit behind two ceilings; proved parts and Assessment)**.** The atoms of [46] form a system with one big prime per event, big cost $\asymp \mathcal{L}$ and conditional mass $P \asymp \mathcal{L}^3$ uniformly in the small coordinates. For such a system, order-k certificates are trivial below $k \asymp P$ on *both* sides (proved, no novelty claimed): a minorant of $\mathbf{1}$[no event] depending on at most $k \le 0.6P - 1$ big coordinates has conditional mean ≤ 0 (Lemma 10.1); a majorant of order k saves at most $k\log(C(P+4k)/k) + O(k+\log k)$ in $\log(1/\mathbb{E})$, if $p^* \le 1/4$ (the weighted k-ary comparison and mean cost of [43, Thm. 8.5 and Cor. 8.6], via the thinned law and binomial extrapolation of [43, Lemmas 8.3–8.4], applied fibrewise; their core is [39, 4]); and the Bonferroni truncations $Q_k(H)$ of order $k \ge e^2P$ achieve both (as minorant for odd k, as majorant for even k). So the critical level is $\asymp \mathcal{L}\cdot\mathcal{L}^3 = \mathcal{L}^4$. A budget $\lambda \asymp \log N$ (rounding to integers) then exploits $\mathcal{L} \asymp \lambda^{1/4}$ and saves $\asymp \lambda^{3/4}$: this is the exceptional-set exponent 3/4 of [46], which is sharp for coefficient-sum congruence majorants [43, Thm. 10.14] (for CRT majorants that are ≥ 0 on $\mathbb{Z}$, with coefficient sum $< N$ and all family primes $\le N^A$; for Case-A classes modulo Elsholtz–Tao [11, §7]). A budget $\lambda \asymp \log x$ (transfer to primes $\le x$) exploits $\mathcal{L} \lesssim \lambda^{1/4}$: the exponent 1/4 of Theorem 1.1, sharp in the sense of Proposition 11.3. Within the stated scopes of these two ceilings this is a conjunction of proved statements; as a description of "the" mechanism it is an Assessment. The full avoidance indicator is not a one-big-prime system: on the minorant side the subfamily is used only for the obstruction (as $F \le \mathbf{1}$[no event of $\mathcal{F}$]), while the positive minorant of Theorem 9.1 is built differently.

12. The distribution of W over primes

Theorem 11.2 bounds the number of primes $p \le x$ with $W(p) > T$ from above. Here we prove a matching lower bound in almost the same range, so that the Haar exponent 3 of Theorem 1.4 is also the tail exponent of W over primes. Write

$$N(x,T) = \#\{p \le x \text{ prime} : \ W(p) > T\},$$
$$N_h(x,T) = \#\{p \le x \text{ hard prime} : \ W(p) > T\}.$$

Theorem 12.1 (Lower tail; proved modulo Theorems 5.2 and 3.1)**.** *There are absolute constants C, T_0 such that for $T \ge T_0$ and every x with $\log x \ge C\mathcal{L}^4 \log \mathcal{L}$,*

$$N(x,T) \ \ge \ N_h(x,T) \ \ge \ \pi(x) \exp\big(-C\mathcal{L}^3 (\log \mathcal{L})^3\big).$$

The threshold $\log x \ge C\mathcal{L}^4 \log \mathcal{L}$ is the threshold of Theorem 9.1 for the existence of one such prime: counting costs nothing in range. The constants C, T_0 are effective, given that the implied constants of Theorem 3.1 are.

Corollary 12.2 (Tail exponent 3 over primes; proved modulo Theorems 5.2, 3.1, [46] and Page's theorem)**.** *There are absolute constants $c, C, c' > 0$ and T_0 such that for $T \ge T_0$ and $\mathcal{L} = \log T \le c'(\log x / \log\log x)^{1/4}$,*

$$c\,\mathcal{L}^3 \ \le \ \log \frac{\pi(x)}{N(x,T)} \ \le \ C\,\mathcal{L}^3 (\log \mathcal{L})^3.$$

In particular $\log\log(\pi(x)/N(x,T)) = (3+o(1))\log\log T$ in this range.

Proof. The lower inequality is Theorem 11.2, whose range $\mathcal{L} \le c_1(\log x)^{1/4}$ contains ours if $c' \le \min(1, c_1)$; its implied constant is absorbed for $T \ge T_0$. The upper inequality is Theorem 12.1: if $c' \le \min(1, (4/C)^{1/4})$, then $\log \mathcal{L} \le \frac{1}{4}\log\log x$ and $C\mathcal{L}^4 \log \mathcal{L} \le \frac{C}{4} c'^4 \log x \le \log x$. □

The constant c comes from [46] and is not effective; C is. The two ranges differ by the factor $(\log\log x)^{1/4}$, which is the factor $\log \mathcal{L}$ in the junta bound $\log Z \ll \mathcal{L}^4 \log \mathcal{L}$; by Proposition 11.3 no minorant on a fibre can work below $\log x \asymp \mathcal{L}^4$ (Assessment: this is the natural range limit of the method). For comparison, [11, Remark 1.2] suggest heuristically that a prime p has an Erdős–Straus representation with probability $1 - O(\exp(-c\log^3 p))$; Theorem 1.4 and Corollary 12.2 are rigorous statements of this exponent 3 for the witness modulus W, on the Haar side and over primes; the latter only in the range $\log T \le (\log x)^{1/4 - o(1)}$, far from the scale $\log T \asymp \log p$ of the heuristic.

Counting primes on a fibre. The proof of Theorem 6.1 gives more than one prime.

Lemma 12.3 (Quantitative transfer; proved modulo Theorem 5.2)**.** *Assume the hypotheses of Theorem 6.1. Let x satisfy $\log x \ge C_2(1 + \log A)\log Z$, and let β_1, χ_1, q_1 be the possible exceptional zero of Theorem 5.2 for the*

parameter $Q_G = x^{1/(\kappa L)}$ of that proof, its character and the conductor of χ_1. Then

$$\#\{p \le x : \ p \equiv r \ (Q), \ W(p) > T\} \ \ge \ \frac{\lambda\, \mu\, x}{3\varphi(Q)\log x},$$

where $\lambda = 1$ unless β_1 exists and $q_1 \mid Q$, and in that case

$$\lambda = \min\,(1,\, 8c_2^{-1}q_1^{-1/2}(\log q_1)^{-2}),$$

with c_2 from Theorem 5.1.

Proof. Nothing in the proof of Theorem 6.1 uses more about x than $\log x \ge C_2(1+\log A)\log Z$, and it bounds $S(x)$ from below in each case. In Case 0, and in the exceptional case when no χ with $c(\chi) \ne 0$ has $\chi^* = \chi_1$, it gives $S(x) \ge (1 - 1/400 - 1/200)\mu x/\varphi(Q)$. In Case B it gives $S(x) \ge \mu x/(3\varphi(Q))$. In Case A (where χ is trivial on H, so $q_1 \mid Q$) it gives $S(x) \ge 0.98\lambda'\mu x/\varphi(Q)$ with $\lambda' \ge \min(u,1)/2$, $u = (1-\beta_1)\log x \ge 16(1-\beta_1)$; by Theorem 5.1, $u \ge 16c_2^{-1}q_1^{-1/2}(\log q_1)^{-2}$, so $0.98\lambda' \ge \lambda/3$. (If $q_1 \nmid Q$, the character χ with $\chi^* = \chi_1$ is not trivial on H, so we are in Case B or χ does not occur.) In all cases $S(x) \ge \lambda\mu x/(3\varphi(Q))$. Every term of $S(x) = \sum_{p\le x,\, p\nmid N_0,\, p\equiv r\,(Q)} B(p)\log p$ is at most $\log x$, and a term with $B(p) > 0$ has $W(p) > T$, because $B(p) \le \mathbf{1}[W(p) > T]$ for these p (they are coprime to all d_i). □

Applied to the quarantine of Theorem 3.3, with the sandwich of Theorem 9.1 but *without* the auxiliary prime ℓ_{aux} (whose only role was to force $p > T$), Lemma 12.3 already gives $N_h(x,T) \ge \pi(x)\exp(-C\mathcal{L}^3(\log\mathcal{L})^5)$ for $\log x \ge C\mathcal{L}^4\log\mathcal{L}$ (proved modulo Theorems 5.2 and 3.1): the losses are $\log\varphi(Q)$, $\log(1/\mu) \le \frac{4}{3}S_\beta + 0.02$ and $\log(1/\lambda) \le \frac{1}{2}\log Q + O(\log\log Q)$, all $\ll \mathcal{L}^3(\log\mathcal{L})^5$ by (4). So a single fibre has relative density $\exp(-O(\mathcal{L}^3(\log\mathcal{L})^5))$: the expensive part $\mathcal{L}S_\beta$ of the construction (the junta) enters only the threshold for x. To lower the power of $\log\mathcal{L}$ we sum over all outcomes of the square-class process.

The leaves of the square-class process. Run the square-class process of §2 with the parameters of Theorem 3.3 ($Y = \mathcal{L}^{C_0+4}$). The rule "step at the least heavy ℓ" is deterministic, so the process is a finite decision tree; its leaves are the final quarantines $L = (Q_L, r_L)$. Let k_L be the number of odd primes dividing Q_L (the number of steps with $a = 0$, the forced ones included), and $\mathbb{P}_{\mathrm{proc}}$ the law of the process.

Lemma 12.4 (Leaves; proved)**.** *The fibres $\{n \equiv r_L \pmod{Q_L}\}$ of distinct leaves are disjoint, and $\mathbb{P}_{\mathrm{proc}}(L) = 4\cdot 2^{k_L}/\varphi(Q_L)$.*

Proof. Each step reveals one more digit of n, by a rule depending only on the digits already revealed. Two distinct leaves first differ at a node where they take different values of the same digit, so their fibres are disjoint. The start $Q = 8$, $r = 1$ has Haar mass $1/\varphi(8) = 1/4$ among the units. A step at $(\ell, 0)$ picks one of the $(\ell-1)/2$ nonzero squares modulo ℓ, with

probability $2/(\ell-1)$, and the chosen subfibre has conditional Haar mass $1/(\ell-1)$; a step at (ℓ,a), $a\ge 1$, picks one of ℓ lifts, with probability $1/\ell$ equal to the conditional Haar mass. (The forced step at 3 picks the unique square 1, with probability $1 = 2/(3-1)$.) Multiplying along the path, $\mathbb{P}_{\mathrm{proc}}(L) = 2^{k_L}\cdot\varphi(Q_L)^{-1}/(1/4)$. □

Lemma 12.5 (Number and size of the level-0 steps; proved modulo Theorem 3.1)**.** *Let Q_{end} be the final modulus and k the number of odd primes dividing it. Then*

$$\mathbb{E}[k] \ll \mathcal{L}^3(\log\mathcal{L})^2\log\log\mathcal{L}, \qquad \mathbb{E}[\log\mathrm{rad}(Q_{\mathrm{end}})] \ll \mathcal{L}^3(\log\mathcal{L})^3.$$

Proof. As in Lemma 2.9(b), but only for the steps at level $a=0$ and with the weight 1, respectively $\log\ell$, in place of $\log\ell$ for every level: optional stopping for $G^{(\ell,0)}$ gives

$$\mathbb{E}[k] \le 3 + \frac{1}{\eta}\sum_E P_H(E)2^{\omega_Y(M)}\beta^{\omega(M)}\omega_Y(M),$$

$$\mathbb{E}[\log\mathrm{rad}(Q_{\mathrm{end}})] \le \log 210 + \frac{1}{\eta}\sum_E P_H(E)2^{\omega_Y(M)}\beta^{\omega(M)}\log\mathrm{rad}(M_Y),$$

as $\sum_{\ell\le Y}\mathbf{1}[\ell\mid M] = \omega_Y(M)$ and $\sum_{\ell\mid M,\,\ell\le Y}\log\ell = \log\mathrm{rad}(M_Y)$; the forced steps contribute 3 and $\log 105$, and the prime 2 contributes $\log 2$.

For the first sum put $t = 1+1/\log\log Y$. Since $y\le t^y/(e\log t)$ for $y\ge 0$ and $1/\log t\le t/(t-1)$, we have $\omega_Y(M)\le(1+\log\log Y)\,t^{\omega_Y(M)}$. Repeat the proof of Lemma 3.2(A) with $f_2t^{\omega_Y}$ in place of f_2. Its values at prime powers are $\le 6t\le 7$ for large T, so the class condition of Theorem 3.1 holds uniformly in T, and its Euler factor at $p\le Y$ grows by the factor $1+O((t-1)/p)$; the product of these factors is $\exp(O((t-1)\log\log Y)) = O(1)$. So the first sum is $\ll\log\log Y\cdot\mathcal{L}^3\log Y$, and $\mathbb{E}[k]\ll\eta^{-1}\mathcal{L}^3\log Y\log\log Y$.

For the second sum put $s = 1/\log Y$. Since $y\le e^{sy}/(es)$ for $y\ge 0$, $\log\mathrm{rad}(M_Y)\le(\log Y)\,\mathrm{rad}(M_Y)^s$. The function $\mathrm{rad}(m_Y)^s = \prod_{\ell\mid m,\,\ell\le Y}\ell^s$ is multiplicative with values in $[1,e]$ at prime powers, so $f_2\,\mathrm{rad}(\cdot_Y)^s$ has values $\le 6e<17$ and the class condition again holds uniformly in T; the Euler factors grow by $1+O((\ell^s-1)/\ell)$, and $\ell^s-1\le(e-1)s\log\ell$, so their product is $\exp(O(s\sum_{\ell\le Y}\log\ell/\ell)) = O(1)$. So the second sum is $\ll\mathcal{L}^3(\log Y)^2$. As $1/\eta\asymp\log\mathcal{L}$ and $\log Y\asymp\log\mathcal{L}$, both bounds follow. □

Proof of Theorem 12.1. Call a leaf L *good* if (i) $\tilde w_\ell\le\eta$ for every $\ell>Y$; (ii) $\log Q_L\le 8\mathbb{E}[\log Q_{\mathrm{end}}]$; (iii) $S_\beta(L)\le 8\mathbb{E}[S_\beta(\mathrm{end})]$; (iv) $k_L\le 8\mathbb{E}[k]$; (v) $\log\mathrm{rad}(Q_L)\le 8\mathbb{E}[\log\mathrm{rad}(Q_{\mathrm{end}})]$. As in the proof of Theorem 3.3, (i) fails with probability $\le 1/4$, and by Markov's inequality each of (ii)–(v) fails with probability $\le 1/8$. So $\mathbb{P}_{\mathrm{proc}}(\mathrm{good})\ge 1/4$.

A good leaf satisfies (4) with C replaced by $2C$ (by Lemma 2.9 and Lemma 3.2, as in Theorem 3.3). Build B_L on its fibre exactly as in the proof of Theorem 9.1, with $S_1 = 2C\mathcal{L}^3\log\mathcal{L}$, and apply Lemma 12.3 to (Q_L,r_L) itself (no auxiliary prime). The hypotheses of Theorem 6.1 were checked in that proof for (Q',r'); the same checks apply to (Q_L,r_L), as no

cell, event or twist character involves ℓ_{aux}. So $\mu_L \ge 0.99\, e^{-4S_\beta(L)/3}$, $A \le 1.03$ and $\log Z_L \le \log Q_L + 2(3\mathcal{L} + 2\tau) \le C_3\mathcal{L}^4 \log \mathcal{L}$ with one absolute C_3 for all good leaves. Take $C \ge C_2(1 + \log 1.03)C_3$; then Lemma 12.3 applies to every good leaf at the same x, and $\lambda_L < 1$ only if the exceptional character of that application has conductor $q_1 \mid Q_L$. A real primitive conductor is $2^e f'$ with $e \le 3$ and f' odd and squarefree, so then $q_1 \le 8\,\mathrm{rad}(Q_L)$, and $\log(1/\lambda_L) \le \frac{1}{2}\log \mathrm{rad}(Q_L) + O(\log\log Q_L + 1)$.

The fibres of distinct leaves are disjoint (Lemma 12.4), and the primes found on them are hard by Lemma 2.5(iii), as $840 \mid Q_L$. So

$$N_h(x,T) \;\ge\; \sum_{L \text{ good}} \frac{\lambda_L \mu_L x}{3\varphi(Q_L)\log x} = \frac{x}{12\log x} \sum_{L \text{ good}} \mathbb{P}_{\mathrm{proc}}(L)\, 2^{-k_L} \lambda_L \mu_L \;\ge\; \frac{x}{48\log x} \min_{L \text{ good}} 2^{-k_L}\lambda_L\mu_L.$$

On good leaves, $k_L \log 2 \ll \mathcal{L}^3(\log\mathcal{L})^2 \log\log\mathcal{L}$ and $\log(1/\lambda_L) \ll \mathcal{L}^3(\log\mathcal{L})^3$ by Lemma 12.5, and $\log(1/\mu_L) \le \frac{4}{3}S_\beta(L) + 0.02 \ll \mathcal{L}^3 \log\mathcal{L}$. As $x/\log x \ge \pi(x)/1.26$, the theorem follows. □

Remark 12.6 (Where the logarithms go; Assessment, and a conditional improvement)**.** The three losses are the square-class restriction 2^{-k_L} (the Haar price of never firing an event), the residual mass S_β (the Haar truth $\mathcal{L}^3$ up to $\log\mathcal{L}$), and the possible exceptional zero. Only the last is specific to primes: majorants absorb the exceptional term as a factor $1 + \epsilon \le 3$ (Theorem 11.2), minorants pay $1 - \epsilon$. If every real zero of every $L(s,\chi)$, χ real primitive of conductor $q \ge 3$, satisfies $1 - \beta \ge c_0/\log q$ (as on GRH for real characters), then in Case A $u \ge c_0 \log x/\log Q_L \ge 1$, so $\lambda_L \ge 1/2$ and the bound of Theorem 12.1 improves to $\pi(x)\exp(-C\mathcal{L}^3(\log\mathcal{L})^2\log\log\mathcal{L})$ (conditional on this hypothesis). Siegel's ineffective bound $1 - \beta \ge c(\varepsilon)q^{-\varepsilon}$ still costs $\varepsilon \log \mathrm{rad}(Q_L)$ and does not remove the third logarithm.

13. Beyond bounded level: a Wiener-norm barrier for linear certificates

Theorem 10.6 concerns minorants of bounded level. Here we show that the planted law behind it is exponentially pseudorandom against residue classes and characters of *every* modulus, and deduce that a large class of linear certificates, whatever the prime input, cannot beat the exponent $1/4$. The results are from the working note [35] (internally reviewed); we give the two short key proofs and state the rest with pointers.

Setting. Fix Q with $\log Q \le T^{0.05}$, a unit r, the fibre $H = \{n \equiv r\ (Q)\}$ with Haar probability $\mathbb{P} = \mathbb{P}_H$, and $F = \mathbf{1}[W > T]$. In this section the big coordinate X_ℓ ($\ell > T^{0.6}$ prime, $\ell \nmid Q$) is the whole ℓ-adic component of n (modulo a power of ℓ large enough for all functions considered); the events of $\mathcal{F}^*$ read it only modulo ℓ, so the quantities $p_\ell(x)$ and the bound $R(x) \ge \mu^*$ of the proof of Theorem 10.6 are unchanged, with $p^* \le T^{-1/2}$. Put $k + 1 = \lfloor \mu^*/2 \rfloor$; then $R(x) \ge (k+1) + (2k+1)r^*$ for every small

configuration x. Let ν be the planted law of the proof of Lemma 10.2 for $\mathcal{F}^*$ and this k, and $\rho = \nu - \mathbb{P}$. Then $F = 0$ ν-almost surely.

A *reduced product* is a function $h = h_s(x_s)\prod_{b\in I} h_b(X_b)$, I a finite set of big coordinates, with $|h_s|, |h_b| \le 1$ and $|\mathbb{E}h_b| \le 1/4$ for $b \in I$. Restricted to H, the indicator of a residue class, a Dirichlet character and an additive character $n \mapsto e(an/q)$ of any modulus are reduced products (up to a unimodular constant): the local factor at a big $b \mid q$ is a class indicator of mean $\le 1/(b-1)$, a non-trivial character of mean 0, or a non-trivial additive character of mean $c_{b^v}(a')/\varphi(b^v)$, of modulus $\le 1/(b-1)$; trivial factors are dropped from I.

Lemma 13.1 (Pseudorandomness of the planted law; proved)**.** *For every reduced product h, $|\mathbb{E}_\rho h| \le (4r^*)^{k+1}$, and $\mathbb{E}_\rho h = 0$ if $|I| \le k$.*

Proof. Fix x_s and use the notation of Lemma 10.1 for the bits $\mathbf{1}[X_b \in \Omega_b(x_s)]$. Given the bits the big coordinates are independent; put $\beta_b = \mathbb{E}[h_b \mid X_b \in \Omega_b]$ and $\gamma_b = \mathbb{E}[h_b \mid X_b \notin \Omega_b]$ ($\beta_b = \gamma_b = 1$ for $b \notin I$). The conditional perturbation is $P_0\sum_{|J|=k+1} w_J\sigma_J$, and

$$\mathbb{E}_{\sigma_J}\Big[\prod_{b\in I} h_b\Big] = \sum_{y\subseteq J}(-1)^{|y|+1}\prod_{b\in y}\beta_b\prod_{b\in J\setminus y}\gamma_b\prod_{b\in I\setminus J}\gamma_b$$
$$= -\prod_{b\in J}(\gamma_b-\beta_b)\prod_{b\in I\setminus J}\gamma_b.$$

For $b \in J \setminus I$ the factor $\gamma_b - \beta_b$ vanishes, so only $J \subseteq I$ contribute; none if $|I| \le k$. Now $|\gamma_b - \beta_b| \le 2$, and $\mathbb{E}h_b = (1-p_b)\gamma_b + p_b\beta_b$ gives $|\gamma_b| \le (1/4 + p_b)/(1-p_b) < 1/2$. Next $w_J \le r^{*\,k+1}/e_{k+1}(r)$, and $e_{k+1}(r) \ge 1$: the chain in the proof of Lemma 10.1 with $s = r$ gives $a\,e_a(r) \ge (k+1)e_{a-1}(r)$ for $a \le k+1$. With $m = |I|$, $\sum_{J\subseteq I,|J|=k+1}\prod_{b\in I\setminus J}|\gamma_b| \le \binom{m}{k+1}2^{-(m-k-1)} \le 2^{k+1}$. As $P_0 \le 1$ and $|h_s| \le 1$, the conditional bound is $r^{*\,k+1}2^{k+1}2^{k+1}$; average over x_s. □

Theorem 13.2 (Wiener-norm barrier; proved modulo Theorems 5.1, 5.2 and the fundamental lemma)**.** *There are absolute $c, T_0 > 0$ such that for $T \ge T_0$, every Q with $\log Q \le T^{0.05}$ and every unit r: if $B \le F$ on H and $B = \sum_i c_i h_i$ is any finite combination of reduced products, then*

$$\mathbb{E}_H B \;\le\; \eta\sum_i |c_i|, \qquad \eta = \exp\big(-c\mathcal{L}^4/\log\mathcal{L}\big).$$

Only the h_i involving at least $k+1 \asymp \mathcal{L}^3/\log\mathcal{L}$ big primes, hence of modulus $> T^{0.6(k+1)}$, contribute.

Proof. $\mathbb{E}_H B = \mathbb{E}_\nu B - \mathbb{E}_\rho B \le \mathbb{E}_\nu F - \mathbb{E}_\rho B = -\mathbb{E}_\rho B \le \sum_i |c_i|(4r^*)^{k+1}$ by Lemma 13.1, and $(4r^*)^{k+1} \le (8T^{-1/2})^{\mu^*/2-1} \le \exp(-c\mathcal{L}\mu^*)$. □

In particular $\|F\|_\times \ge \mathbb{E}_H F/\eta$ for the ℓ^1 norm $\|\cdot\|_\times$ over such representations: the avoidance indicator costs $e^{c\mathcal{L}^4/\log\mathcal{L}}$ times its mean. Theorem 10.6 is the case $|I| \le k$. (With the atoms of §11 in place of $\mathcal{F}^*$, the same argument gives $\eta = e^{-c\mathcal{L}^4}$ modulo [46]; we do not use this.)

Linear certificates [35, Def. 2.1]. Let $\mathfrak{h}$ be a family of reduced products on H with moduli in a finite set, N_x the number of primes $p \le x$ in H coprime to these moduli, and m_x their counting measure. A nonnegative measure m on H of mass N_x is $\mathfrak{h}$-*consistent at accuracy* $\mathfrak{E}$ if $|\int h\,dm - N_x \mathbb{E}_H h| \le \mathfrak{E}$ for all $h \in \mathfrak{h}$. A *linear certificate* concludes "$\int F\,dm_x > 0$" from the facts that m_x is nonnegative of mass N_x and $\mathfrak{h}$-consistent at accuracy $\mathfrak{E}$, and from nothing else (in particular not from its atomicity, integrality or support); it is valid iff every consistent m has $\int F\,dm > 0$. Every minorant transfer is one: from $B = \sum c_i h_i \le F$ it concludes $\sum_p F(p) \ge N_x \mathbb{E}_H B - \mathfrak{E} \sum |c_i|$. By Lemma 13.1 the diffuse measure $N_x \nu$ is consistent at every accuracy $\mathfrak{E} \ge N_x (4r^*)^{k+1}$, for every family $\mathfrak{h}$, and has $\int F\,d(N_x\nu) = 0$; so valid linear certificates need $\mathfrak{E} < N_x \eta$ [35, Thm. 2.2]. On the other hand, for a modulus $q > x$ no bound that is uniform over a translation orbit can be small: if $\mathfrak{h}$ contains all classes in the fibre modulo q, then $\mathfrak{E} \ge 1/2$; if it contains all Dirichlet characters (or all additive characters) modulo q, then $\mathfrak{E}^2 \ge N_x/2$ ([35, Lemma 2.3], by orthogonality, as soon as N_x is at most half the number of units of the fibre modulo q).

Corollary 13.3 (Full-orbit uniform linear certificates are capped at 1/4; proved implication in [35, Cor. 2.4], same inputs; sketch here)**.** *Let a linear certificate use a family $\mathfrak{h}$ of residue classes, Dirichlet characters or additive characters which contains, with each member of modulus $> x$, its whole translation orbit, with one error bound per orbit. If it certifies a prime $p \le x$ in H with $W(p) > T$, then $\log x \gg \mathcal{L}^4/\log \mathcal{L}$. Hence no such certificate, whatever true prime input it uses (Bombieri–Vinogradov, Elliott–Halberstam, GRH for the characters it uses, Siegel-zero main terms as below), gives $W(p) \ge \exp((\log p)^{1/4+\delta})$.*

The proof [35, Cor. 2.4] splits B into members with at most k big primes, on which $\mathbb{E}_\rho = 0$, and deep members, whose moduli exceed x^2 if $\log x \le c\mathcal{L}^4/\log \mathcal{L}$; each deep orbit costs at least 1/2 (classes) or $\sqrt{N_x/2}$ (characters), while by Theorem 13.2 it contributes at most η per unit of coefficient to the mean, so positivity needs $N_x \eta > 1/2$. One-sided accounting (dropping deep classes with positive coefficients) and Hölder-averaged accounting over an orbit do not help [35, Cor. 2.4 and Prop. 2.5]. *Siegel zeros* (proved in [35, Thm. 3.1], same inputs; statement only here): for any real character χ_1 and $\epsilon \in [0,1]$, the reweighted fake $(1 - \epsilon\chi_1)\nu$ is nonnegative, has $F = 0$ almost surely, and agrees with the Siegel-model law $(1 - \epsilon\chi_1)\mathbb{P}$ to within $2(4r^*)^{k+1}$ on every class, Dirichlet character and additive character. Let s be the number of big primes dividing the conductor of χ_1. The total masses differ by $\epsilon|\mathbb{E}_\rho \chi_1| \le (4r^*)^{k+1}$, and agree exactly when $s \le k$, the relevant case; otherwise the certificate framework must let the total mass vary within the accuracy. If B has level $\le T^{0.6(k-s)}$, then $\mathbb{E}_{(1-\epsilon\chi_1)\mathbb{P}} B \le 0$ exactly. So an exceptional term in the prime law does not rescue a minorant transfer of this kind.

Proposition 13.4 (Equidistribution hypotheses cap at 1/4; proved implication, same inputs)**.** *Let N be the number of primes $p \le x$ in H, and for a modulus q let $N_{x,q}$ be the number of those coprime to q. Let $\mathcal{I}$ be any set of statements $|\sum_{p\le x,\, p\in H,\, p\nmid q} h(p) - N_{x,q}\mathbb{E}_H h| \le \mathfrak{E}_h$ for residue classes, Dirichlet characters or additive characters h of moduli $q \le x$, individually with $\mathfrak{E}_h \ge 2\log x$, or summed over a set of moduli with an allowance at least $2\log x$ times the number of terms (for example Bombieri–Vinogradov, the Elliott–Halberstam conjecture at any level $\theta < 1$, GRH restricted to moduli $\le x$), in the form of their consequences for the unweighted counts at the given x (information about several scales or the weight $\log p$ uses the position of the primes in $[1, x]$ and is outside the framework; see the Scope paragraph below). A linear certificate (with N as the total mass) that uses $\mathcal{I}$ and produces a prime $p \le x$ in H with $W(p) > T$ needs $\log x \ge c\mathcal{L}^4/\log\mathcal{L}$.*

Proof. Let $\log x < 0.6\mathcal{L}(k+1)$. A modulus $q \le x$ has at most k big prime factors, so every h occurring in $\mathcal{I}$ involves at most k big coordinates on H, and $\mathbb{E}_\nu h = \mathbb{E}_H h$ (Lemma 13.1). The measure $m = N\nu$ on H is nonnegative of mass N and gives h the value $N\mathbb{E}_H h$, which differs from the centring $N_{x,q}\mathbb{E}_H h$ by at most $N - N_{x,q} \le \omega(q) \le 2\log x$, as $|h| \le 1$. Summing gives the averaged statements. Thus m satisfies $\mathcal{I}$, while $\int F\,dm = 0$. □

For Bombieri–Vinogradov or Elliott–Halberstam at level $\theta < 1$ the extra allowance $2x^\theta \log x$ is far below the usual $x(\log x)^{-A}$. Raising the level of distribution only changes c: beyond level x the primes are not equidistributed in the orbit-uniform sense above. Full GRH is *not* covered by Proposition 13.4: it also asserts $|\sum_{p\le x}\chi(p)| \ll \sqrt{x}\log^2(qx)$ for moduli $q > x$, and whether the planted measure satisfies these bounds is open; Corollary 13.3 covers such characters only when they are used orbit-uniformly.

Scope [35, §6]. Not covered: certificates that use the atomicity, integrality or support of the prime measure (for instance that classes modulo $q > x$ whose least positive member exceeds x are empty, i.e. minorants valid only on $[1, x]$ or only on primes; single moduli $\le x^{1/5}/(QT)$ are excluded in [35, Prop. 5.2]); non-periodic families inside the avoiding set; non-linear arguments (pair correlations, density increments); and whether Type II information can help. On the integer side, the Type I sums produced by Vaughan- or Heath-Brown-type identities are blocked only under full-orbit uniform accounting [35, Prop. 4.1], which is not the natural accounting for integers. Our reading (Assessment) is that 1/4 is a sieve-*dimension* barrier (dimension $\kappa \asymp \mathcal{L}^3/\log\mathcal{L}$, sifting limit $\beta_\kappa \asymp \kappa$, $\log z \asymp \mathcal{L}$), not a parity barrier.

14. The exponent 1/3 under a Linnik-type hypothesis for sifted sets

By §13, the heuristic exponent 1/3 (Remark 10.9) is out of reach of linear certificates with standard prime input. Here we identify a natural hypothesis that gives it. The material is from [36] (internally reviewed).

A *unit-class sieve system* with moduli $\le T$ is a finite family $\mathcal{E}$ of events $\{n \equiv a_E \pmod{m_E}\}$ with $2 \le m_E \le T$ and $(a_E, m_E) = 1$; no condition is imposed on their number, overlaps or dimension. Its *avoiding set* is $S(\mathcal{E}) = \{n \in \hat{\mathbb{Z}}^\times : \ n \not\equiv a_E \ (m_E) \ \forall E\}$, and $\delta(\mathcal{E})$ is its Haar measure (the proportion of units modulo $\operatorname{lcm}(m_E)$ in no event).

Hypothesis LS(C) (Linnik's theorem for sifted sets; a conjecture). For every $T \ge 3$ and every unit-class sieve system $\mathcal{E}$ with moduli $\le T$ and $\delta(\mathcal{E}) > 0$ there is a prime $p > T$ avoiding $\mathcal{E}$ with

$$\log p \ \le \ C(\log T + \log(1/\delta(\mathcal{E}))).$$

For a single residue class (all other classes modulo q removed, $T = q$, $\delta = 1/\varphi(q)$) this is Linnik's theorem. LS is the log-scale form, not the Cramér form, of the model in which a sifted set behaves like a residue class of the same density, for the purpose of containing one prime. The term $C \log T$ cannot be dropped: shifted Jacobsthal configurations (one class modulo each prime $\le z$) have $\delta \asymp 1/\log z$ and least avoiding integers of size $z^{1+o(1)}$. The restriction to moduli $\le T$ matters: for an arbitrary set of density δ modulo Q no bound $p \ll \delta^{-1}(\log Q)^{O(1)}$ holds (take the units in $(Q(1-\delta), Q)$).

Theorem 14.1 (LS implies exponent 1/3; proved modulo Theorem 3.1, conditional on LS(C))**.** *Assume LS(C). For every large T there is a hard prime $p > T$ with $W(p) > T$ and $\log p \ll_C \mathcal{L}^3(\log\mathcal{L})^5$. Consequently, for infinitely many hard primes,*

$$W(p) \ \ge \ \exp\left(c_C(\log p)^{1/3}(\log\log p)^{-5/3}\right).$$

LS(C) is used only for one system $\mathcal{E}_T$ for each T.

Proof. Let $\mathcal{E}_T$ consist of the events $n \equiv -4D \pmod{M}$ for all atoms (M, D) with $M \le T$ (unit classes, as $\gcd(4D, M) = 1$), and the events $n \equiv c \pmod{840}$ for the unit classes c that are not squares modulo 840. Its moduli are $\le T$. Let (Q, r) be the quarantine of Theorem 3.3. By Lemma 2.5(i) the units $n \equiv r \pmod{Q}$ with $F(n) = 1$ have $W(n) > T$, so by Lemma 2.1 they avoid every atom; as $840 \mid Q$ and r is a square modulo 840 (Lemma 2.5(iii)), they lie in $S(\mathcal{E}_T)$. The class r mod Q has Haar measure $1/\varphi(Q)$ and within it $\mathbb{E}F = \delta \ge e^{-4S_\beta/3}$, so by (4)

$$\log\frac{1}{\delta(\mathcal{E}_T)} \ \le \ \log\varphi(Q) + \tfrac{4}{3}S_\beta \ \ll \ \mathcal{L}^3(\log\mathcal{L})^5.$$

LS(C) gives a prime $p > T$ avoiding $\mathcal{E}_T$ with $\log p \ll_C \mathcal{L}^3(\log\mathcal{L})^5$. It is a square modulo 840, hence hard, and avoids every atom with $M \le T$, so $W(p) > T$ by Lemma 2.1. As $p > T$, letting $T \to \infty$ gives infinitely many primes, and $\mathcal{L} \gg (\log p)^{1/3}(\log\log p)^{-5/3}$. □

More generally, LS with $C(\cdot)$ replaced by an increasing function $\Phi(\log T + \log(1/\delta))$ gives $\log p \le \Phi(C_0\mathcal{L}^3(\log\mathcal{L})^5)$; for instance $\Phi(\lambda) = \lambda^\theta$ gives the exponent $1/(3\theta)$. LS is used far below the sieve limit: the system has dimension $\kappa \asymp \mathcal{L}^3/\log\mathcal{L}$ on the big primes, LS asks for $\log x \asymp \log(1/\delta) \asymp$

$\kappa \log \mathcal{L}$ (up to logarithms), while a lower-bound sieve needs $\log x \gg \kappa \log T$. This ratio $\log T$ versus $\log\log T$ per unit of dimension is the gap between 1/4 and 1/3. By Proposition 13.4 and Corollary 13.3, LS cannot be derived through linear certificates from equidistribution hypotheses; it is a beyond-the-barrier hypothesis. The planted measures of §13 do not refute it: they are reweightings of Haar measure onto the complement of the avoiding set, not sequences of primes.

Other strengths [36, §§2,5]. (i) A Cramér form ($\log p \le \log(1/\delta) + A\log T$) implies $\mathrm{LS}(\max(1, A))$; a log-scale counting form $\#\{T < p \le x : p \in S(\mathcal{E})\} \ge c_0\delta^C \pi(x)$ for $\log x \ge C(\log T + \log(1/\delta))$ implies $\mathrm{LS}(C')$ for some $C' = C'(C, c_0)$. (ii) Main-term forms are too strong: in the integer analogue (events $n \equiv 0 \pmod{\ell}$, $\ell \le z$, at $x = z^u$) the count is $(e^\gamma \omega(u) + o(1))\delta x$ with Buchstab's function ω, not $\sim \delta x$ (proved, classical); for primes a uniform κ-tuple heuristic predicts a compounding deficit $(e^\gamma\omega(u))^\kappa$ (Assessment). (iii) LS holds, by the Brun–Titchmarsh inequality, for systems of total event mass at most $(1-\varepsilon)/2$ (proved); the ES system is very far from this case.

Hypotheses for product sets do not suffice. A *product system* has only prime-power moduli, so that $S(\mathcal{E}) = \prod_\ell A_\ell$ is a CRT product with $\delta = \prod_\ell |A_\ell|/\varphi(\ell^{e_\ell})$, a singular series. Let $\mathrm{HL}_{\mathrm{prod}}(C)$ be $\mathrm{LS}(C)$ restricted to product systems; it has the shape of a Hardy–Littlewood or Bateman–Horn statement, but is a least-prime hypothesis for growing systems, not a standard fixed-polynomial statement. Let $F_T = \{n \in \hat{\mathbb{Z}}^\times : W(n) > T\}$.

Proposition 14.2 (Product subsets of the avoiding set are thin [36, Prop. 4.1, Cor. 4.2])**.** *(a) (Proved.) There are product sets $S \subseteq F_T$, inside the hard classes, with $\log(1/\delta(S)) \le T^{1/2+o(1)}$. (b) (Proved in [36] modulo the Barban–Davenport–Halberstam theorem [8, Ch. 29]; ineffective; sketch below.) Every product set $S \subseteq F_T$ has $\log(1/\delta(S)) \gg T^{1/2}(\log T)^{-8}$. Consequently $HL_{\mathrm{prod}}(C)$ implies $W(p) \ge (\log p)^{2-o(1)}$ for infinitely many hard p, and no application of HL_{prod} to a product subset of F_T certifies more than $W(p) \ge (\log p)^{2+o(1)}$.*

Proof of (a). Fix $\varepsilon > 0$ and $y = T^{1/2+\varepsilon}$. Take $A_2 = \{1 \bmod 8\}$ and, for odd $\ell \le y$, the quadratic residues (lifted to ℓ^{e_ℓ}). An atom (M, D) all of whose primes are $\le y$ is avoided: $(n/M) = 1$ for $n \in S$ while $(-4D/M) = -1$ (Lemma 2.2). Every other atom has exactly one prime $\ell > y$, to the first power; let B_ℓ be the set of its classes $-4D \bmod \ell$ and $A_\ell = (\mathbb{Z}/\ell)^\times \setminus B_\ell$ for $y < \ell \le T$. Then S avoids all atoms, and $|B_\ell| \le \sum_{v \le T/\ell} \tau(A_{v\ell}^2) \le T^{1/2-\varepsilon+o(1)} \le \ell/2$, so $\log(1/\delta(S)) \le \pi(y)\log 2 + O(1) + \sum_{\ell > y} 2|B_\ell|/(\ell-1) \le T^{1/2+\varepsilon+o(1)}$. □

Part (b) [36, Prop. 4.1(b)] uses only the atoms $(\ell_1\ell_2, q)$ with primes $\ell_1 \equiv 1$, $\ell_2 \equiv 3 \pmod 4$ in $(\sqrt{T}/2, \sqrt{T}]$ and a prime $q \mid (\ell_1\ell_2 + 1)/4$: since S is a product, each such atom must be blocked at ℓ_1 or at ℓ_2, which forces $\gg T^{1/2}(\log T)^{-2}$ blocked classes per prime on average once the primes ℓ are equidistributed modulo $4q$ on average over $q \le \sqrt{T}(\log T)^{-6}$. So the

ES avoiding set, of density $e^{-\mathcal{L}^{3+o(1)}}$, exceeds its best product subset by a factor $e^{T^{1/2+o(1)}}$: hypotheses that see only prime-by-prime local structure are useless here, and the hypothesis must apply to non-product sifted sets, as LS does.

15. Other numerators m/n

The multiplier family exists for every numerator. Fix an integer $m \geq 4$ ($m = 5$ is Sierpiński's equation). For $M \geq 3$ with $M \equiv -1 \pmod m$ put $A = A_M = (M+1)/m$. If $uvw = A$ and $n \equiv -uv^{-1} \pmod M$, then $s = (nv+u)/M$ is a positive integer and

$$\frac{m}{n} = \frac{1}{suw} + \frac{1}{nsvw} + \frac{1}{nuvw} \tag{13}$$

(as $\frac{1}{suw} + \frac{1}{nsvw} = \frac{M}{nuvw}$ and $M + 1 = mA$). As $mA \equiv 1 \pmod M$, the proof of Lemma 2.1 with 4 replaced by m shows $\{-uv^{-1} \bmod M : \ uvw = A\} = \mathcal{R}_m(M) := \{-mD \bmod M : \ D \mid A^2\}$. An *$m$-atom* is a pair (M, D) with $M \equiv -1 \pmod m$, $M \geq 3$, $D \mid A_M^2$; its event is $n \equiv -mD \pmod M$. Let $W_m(n)$ be the least M with $n \bmod M \in \mathcal{R}_m(M)$, so $W_4 = W$, and let $\delta_m^*(T)$ be the Haar measure of $\{n \in \hat{\mathbb{Z}}^\times : \ W_m(n) > T\}$. A unit class $r \bmod Q$ is *Type-II-hard* if $r \bmod M \notin \mathcal{R}_m(M)$ for every $M \mid Q$ with $M \equiv -1 \pmod m$: no identity (13) with $M \mid Q$ covers it. In this section "hard" always refers to the family (13) alone; we make no statement about other identities for m/p, and nothing here bears on the Erdős–Straus or Sierpiński conjectures. Full proofs of the statements below that are only summarised here are in the working notes [29, 30] (internally reviewed, not externally refereed).

The Jacobi symbol of the event classes. For an m-atom (M, D) with M odd let e be the squarefree part of mD, and write $e = 2^t e_o$ with $t \in \{0, 1\}$ and e_o odd.

Lemma 15.1 (Jacobi dichotomy; proved)**.** *Let (M, D) be an m-atom with M odd.*

(a) *Every prime $\ell \mid M$ has $\ell \nmid mD$ and $\left(\frac{-mD}{\ell}\right) = \left(\frac{-e}{\ell}\right)$.*
(b) *If $M \equiv 3 \pmod 4$, then $\left(\frac{-mD}{M}\right) = -\left(\frac{2}{M}\right)^t$.*
(c) *If $M \equiv 1 \pmod 4$, then $\left(\frac{-mD}{M}\right) = \left(\frac{2}{M}\right)^t (-1)^{(e_o-1)/2}$.*
(d) *If $m \equiv 0 \pmod 4$, then $\left(\frac{-mD}{M}\right) = -1$ for every m-atom.*

Proof. (a) $\gcd(M, mA) = 1$ (as $mA = M + 1$) and every prime of D divides A, so $\ell \nmid mD$; and $-mD = -e \cdot k^2$ with $\gcd(k, \ell) = 1$. (b), (c) By (a), $\left(\frac{-mD}{M}\right) = \left(\frac{-1}{M}\right)\left(\frac{2}{M}\right)^t \prod_{q \mid e_o} \left(\frac{q}{M}\right)$. Every prime $q \mid e$ divides $mA = M + 1$, so $M \equiv -1 \pmod q$, and for odd q reciprocity gives $\left(\frac{q}{M}\right) = \left(\frac{M}{q}\right)(-1)^{\frac{q-1}{2}\frac{M-1}{2}} = \left(\frac{-1}{q}\right)(-1)^{\frac{q-1}{2}\frac{M-1}{2}}$. This is 1 if $M \equiv 3 \pmod 4$, and $\left(\frac{-1}{q}\right)$ if $M \equiv 1 \pmod 4$; and $\prod_{q \mid e_o} \left(\frac{-1}{q}\right) = (-1)^{(e_o-1)/2}$. (d) Now $M = mA - 1 \equiv 3 \pmod 4$. If $t = 1$, then $v_2(mD)$ is odd; if $8 \mid m$, then $8 \mid mA$, and if $m \equiv 4 \pmod 8$, then

$v_2(D)$ is odd, so $2 \mid A$ and $8 \mid mA$. Either way $M \equiv 7 \pmod 8$, so $\left(\frac{2}{M}\right) = 1$, and (b) gives -1. □

For $m = 4$ this is Lemma 2.2. We checked (a)–(c) by computer for $M \le 5 \cdot 10^4$ and $m \in \{4, 5, 6, 7, 8, 9, 10, 11, 12, 14, 18\}$ (no failures; not used). For $m \not\equiv 0 \pmod 4$ the symbol takes both values, and the square-class process breaks down at a fixed small prime.

Proposition 15.2 (Square classes fire for $m \not\equiv 0$ (4); proved)**.** *Let $m \not\equiv 0 \pmod 4$. There are infinitely many primes $\ell \equiv -1 \pmod m$ such that the classes of the m-atoms (ℓ, D) meet both cosets of the squares modulo ℓ. For such ℓ, every rule that chooses $r \bmod \ell$ uniformly in a coset of the squares (either coset, by any rule) fires an event with probability at least $2/(\ell - 1)$.*

Proof. It suffices to find a prime $\ell \equiv -1 \pmod m$ with $\ell \equiv 1 \pmod 4$ and a prime $q \mid A_\ell$ with $\left(\frac{q}{\ell}\right) = -1$: then $D = 1$ and $D = q$ give the classes $-m$ and $-mq$, whose Legendre symbols differ. If m is odd, take $q = 2$ and $\ell \equiv -1 \pmod m$, $\ell \equiv 5 \pmod 8$; then $\ell + 1 \equiv 6 \pmod 8$, so $2 \mid A_\ell$, and $\left(\frac{2}{\ell}\right) = -1$. If $m \equiv 2 \pmod 4$, take an odd prime $q \equiv 3 \pmod 4$ with $q \nmid m$ and $\ell \equiv -1 \pmod{mq}$, $\ell \equiv 1 \pmod 4$ (compatible, as $v_2(mq) = 1$); then $q \mid A_\ell$ and, by the reciprocity step above, $\left(\frac{q}{\ell}\right) = \left(\frac{-1}{q}\right) = -1$. Dirichlet's theorem gives infinitely many such ℓ. □

For example $m = 5$, $\ell = 29$ ($A = 6$). The Haar mass of the atoms through a fixed small prime ℓ is $\gg \mathcal{L}^3/\ell$, so such an ℓ is always stepped by the process of §2; that process therefore does not run for $m \not\equiv 0 \pmod 4$. For every m, atoms with $M \equiv 7 \pmod 8$ are never fired by square classes (Lemma 15.1(b)); this condition is sufficient, not necessary (e.g. $m = 5$, $M = 19$, $D = 1$).

Results.

Theorem 15.3 ($m \equiv 0$ (4); proved in [29] modulo Theorems 5.2 and 3.1, and the fundamental lemma for the lower bound in (i))**.** *Let $m \equiv 0 \pmod 4$. Constants depend on m.*

(i) $\mathcal{L}^3/\log\mathcal{L} \ll \log(1/\delta_m^*(T)) \ll \mathcal{L}^3(\log\mathcal{L})^5$.

(ii) *For infinitely many primes p we have*

$$W_m(p) \ge \exp\left(c_m(\log p)^{1/4}(\log\log p)^{-1/4}\right);$$

these p are quadratic residues modulo 840 and Type-II-hard modulo their quarantine modulus Q.

Summary of the proof ([29, Thm. 3.1]). The proofs of §§2–9 use the shape of the atoms only in the following places, each of which holds for $m \equiv 0 \pmod 4$. (1) Lemma 2.2 is replaced by Lemma 15.1(a),(d), with d replaced by e; the drift at a level-0 step in Lemma 2.9(a) is $1 + \left(\frac{-e}{\ell}\right) \le 2$. (2) $M \equiv 3 \pmod 4$ is odd, so the start $Q = 8$, $r = 1$ has $p_0 = P_H$, and the forced steps at $3, 5, 7$ are square steps. (3) In Lemma 3.2(A), Theorem 3.1 is applied

to $Q_1(n) = n$, $Q_2(n) = mn - 1$: coprime linear forms without fixed prime divisor, with $\rho(p) = 2$ for $p \nmid m$ and $\rho(p) = 1$ for $p \mid m$, so the Euler product changes by a factor $O_m(1)$; Lemma 3.2(B) is an elementary divisor sum and is unchanged. (4) Theorem 3.3, the sandwich (§§7–8) and the transfer concern an abstract system of unit-class events and need only (1)–(3); the hypotheses on r in Theorem 6.1 are those produced by the process. (5) The lower bound in (i) is Proposition 15.4 below. □

Proposition 15.4 (Haar lower bound for every m; proved in [29] modulo the fundamental lemma)**.** *For every $m \geq 4$, $\log(1/\delta^*_m(T)) \gg_m \mathcal{L}^3/\log \mathcal{L}$.*

Summary of the proof ([29, Prop. 3.2])*.* The proof of Theorem 4.4 with 4 replaced by m: in the family used there replace the conditions $M \equiv 3 \pmod 4$, $M \equiv -1 \pmod{4n}$ by $M \equiv -1 \pmod{mn}$, so that $n \mid A_M$; then $mD \leq mn^2 < M$ for $M \geq \sqrt{T}$, so the classes $-mD \bmod M$ are distinct, and the fundamental lemma sifts progressions modulo mn instead of $4n$. The Janson-type inequality (Theorem 4.3) is abstract. No Jacobi input is used. □

For $m \not\equiv 0 \pmod 4$ the exponent $1/5$ of the third version of this note extends, because that version quarantines n to the class of one, which is hard for every m, and never uses quadratic residues.

Theorem 15.5 (Every m, exponent $1/5$; proved in [29] modulo Theorem 5.2 and the Elsholtz–Tao estimates [11, Prop. 1.4, Thm. 7.1, Cor. 7.4, (7.10)])**.** *For every fixed $m \geq 4$, $\log(1/\delta^*_m(T)) \ll_m \mathcal{L}^5 \log \mathcal{L}$, and for infinitely many primes p, $W_m(p) \geq \exp\left(c_m(\log p)^{1/5}(\log\log p)^{-1/5}\right)$; these p satisfy $p \equiv 1 \pmod Q$ with $840 \mid Q$ and are Type-II-hard modulo Q.*

The proof [29, Cor. 6.1 and §6b] is a written-out substitution $4 \mapsto m$ in the graded class-of-one argument (Remark 2.6), with the energy bound of §7 and Theorem 6.1. The changes are: the Elsholtz–Tao estimates are stated for general coefficients, so $4 \mapsto m$ is admissible; the quadratic $mdx^2 + 1$ can have 4 roots modulo 2^k when md is odd, which changes one Euler factor; for odd m, M can be even, so the prime 2 becomes a graded coordinate like the others; the class of one is excluded by an archimedean argument valid for every m ($1 \equiv -mD \pmod M$ iff $M \mid mD+1$ iff $M \mid D+A$, as $mA \equiv 1$; the involution $D \mapsto A^2/D$ preserves this, as $\gcd(AD, M) = 1$, so we may take $D \leq A$, and then $0 < D + A \leq 2A < M$ since $(m-2)A > 1$); and the parity assertions of the original proof are replaced by $\ell \nmid mad$. This improves the exponent $1/7$ that the architecture of the second version gives for general m, in particular for Sierpiński's $5/n$, but only for the witness W_5 of the family (13).

The case $m \not\equiv 0$ (4) at exponent $1/4$. Two ingredients survive. First, the transfer does not need r to be a square:

Lemma 15.6 (Transfer for arbitrary classes; proved modulo Theorem 5.2)**.** *Theorem 6.1 holds for every unit $r \bmod Q$, $8 \mid Q$, if the cells are consistent with r; no quadratic condition on r is needed.*

Proof. The value of a real character at r enters only in Case A of the proof, through $c(\chi) = \chi(r)\mu/\varphi(Q)$ for the real χ with $\chi^* = \chi_1$ trivial on $\{n \equiv 1\ (Q)\}$ (and through the corresponding identity in (c) when $f_2 = 1$). If $\chi_1(r) = 1$, the proof is unchanged. If $\chi_1(r) = -1$, the exceptional main term $-c(\chi)x^{\beta_1}/\beta_1$ is positive, the main terms add up to $\lambda'\mu x/\varphi(Q)$ with $\lambda' = 1 + x^{\beta_1 - 1}/\beta_1 \geq 1$, and the errors are bounded as in Case 0. In (c) and Case B, $\psi_1(r)$ is replaced by a sign, and the bound $|c(\chi)| \leq \mu/(4\varphi(Q))$ is unchanged; non-real characters enter only through $|c(\chi)|$. □

Second, the bookkeeping survives if the square-class process is replaced by the *admissible unit process*: at each step reveal the next digit of n uniformly among the classes that do not complete an event (the *non-forbidden* classes), after a fixed prefix in which $r_0 \bmod Q_0$ is uniform on the Type-II-hard classes modulo a constant modulus $Q_0 = Q_0(m)$. The prime 2 is an ordinary coordinate for odd m. We always take $8 \mid Q_0$, as Lemma 15.6 requires (for $m \equiv 2 \pmod 4$ no M is even, so this changes no mass; for odd m the class of one is hard, so the hard set modulo Q_0 is non-empty). If f is the forbidden fraction at a step, put $\Lambda(\ell) = \prod(1-f)^{-1}$ over the steps at ℓ after the prefix ($\Lambda = \infty$ if some $f = 1$). The process may be run adaptively (step at the least heavy $\ell \leq Y$, as in §2), or with a forced first stage that reveals the digits modulo all powers $q \leq Z = \mathcal{L}^3(\log \mathcal{L})^B$ of primes not dividing m, in increasing order of q, before the adaptive stage; the forced steps are ordinary steps for the supermartingales and cost $\log \operatorname{lcm}\{q \leq Z\} \ll \mathcal{L}^3(\log \mathcal{L})^B$.

Hypothesis $\mathrm{ADM}_m(K, Q_0)$: uniformly in large T (with β, η as before and $Y = \mathcal{L}^C$ for a suitable $C = C(K)$), with probability at least $7/8$ the admissible unit process (in either variant) satisfies $\Lambda(\ell) \leq K$ for every $\ell \leq Y$.

For $m \equiv 0 \pmod 4$ the square-restricted version holds with $K = 2$, by Lemma 15.1(d).

Theorem 15.7 (Conditional on ADM_m; the implication is proved in [29] modulo Theorems 5.2 and 3.1 for even m, and for odd m modulo also the parity split described below, not written out in [29])**.** *Assume* $\mathrm{ADM}_m(K, Q_0)$ *for either variant of the process. Then* $\log(1/\delta_m^*(T)) \ll \mathcal{L}^3(\log \mathcal{L})^{K+4}$*, and* $W_m(p) \geq \exp\left(c(\log p)^{1/4}(\log\log p)^{-B_K}\right)$ *for infinitely many primes p, Type-II-hard modulo their quarantine modulus.*

In the proof [29, Thm. 5.1] only Lemma 2.9 changes: the factor $2^{u_i(E)}$ is replaced by $\prod_{\ell \mid M,\, \ell \leq Y} K/\Lambda_i(\ell)$, the process is stopped before a step that would make some $\Lambda(\ell) > K$, and the prefix costs a factor $1/\delta_0$ (the density of the hard classes modulo Q_0) in all initial masses; the transfer uses Lemma 15.6. Success probability $7/8$ can be replaced by any fixed $s_0 > 0$ [30, Lemma 1.1], and the forced stage changes nothing in this argument [30, §1, "Order"]. For odd m the pair of forms n, $mn - 1$ of Lemma 3.2(A) has the fixed prime

divisor 2, so Theorem 3.1 does not apply verbatim; one splits by the parity of n (forms t, $2mt-1$ for $n=2t$, and $2t+1$, $mt+(m-1)/2$ for $n=2t+1$, neither pair having a fixed prime divisor) and absorbs the bounded factors at 2. This step is not written out in [29].

Towards ADM_m. ADM_m is open for every $m \not\equiv 0 \pmod 4$. Numerically the drift is small: for $m \in \{5,6,7,9,10,11,13\}$ and $T \le 3\cdot 10^5$, a simplified full-depth version of the process with a 2- and 3-adic prefix never died and had $\max_\ell \Lambda(\ell) \le 4.3$, the maximum being at $\ell \le 19$ (Evidence, [29, §6]). Rigorously, [30] uses the ordered variant of the process: after a prefix modulo $Q(q_0) = \mathrm{lcm}\{q \le q_0\}$ (over the powers q of primes not dividing m, with 8 adjoined as above), a forced stage reveals the digits r mod q for all such prime powers $q_0 < q \le Z = \mathcal{L}^3(\log\mathcal{L})^B$ in increasing order of q (so that an atom completed at q has its cofactor composed of smaller prime powers), followed by the adaptive threshold stage, and the process is stopped before any step that would make some $\Lambda(\ell) > K$. For this process [30, Thm. 3.1] (proved modulo Henriot's coefficient-uniform form of Theorem 3.1 [17, Thm. 5]) shows that the sums of the masses of the atoms completed at a step are uniform in T, and that after any prefix law ν on the hard classes modulo $Q(q_0)$ with density at most C_ν relative to Haar measure, the process fails with probability at most $C_\nu(c_1 q_0^{-1/2+\varepsilon} + o(1))$, with c_1 not explicit. So ADM_m for the ordered variant reduces to the prefix: a hypothesis SI on ν-weighted completed-atom sums (a conjecture) implies ADM_m, and hence exponent $1/4$ for W_m [30, Prop. 5.1]. For the uniform prefix law, the data for $q_0 \le 19$ suggest that C_ν grows faster than $q_0^{1/2}$, so the plan "certified finite prefix plus moments" does not obviously close (Assessment, [30, §4]); the threshold $q_0^{1/2}$ comes from a Markov step and is method-dependent (a second moment might replace it by $q_0^{1-\varepsilon}$).

16. Summary of status

- Proved: Lemmas 2.1, 2.2, 2.3, 2.5, 2.8 (with the classical Lemma 2.7), 2.9, Lemma 3.2(B); the Janson-type inequality (Lemmas 4.1, 4.2, Theorem 4.3) and Lemmas 4.6, 4.7; the energy bound (Lemmas 7.2–7.4, Theorems 7.5 and 7.9, Corollaries 7.6 and 7.10, Remark 7.7, and Remark 7.8 except the cited switching-lemma constant); Lemmas 8.1, 8.2, 8.3, 8.4, 8.5, 8.6; the planting and level-barrier Lemmas 10.1, 10.2, and Lemmas 10.3, 10.5.
- Proved modulo the Nair–Tenenbaum theorem (Theorem 3.1): part (A) of Lemma 3.2 and the Haar upper bound Theorem 3.3.
- Proved modulo the fundamental lemma of sieve theory: Lemma 4.5 and the Haar lower bound Theorem 4.4. With the previous item: Theorem 1.4 with $(\log T)^3/\log\log T$ in the lower bound (the Haar exponent is 3); the lossless lower bound needs [46] (below).
- Proved modulo Gallagher's theorem (Theorems 5.2 and 5.1, quoted from the draft [27]): the linear transfer Theorem 6.1 and Lemma 10.4.

- Proved modulo Gallagher's theorem and Theorem 3.1: Theorem 9.1 and Theorem 1.1.
- Proved modulo Gallagher's theorem and the fundamental lemma: Theorem 10.6 (= Theorem 1.5). Proved implications: Corollary 10.7 (within its stated scope) and Corollary 10.8 (which also uses Theorem 3.1).
- Proved modulo [46] (internally proved, not externally refereed): Proposition 11.1 (Haar lower bound $\gg \mathcal{L}^3$; with Theorem 3.3 it gives Theorem 1.4 in the stated form) and Proposition 11.3 (ceiling at level $c\mathcal{L}^4$). Proved modulo [46] and Page's theorem (12): Theorem 11.2 (= Theorem 1.2). Remark 11.4: a conjunction of proved statements within their scopes, Assessment as a description of the mechanism.
- Proved: Lemma 13.1. Proved modulo Gallagher's theorem and the fundamental lemma: Theorem 13.2. Proved implications, same inputs: Proposition 13.4; Corollary 13.3 and the results of [35] quoted after it (Siegel model, one-sided and averaged accounting) are proved in [35] and only sketched here; within the scope stated at the end of §13 (linear certificates; full GRH, support-aware, non-periodic and non-linear arguments are not covered).
- Conditional on Hypothesis LS, proved modulo Theorem 3.1: Theorem 14.1 (= Theorem 1.6). Hypothesis LS itself is a conjecture. Proposition 14.2: part (a) is proved; part (b) is proved in [36] modulo the theorem of Barban, Davenport and Halberstam. In the remarks on other strengths of LS after Theorem 14.1, the integer Buchstab statement is classical and the compounding for primes is an Assessment.
- Proved modulo Gallagher's theorem: Lemma 12.3 (quantitative transfer) and Lemma 15.6. Proved: Lemma 12.4. Proved modulo Theorem 3.1: Lemma 12.5. Proved modulo Gallagher's theorem and Theorem 3.1: the lower tail Theorem 12.1 and the one-fibre bound stated after Lemma 12.3. Proved modulo these, [46] and Page's theorem: Corollary 12.2 (= Theorem 1.3). Remark 12.6: the improved bound is conditional on the stated zero-free bound for real characters; the rest is an Assessment.
- Other numerators: Lemma 15.1 and Proposition 15.2 are proved. Proved in [29] (proofs summarised here): Theorem 15.3 modulo Gallagher's theorem and Theorem 3.1 (the lower bound in (i) modulo the fundamental lemma), Proposition 15.4 modulo the fundamental lemma, Theorem 15.5 modulo Gallagher's theorem and the Elsholtz–Tao estimates. Theorem 15.7 is conditional on the open hypothesis ADM_m (the implication is proved in [29] for even m; for odd m it also needs the parity split described after the theorem, which is not written out in [29]); the reduction of ADM_m to the prefix is proved in

[30] modulo Henriot's form of the Nair–Tenenbaum theorem, and the hypothesis SI is a conjecture. The numerics for ADM_m are Evidence.

- Evidence, not used: the computer check of Lemma 2.2 for $M \le 2\cdot 10^5$; the computer check of Lemma 15.1 for $M \le 5\cdot 10^4$ and eleven values of m; the exact check of Theorem 4.3 on small instances; the Monte Carlo fit of $\delta^*(T)$ for $T \le 32767$.
- Assessment: Remarks 6.2, 9.2 and 10.9; the range comparison after Corollary 12.2; the heuristic two-sided typical size after Theorem 11.2; the reading of $1/4$ as a sieve-dimension barrier (§13).
- Not claimed: anything about the Erdős–Straus or Sierpiński conjectures, or about representations of m/p outside the family (13); ADM_m or SI; the exponent $1/3$ unconditionally, or the truth of Hypothesis LS; anything about arguments outside the scopes of Corollaries 10.7 and 13.3 and Proposition 13.4; any numerical instance; priority for Theorem 7.5 or Theorem 4.3 (literature searches partial).
- The Elsholtz–Tao estimates, Håstad's switching lemma and the Thorner–Zaman theorem are not used, except that the Elsholtz–Tao estimates enter Theorem 15.5 (through the class-of-one architecture of the third version); otherwise they are cited only for comparison.

References

[1] N. Alon and J. H. Spencer, The Probabilistic Method, 4th ed., Wiley, 2016.

[2] Anonymous, Large Erdős–Straus witnesses: the multiplier parameter exceeds every power of $\log p$, preprint, 2026.

[3] L. M. J. Bazzi, Polylogarithmic independence can fool DNF formulas, SIAM J. Comput. 38 (2009), 2220–2272 (FOCS 2007).

[4] I. Benjamini, O. Gurel-Gurevich and R. Peled, On K-wise independent distributions and Boolean functions, arXiv:1201.3261 (2012).

[5] P. Balister, B. Bollobás, R. Morris, J. Sahasrabudhe and M. Tiba, On the Erdős covering problem: the density of the uncovered set, Invent. Math. 228 (2022), 377–414.

[6] J. Bourgain, On the Fourier–Walsh spectrum of the Moebius function, Israel J. Math. 197 (2013), 215–235.

[7] M. Braverman, Polylogarithmic independence fools AC^0 circuits, J. ACM 57 (2010), no. 5, Art. 28.

[8] H. Davenport, Multiplicative Number Theory, 3rd ed., Springer GTM 74, 2000.

[9] B. Efron and C. Stein, The jackknife estimate of variance, Ann. Statist. 9 (1981), 586–596.

[10] P. Erdős and J. Spencer, Lopsided Lovász local lemma and Latin transversals, Discrete Appl. Math. 30 (1991), 151–154.

[11] C. Elsholtz and T. Tao, Counting the number of solutions to the Erdős–Straus equation on unit fractions, J. Aust. Math. Soc. 94 (2013), 50–105 (Proposition numbering as in arXiv:1107.1010v6).

[12] M. Filaseta, K. Ford, S. Konyagin, C. Pomerance and G. Yu, Sieving by large integers and covering systems of congruences, J. Amer. Math. Soc. 20 (2007), 495–517.

[13] J. Friedlander and H. Iwaniec, Opera de Cribro, Amer. Math. Soc. Colloq. Publ. 57, 2010.

[14] P. X. Gallagher, A large sieve density estimate near $\sigma = 1$, Invent. Math. 11 (1970), 329–339.

[15] S. W. Graham and C. J. Ringrose, Lower bounds for least quadratic nonresidues, in: Analytic Number Theory, Progr. Math. 85, Birkhäuser, Boston, 1990, 269–309.
[16] B. Green, On (not) computing the Möbius function using bounded depth circuits, Combin. Probab. Comput. 21 (2012), 942–951.
[17] K. Henriot, Nair–Tenenbaum bounds uniform with respect to the discriminant, Math. Proc. Cambridge Philos. Soc. 152 (2012), 405–424 (arXiv:1102.1643).
[18] B. Haeupler, B. Saha and A. Srinivasan, New constructive aspects of the Lovász local lemma, J. ACM 58 (2011), no. 6, Art. 28.
[19] J. Håstad, Almost optimal lower bounds for small depth circuits, in: Proc. 18th ACM STOC (1986), 6–20.
[20] R. Hough, Solution of the minimum modulus problem for covering systems, Ann. of Math. (2) 181 (2015), 361–382.
[21] G. H. Hardy and E. M. Wright, An Introduction to the Theory of Numbers, 6th ed., Oxford, 2008.
[22] W. Hoeffding, A class of statistics with asymptotically normal distribution, Ann. Math. Statist. 19 (1948), 293–325.
[23] S. Janson, Poisson approximation for large deviations, Random Structures Algorithms 1 (1990), 221–229.
[24] V. Lecomte and L.-Y. Tan, Sharper bounds on the Fourier concentration of DNFs, in: Proc. 62nd IEEE FOCS (2021), arXiv:2109.04525.
[25] N. Linial, Y. Mansour and N. Nisan, Constant depth circuits, Fourier transform, and learnability, J. ACM 40 (1993), 607–620.
[26] H. L. Montgomery and R. C. Vaughan, Multiplicative Number Theory I. Classical Theory, Cambridge Stud. Adv. Math. 97, Cambridge University Press, 2007.
[27] H. L. Montgomery and R. C. Vaughan, Multiplicative Number Theory III, unpublished draft (course notes, R. C. Vaughan's Penn State web page, accessed 5 October 2026), §28.7–28.8.
[28] J. C. Lagarias, H. L. Montgomery and A. M. Odlyzko, A bound for the least prime ideal in the Chebotarev density theorem, Invent. Math. 54 (1979), 271–296.
[29] Anonymous, The m/n witness modulus at exponent $1/4$, working note (file `POINTWISE_MN.md` of the project repository), 2026, internally reviewed.
[30] Anonymous, Towards the admissibility hypothesis ADM_m, working note (file `POINTWISE_MN2.md` of the project repository), 2026, internally reviewed.
[31] L. J. Mordell, Diophantine Equations, Pure and Applied Mathematics 30, Academic Press, 1969.
[32] S. Mihnea and B. C. Dumitru, Further verification and empirical evidence for the Erdős–Straus conjecture, arXiv:2509.00128 (2025).
[33] M. Nair and G. Tenenbaum, Short sums of certain arithmetic functions, Acta Math. 180 (1998), 119–144.
[34] R. O'Donnell, Analysis of Boolean Functions, Cambridge University Press, 2014.
[35] Anonymous, Beyond low-conductor minorants? A Wiener-norm barrier at $\mathcal{L}^4$, working note (file `POINTWISE_OMEGA15.md` of the project repository), 2026, internally reviewed.
[36] Anonymous, Which standard-type hypothesis gives the exponent $1/3$?, working note (file `POINTWISE_OMEGA16.md` of the project repository), 2026, internally reviewed.
[37] R. Obláth, Sur l'équation diophantienne $4/n = 1/x_1 + 1/x_2 + 1/x_3$, Mathesis 59 (1950), 308–316.
[38] C. Pomerance and A. Weingartner, Exceptions to the Erdős–Straus–Schinzel conjecture, *Ramanujan J.* **69** (2026), no. 2, Paper No. 31, doi:10.1007/s11139-025-01312-2; arXiv:2511.16817 (2025; cited version).
[39] R. Peled, A. Yadin and A. Yehudayoff, The maximal probability that k-wise independent bits are all 1, Random Structures Algorithms 38 (2011), 502–525 (arXiv:0801.0059).

[40] A. Razborov, A simple proof of Bazzi's theorem, ACM Trans. Comput. Theory 1 (2009), no. 1, Art. 3.
[41] J. B. Rosser and L. Schoenfeld, Approximate formulas for some functions of prime numbers, Illinois J. Math. 6 (1962), 64–94.
[42] S. E. Salez, The Erdős–Straus conjecture: new modular equations and checking up to $N = 10^{17}$, arXiv:1406.6307 (2014).
[43] Anonymous, Sieve limits for the Erdős–Straus exceptional set: why 3/4 is sharp for congruence sieves, research note, version 4, 2026 (internally reviewed, not externally refereed).
[44] T. Tao, 254A, Notes 4: Some sieve theory, blog post, 21 January 2015.
[45] S. Kandukoori, S. Ramesh and P. Kumar, An improved bound for the exceptional set of the Erdős–Straus equation, preprint dated 29 July 2026, https://sushaan-k.github.io/erdos-straus/paper.pdf; Lean formalisation https://github.com/sushaan-k/erdos-straus-lean. Unrefereed as far as known.
[46] Anonymous, A three-quarter exceptional-set bound for the Erdős–Straus equation: an internally proved research note, 2026 (internally proved, not externally refereed); numbering as in the version of 5 October 2026 (archived as version 1, `paper/es-threequarter-note-v1.tex`).
[47] J. Thorner and A. Zaman, Refinements to the prime number theorem for arithmetic progressions, Math. Z. 306 (2024), Paper No. 54 (arXiv:2108.10878v2).
[48] J. Thorner and A. Zaman, An explicit bound for the least prime ideal in the Chebotarev density theorem, Algebra Number Theory 11 (2017), 1135–1197.
[49] K. Yamamoto, On the Diophantine equation $4/n = 1/x + 1/y + 1/z$, Mem. Fac. Sci. Kyushu Univ. Ser. A 19 (1965), 37–47 (not accessed; cited via [11]).
[50] R. C. Vaughan, On a problem of Erdős, Straus and Schinzel, Mathematika 17 (1970), 193–198.